

Report Number: R032-17

***MISO DPP 2016 February West Area
Phase 1 Study***

Prepared for

MISO

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08/11/2017

Siemens PTI Project Number 62OT-001424

Revision History

Date	Rev.	Description
08/07/2017	A	Final Draft Report
08/11/2017	B	Final Report

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Executive Summary

This report presents results of a System Impact Study (SIS) performed to evaluate the interconnection of the DPP 2016 February Phase 1 West Area Group (DPP West Area) generating facilities. The results for 2021 scenario are summarized below.

1.1 Project List

The DPP West Area study group has twenty eight (28) generation projects with a combined nameplate rating of 5690 MW. The DPP West Area generating facilities are listed in Table ES-1. The modeling details and projects' slider diagrams are shown in Appendix B.

Table ES-1: Generating Facilities in DPP 2016 February West Area Group

MISO Project #	Service Type	TO	County, State	Point Of Interconnection	Fuel Type	Max Output	SH MW	SPK MW	Stability MW
J432	NRIS	XEL	Brookings, SD	Brookings 345 kV	Wind	98	98	19.6	98
J460	NRIS	GRE	Lincoln, MN	Brookings-H081 345 kV line	Wind	200	200	40	200
J475	NRIS	MEC	Poweshiek, IA	Montezuma 345 kV	Wind	200	200	40	200
J485	NRIS	RPU	Olmsted, MN	West Side 161 kV	CT	46.85	0	46.85	46.85
J488	ERIS	OTP	Grant, Roberts, SD	Big Stone-Ellendale 345 kV	Wind	151.8	151.8	30.36	151.8
J489	ERIS	OTP	Grant, Roberts, SD	Big Stone-Ellendale 345 kV	Wind	151.8	151.8	30.36	151.8
J490	NRIS	MDU	Corson, Sioux, ND, SD	McIntosh Substation	Wind	60	60	12	60
J493	NRIS	OTP	Yellow Medicine, MN	Big Stone-Brookings 345 kV	Wind	150	150	30	150
J495	NRIS	ITCM	Winnebago, IA	Ledyard-Colby 345 kV line	Wind	200	200	40	200
J498	NRIS	MEC	Boone, Greene, IA	Grimes-Lehigh 345 kV	Wind	340	340	68	340
J499	NRIS	MEC	Adair, Madison, IA	Fallow-Grimes 345 kV	Wind	340	340	68	340
J500	NRIS	MEC	Adair, IA	New substation at intersection of Boone-Atchison 345 kV and Rolling Hills-Madison Co 345 kV	Wind	500	500	100	500
J501	NRIS	MEC	Hancock, Kossuth, IA	Kossuth Co 345 kV	Wind	500	500	100	500
J504	NRIS	ITCM	Linn, IA	Bertram-Duane Arnold 161 kV	Solar	50	0	50	50
J506	NRIS	MEC	Plymouth, IA	Raun-Lakefield Jct 345 kV	Wind	200	200	40	200

MISO Project #	Service Type	TO	County, State	Point Of Interconnection	Fuel Type	Max Output	SH MW	SPK MW	Stability MW
J510	NRIS	OTP	Deuel, SD	Scandinavia Township	CT	321.9 sum / 326.9 win	0	326.9	326.9
J514	NRIS	ITCM	Marshall, IA	Marshalltown substation	CC	30 sum / 65 win	32.5	65	65
J523	NRIS	ITCM	Mower, MN	Adams 161 kV	Solar	50	0	50	50
J524	NRIS	MEC	Webster, IA	Webster 161 kV	Solar	100	0	100	100
J525	NRIS	GRE	Murray, MN	Lake Wilson–Hadley 69 kV	Solar	50	0	50	50
J526	NRIS	OTP	Deuel, SD	Brooking Co-Big Stone South 345 kV	Wind	300	300	60	300
J527	NRIS	MEC	Adair, IA	Booneville-Cooper 345 kV	Wind	250	250	50	250
J528	NRIS	MEC	Adair, IA	Rolling Hills-Madison 345 kV	Wind	200	200	40	200
J529	NRIS	MEC	Palo Alto, IA	Obrien-Kossuth 345 kV	Wind	250	250	50	250
J530	NRIS	MEC	Iowa, IA	Montezuma-Hills 345 kV	Wind	250	250	50	250
J531	NRIS	ITCM	Chickasaw, IA	Hazleton-Adams 345 kV	Wind	200	200	40	200
J534	NRIS	MEC	Humboldt, Kossuth, IA	Kossuth–Webster 345 kV	Wind	250	250	50	250
J535	NRIS	MEC	Sac, IA	J411–Lehigh 345 kV	Wind	210	210	42	210

1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2016 February West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Except for generation projects J498, J499, J500, J501, and J529, all other non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

The reactive power requirement analysis results are summarized as following:

- J488, J489, and J493 do not meet the reactive power requirements per FERC Order 827.
- J498, J499, J500, J501, and J529 are exempted from the FERC Order 827 reactive power requirements.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of the System Impact Study report date. The total cost of network upgrades in the interconnection plan required for each generation project is listed in Table ES-2. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table ES-2: Total Cost of Network Upgrades for DPP 2016 February West Area Generation Projects

Project Num	ERIS Network Upgrades (\$)							NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrade (\$)	Total Cost (Exclude TOIF) (\$)	M2 Received (\$)	M3 (\$)
	Steady-State & Voltage Stability ¹	Transient Stability	Short Circuit	GRE LPC	CIPCO Affected System	PJM Affected System	SPP Affected System		TO Network Upgrades	TO - Owned Direct assigned				
J432	\$54,441,050	\$0	\$0	\$0	\$0	\$0	\$13,692,488	\$0	\$2,000,000	NA	\$0	\$56,441,050	\$691,615	\$4,952,490
J460	\$116,254,300	\$0	\$0	\$0	\$0	\$0	\$26,399,241	\$0	\$10,000,000	NA	\$0	\$126,254,300	\$1,281,664	\$11,343,766
J475	\$48,128,818	\$0	\$0	\$0	\$0	\$139,483	\$8,671,754	\$0	\$2,622,000	NA	\$0	\$50,750,818	\$729,210	\$4,345,872
J485	\$0	\$0	\$0	\$0	\$0	\$0	\$1,735,423	\$0	\$2,046,000	\$340,600	\$0	\$2,046,000	\$190,507	\$14,093
J488	\$107,877,355	\$0	\$0	\$0	\$0	\$48,800	\$0	\$0	\$500,000	NA	\$0	\$108,377,355	\$979,992	\$9,857,744
J489	\$107,877,355	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$500,000	NA	\$0	\$108,377,355	\$979,992	\$9,857,744
J490	\$31,911,305	\$0	\$0	\$0	\$0	\$0	\$10,374,124	\$5,750,000	\$4,800,000	NA	\$0	\$42,461,305	\$600,000	\$3,646,130
J493	\$101,371,431	\$0	\$0	\$0	\$0	\$0	\$10,211,502	\$0	\$3,774,376	\$1,073,340	\$0	\$105,145,808	\$1,005,589	\$9,508,992
J495	\$59,300,497	\$0	\$0	\$0	\$207,751	\$0	\$6,611,314	\$0	\$11,000,000	NA	\$0	\$70,300,497	\$1,279,452	\$5,750,598
J498	\$155,251,971	\$0	\$0	\$0	\$0	\$1,890,565	\$18,446,033	\$0	\$10,799,800	\$634,600	\$0	\$166,051,771	\$1,585,468	\$15,019,709
J499	\$101,469,513	\$0	\$0	\$0	\$0	\$2,030,723	\$24,038,065	\$0	\$11,750,200	\$634,600	\$0	\$113,219,713	\$1,572,468	\$9,749,503
J500	\$98,654,106	\$0	\$0	\$0	\$0	\$3,042,802	\$43,847,013	\$0	\$23,714,000	\$634,600	\$0	\$122,368,106	\$2,141,629	\$10,095,182
J501	\$164,258,976	\$0	\$0	\$0	\$0	\$6,848,760	\$20,529,679	\$0	\$1,172,800	\$634,600	\$0	\$165,431,776	\$2,259,629	\$14,283,549
J504	\$0	\$0	\$0	\$0	\$0	\$0	\$435,287	\$433,684	\$4,500,000	NA	\$492,371	\$5,426,055	\$242,863	\$299,743
J506	\$55,845,911	\$0	\$0	\$0	\$0	\$0	\$13,413,685	\$3,000,000	\$11,500,000	NA	\$0	\$70,345,911	\$1,166,452	\$5,868,139
J510	\$0	\$0	\$0	\$0	\$0	\$53,871	\$26,480,073	\$0	\$8,225,624	\$663,687	\$0	\$8,225,624	\$1,145,081	\$0
J514	\$9,138,209	\$0	\$0	\$0	\$20,903,608	\$0	\$0	\$45,566,316	\$15,000	NA	\$45,779	\$54,765,304	\$650,000	\$4,826,530
J523	\$0	\$0	\$0	\$0	\$0	\$0	\$1,944,822	\$0	\$1,200,000	NA	\$0	\$1,200,000	\$177,863	\$0
J524	\$0	\$0	\$0	\$0	\$0	\$0	\$6,231,835	\$0	\$1,000,000	NA	\$0	\$1,000,000	\$355,726	\$0
J525	\$0	\$0	\$0	\$26,900,000	\$0	\$0	\$3,596,792	\$0	\$3,000,000	NA	\$0	\$29,900,000	\$340,363	\$2,649,637
J526	\$213,600,301	\$0	\$0	\$0	\$0	\$1,658,000	\$31,883,855	\$0	\$11,250,000	NA	\$0	\$224,850,301	\$1,487,177	\$20,997,853
J527	\$42,234,467	\$0	\$0	\$0	\$0	\$0	\$25,279,629	\$0	\$11,500,000	NA	\$0	\$53,734,467	\$1,160,315	\$4,213,132
J528	\$33,578,105	\$0	\$0	\$0	\$0	\$0	\$20,255,831	\$0	\$11,500,000	NA	\$0	\$45,078,105	\$969,452	\$3,538,358
J529	\$76,846,489	\$0	\$0	\$0	\$0	\$0	\$15,978,543	\$0	\$11,328,700	\$634,600	\$0	\$88,175,189	\$1,440,315	\$7,377,204
J530	\$56,557,918	\$0	\$0	\$0	\$0	\$62,796	\$7,621,199	\$0	\$11,500,000	NA	\$0	\$68,057,918	\$959,315	\$5,846,477
J531	\$189,358,946	\$0	\$0	\$0	\$4,802,921	\$0	\$5,671,066	\$0	\$11,000,000	NA	\$1,262,944	\$201,621,890	\$968,452	\$19,193,737
J534	\$78,454,357	\$0	\$0	\$0	\$0	\$6,000,000	\$16,266,931	\$0	\$11,500,000	NA	\$0	\$89,954,357	\$1,890,315	\$7,105,121
J535	\$66,658,618	\$0	\$0	\$0	\$0	\$0	\$14,833,819	\$0	\$11,500,000	NA	\$0	\$78,158,618	\$1,522,177	\$6,293,685
Total (\$)	\$1,969,070,000	\$0	\$0	\$26,900,000	\$25,914,280	\$21,775,800	\$374,450,003	\$54,750,000	\$205,198,500	\$5,250,627	\$1,801,094	\$2,257,719,594	\$29,773,081	\$196,634,986

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

1.4 Per Project Summary

This section provides estimated cost of Network Upgrades on a per project basis for the 2021 scenario. The shared cost of Network Upgrades for all the generation projects are listed below.

The Interconnection Customers are required to mitigate the constraints observed from the 2021 scenario.

1.4.1 J432 Summary

Network Upgrade	Cost	J432	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$11,088,653	Backbone NU Big Stone South-Alexandria 345 kV
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$1,544,447	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$31,551,797	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$9,202,638	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$8,743	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$473,755	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$146,511	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$72,837	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$43,858	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$37,008	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$55,784	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$192,143	Other Reactive Power NU
Brookings Co-White 345 kV #2	\$60,000	\$9,816	Affected System Upgrade

Executive Summary

Network Upgrade	Cost	J432	NUs Type
Bondurant-Montezuma 345 kV	\$2,100,000	\$22,875	SH MISO
Aurora - Brookings 115kV CKT 1	\$8,800,000	\$879,146	SPP Affected System
Aurora - Flandreau 115kV CKT 1	\$13,600,000	\$1,358,680	SPP Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$5,613,560	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$95,563	SPP Affected System
Meadow Grove - North Petersburg 115kV CKT 1	\$17,500,000	\$2,597,004	SPP Affected System
Meadow Grove 230/115/13kV Transformer CKT 1	\$12,500,000	\$1,855,003	SPP Affected System
Pahoja - Sioux Falls 230kV CKT 1	\$14,000,000	\$1,293,532	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$68,143,354	

1.4.2 J460 Summary

Network Upgrade	Cost	J460	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$22,594,651	Backbone NU Big Stone South-Alexandria 345 kV
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$6,404,796	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$65,782,035	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$19,349,253	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$17,267	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$928,682	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$303,155	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$148,479	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$82,980	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$71,745	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$114,729	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$412,440	Other Reactive Power NU
Bondurant-Montezuma 345 kV	\$2,100,000	\$44,088	SH MISO
Aurora - Brookings 115kV CKT 1	\$8,800,000	\$1,574,608	SPP Affected System
Aurora - Flandreau 115kV CKT 1	\$13,600,000	\$2,433,486	SPP Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$11,203,905	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$185,397	SPP Affected System

Executive Summary

Network Upgrade	Cost	J460	NUs Type
Meadow Grove - North Petersburg 115kV CKT 1	\$17,500,000	\$5,007,802	SPP Affected System
Meadow Grove 230/115/13kV Transformer CKT 1	\$12,500,000	\$3,577,001	SPP Affected System
Pahoja - Sioux Falls 230kV CKT 1	\$14,000,000	\$2,417,042	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$142,653,541	

1.4.3 J475 Summary

Network Upgrade	Cost	J475	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$33,346,570	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$133,900	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$158,405	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$236,166	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$31,282	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$48,387	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$485,219	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$1,031,108	Other Reactive Power NU
J475 substation xfmr	IC Cost	IC Cost	SH MISO
J530 POI-Hills 345 kV	\$600,000	\$229,475	SH MISO
Henry Co-Denmark 161 kV	\$960,000	\$426,728	SH MISO
Wapello-Jeff 161 kV	\$27,000,000	\$12,001,578	SH MISO
Casey-Breed 345 kV	\$700,000	\$139,483	PJM Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$8,377,695	SPP Affected System
Council Bluffs - SUB 3456 345KV CKT 1	\$5,500,000	\$294,059	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$56,940,056	

1.4.4 J485 Summary

Network Upgrade	Cost	J485	NUs Type
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$1,568,967	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$19,100	SPP Affected System
Pahoja - Sioux Falls 230kV CKT 1	\$14,000,000	\$147,356	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for		\$1,735,423	

Executive Summary

Network Upgrade	Cost	J485	NUs Type
each Project			

1.4.5 J488 Summary

Network Upgrade	Cost	J488	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$29,088,417	Backbone NU Big Stone South-Alexandria 345 kV
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$2,229,731	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$57,188,398	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$16,414,337	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$12,381	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$690,231	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$221,299	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$102,938	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$49,804	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$51,278	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$84,968	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$312,338	Other Reactive Power NU
Brookings Co-White 345 kV #2	\$60,000	\$6,587	Affected System Upgrade
Hankinson-Forman 230 kV	\$500,000	\$250,000	SH MISO
Oakes-Forman 230 kV	\$750,000	\$375,000	SH MISO
Oakes-Ellendale 230 kV	\$1,550,000	\$775,000	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$31,237	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$48,800	PJM Affected System
Total Cost Per Project for ERIS for each Project		\$107,932,742	

1.4.6 J489 Summary

Network Upgrade	Cost	J489	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$29,088,417	Backbone NU Big Stone South-Alexandria 345 kV
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$2,229,731	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV

Executive Summary

Network Upgrade	Cost	J489	NUs Type
N Rochester-Rocky Run 345 kV	\$798,787,928	\$57,188,398	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$16,414,337	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$12,381	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$690,231	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$221,299	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$102,938	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$49,804	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$51,278	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$84,968	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$312,338	Other Reactive Power NU
Brookings Co-White 345 kV #2	\$60,000	\$6,587	Affected System Upgrade
Hankinson-Forman 230 kV	\$500,000	\$250,000	SH MISO
Oakes-Forman 230 kV	\$750,000	\$375,000	SH MISO
Oakes-Ellendale 230 kV	\$1,550,000	\$775,000	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$31,237	SH MISO
Total Cost Per Project for ERIS for each Project		\$107,883,942	

1.4.7 J490 Summary

Network Upgrade	Cost	J490	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$530,588	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$19,012,701	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$5,217,113	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$5,366	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$350,749	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$77,728	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$36,011	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$23,837	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$24,883	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$30,276	Other Reactive Power NU

Executive Summary

Network Upgrade	Cost	J490	NUs Type
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$86,476	Other Reactive Power NU
Bondurant-Montezuma 345 kV	\$2,100,000	\$15,576	SH MISO
Glenham 230-115-41.6 kV xfmr	\$6,500,000	\$6,500,000	SH MISO
Sheynne-Fargo 230 kV	\$750,000	\$750,000	NRIS
Merricourt-Ellendale 230 kV	\$5,000,000	\$5,000,000	NRIS
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$3,676,957	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$112,334	SPP Affected System
Meadow Grove - North Petersburg 115kV CKT 1	\$17,500,000	\$3,071,183	SPP Affected System
Meadow Grove 230/115/13kV Transformer CKT 1	\$12,500,000	\$2,193,702	SPP Affected System
Pahoja - Sioux Falls 230kV CKT 1	\$14,000,000	\$769,948	SPP Affected System
Fort Randall - Fort Thompson 230kV CKT 1	\$550,000	\$550,000	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$48,035,429	

1.4.8 J493 Summary

Network Upgrade	Cost	J493	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$32,376,892	Backbone NU Big Stone South-Alexandria 345 kV
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$2,325,313	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$50,424,832	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$14,651,349	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$13,077	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$713,132	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$222,951	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$109,038	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$62,431	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$55,039	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$85,072	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$298,409	Other Reactive Power NU
Brookings Co-White 345 kV #2	\$60,000	\$12,863	Affected System Upgrade
Bondurant-Montezuma 345 kV	\$2,100,000	\$33,897	SH MISO

Executive Summary

Network Upgrade	Cost	J493	NUs Type
Aurora - Brookings 115kV CKT 1	\$8,800,000	\$1,240,553	SPP Affected System
Aurora - Flandreau 115kV CKT 1	\$13,600,000	\$1,917,218	SPP Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$5,236,961	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$97,735	SPP Affected System
Pahoja - Sioux Falls 230kV CKT 1	\$14,000,000	\$1,719,035	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$111,595,796	

1.4.9 J495 Summary

Network Upgrade	Cost	J495	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$12,974,127	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$33,302,155	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$9,329,679	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$18,514	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$732,522	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$501,643	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$577,205	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$534,039	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$60,610	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$188,945	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$764,687	Other Reactive Power NU
Emery-Colby 161 kV	\$500,000	\$235,625	SH MISO
Franklin-Wall Lake 161 kV	\$500,000	\$80,747	SH MISO
Hazleton S-Dundee 161 kV	\$100,000	\$36,704	SH CIPCO LPC
Liberty-Hickory Crk 161 kV	\$500,000	\$171,047	SH CIPCO LPC
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$6,611,314	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$66,119,563	

Executive Summary

1.4.10 J498 Summary

Network Upgrade	Cost	J498	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$84,283,094	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$35,378,401	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$10,057,108	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$74,457	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$1,853,630	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$475,857	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$219,702	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$318,156	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$357,662	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$480,969	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$305,560	Other Reactive Power NU
J498 POI-Grimes 345 kV	\$2,500,000	\$894,680	SH MISO
Boone Jct-Sub T Fort Dodge 161 kV	\$7,500,000	\$7,500,000	SH MISO
Poweshiek-Reasnor 161 kV	\$25,000,000	\$6,605,378	SH MISO
Raun-S3451 345 kV	\$72,000,000	\$72,000,000	Affected System Upgrade
Raun-Tekamah MISO portion	\$5,000,000	\$2,933,428	SH MISO
Grimes-Sycamore 345 kV #2	\$3,100,000	\$1,371,146	SH MISO
Grimes-Granger Tap 161 kV	\$2,000,000	\$861,915	SH MISO
Granger Tap-108th & 54th 161 kV	\$1,250,000	\$565,844	SH MISO
Bondurant-Sycamore 345 kV	\$1,000,000	\$197,152	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$232,933	SH MISO
Franklin-Wall Lake 161 kV	\$500,000	\$284,898	SH MISO
Benton-T-094 345 kV	\$19,600,000	\$1,830,000	PJM Affected System
Walto-Electric Junction 345 kV	\$350,000	\$60,565	PJM Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$17,980,322	SPP Affected System
Council Bluffs - SUB 3456 345KV CKT 1	\$5,500,000	\$465,711	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$247,588,569	

1.4.11 J499 Summary

Network Upgrade	Cost	J499	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$56,802,663	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$22,427,562	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$6,319,874	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$86,489	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$2,284,772	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$353,759	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$83,309	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$201,206	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$437,774	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$624,773	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$138,590	Other Reactive Power NU
Poweshiek-Reasnor 161 kV	\$25,000,000	\$7,612,642	SH MISO
Council Bluffs-S3456 345 KV	\$2,000,000	\$990,401	Affected System Upgrade
Grimes-Sycamore 345 kV #2	\$3,100,000	\$1,728,854	SH MISO
Grimes-Granger Tap 161 kV	\$2,000,000	\$1,138,085	SH MISO
Granger Tap-108th & 54th 161 kV	\$1,250,000	\$684,156	SH MISO
Bondurant-Sycamore 345 kV	\$1,000,000	\$245,400	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$299,606	SH MISO
Benton-T-094 345 kV	\$19,600,000	\$1,806,000	PJM Affected System
Casey-Breed 345 kV	\$700,000	\$224,723	PJM Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$22,502,979	SPP Affected System
Council Bluffs - SUB 3456 345KV CKT 1	\$5,500,000	\$1,535,086	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$128,528,702	

1.4.12 J500 Summary

Network Upgrade	Cost	J500	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$41,583,335	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$30,533,022	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$8,614,761	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$118,703	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$3,775,465	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$485,996	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$114,496	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$274,134	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$607,533	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$882,508	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$175,753	Other Reactive Power NU
Poweshiek-Reasnor 161 kV	\$25,000,000	\$10,781,980	SH MISO
Council Bluffs-S3456 345 KV	\$2,000,000	\$1,009,599	Affected System Upgrade
Bondurant-Sycamore 345 kV	\$1,000,000	\$290,204	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$416,216	SH MISO
Benton-T-094 345 kV	\$19,600,000	\$2,627,000	PJM Affected System
Casey-Breed 345 kV	\$700,000	\$335,793	PJM Affected System
Walto-Electric Junction 345 kV	\$350,000	\$80,009	PJM Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$42,312,046	SPP Affected System
Council Bluffs - SUB 3456 345KV CKT 1	\$5,500,000	\$1,534,967	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$146,553,520	

1.4.13 J501 Summary

Network Upgrade	Cost	J501	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$34,558,377	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$89,694,530	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$25,853,219	Other NUs for MWEX Voltage Stability

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Network Upgrade	Cost	J501	NUs Type
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$61,607	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$2,305,291	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$965,020	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$789,162	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$742,881	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$255,018	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$348,843	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$1,162,102	Other Reactive Power NU
J498 POI-Grimes 345 kV	\$2,500,000	\$595,672	SH MISO
Killdeer 345-161 kV xfmr	\$6,500,000	\$6,500,000	SH MISO
Emery-Colby 161 kV	\$500,000	\$264,375	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$162,879	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$159,200	PJM Affected System
Benton–T-094 345 kV	\$19,600,000	\$2,741,000	PJM Affected System
Wilton–Dumont 765 kV	\$9,000,000	\$1,109,000	PJM Affected System
Crete–St John 345 kV	\$18,000,000	\$2,452,300	PJM Affected System
Wilton 765-345 kV transformers	\$11,000,000	\$294,500	PJM Affected System
Walto–Electric Junction 345 kV	\$350,000	\$92,760	PJM Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$20,529,679	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$191,637,415	

1.4.14 J504 Summary

Network Upgrade	Cost	J504	NUs Type
Arnold-Morgan Valley 161 kV	\$2,245,000	\$492,371	SNU
Marshalltown-Laurel-Jasper 161 kV	\$37,500,000	\$433,684	NRIS
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$435,287	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,361,342	

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1.4.15 J506 Summary

Network Upgrade	Cost	J506	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$8,181,832	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$34,813,691	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$10,087,492	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$25,403	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$1,246,724	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$301,358	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$186,569	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$184,540	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$117,242	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$144,504	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$286,584	Other Reactive Power NU
Raun-Tekamah MISO portion	\$5,000,000	\$194,723	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$75,250	SH MISO
Dickenson Co-Triboji 161 kV	\$3,000,000	\$3,000,000	NRIS
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$13,413,685	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$72,259,596	

1.4.16 J510 Summary

Network Upgrade	Cost	J510	NUs Type
Walto–Electric Junction 345 kV	\$350,000	\$53,871	PJM Affected System
Aurora - Brookings 115kV CKT 1	\$8,800,000	\$2,662,227	SPP Affected System
Aurora - Flandreau 115kV CKT 1	\$13,600,000	\$4,114,350	SPP Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$15,804,709	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$209,738	SPP Affected System
Pahoja - Sioux Falls 230kV CKT 1	\$14,000,000	\$3,689,049	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$26,533,944	

1.4.17 J514 Summary

Network Upgrade	Cost	J514	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$1,844,385	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$1,224,193	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$332,839	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$5,041	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$89,159	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$60,561	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$2,830	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$46,428	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$11,388	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$31,494	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$82,500	Other Reactive Power NU
Marshalltown-Blairstown 115 kV	\$2,700,000	\$2,700,000	SH MISO
Blairstown-Prairie Crk 115 kV	\$2,700,000	\$2,700,000	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$7,391	SH MISO
CIPCO IR20-Dundee 161 kV	\$6,000,000	\$139,280	SH CIPCO LPC
Prairie Crk-Marion 115 kV	\$50,000	\$50,000	SH CIPCO LPC
Prairie Crk-Bertram 115 kV	\$16,000,000	\$16,000,000	SH CIPCO LPC
Prairie Crk-Squared 115 kV	\$50,000	\$50,000	SH CIPCO LPC
Bertram 161-115 kV xfmr	\$4,000,000	\$4,000,000	SH CIPCO LPC
Rosehollow-Bertram 161 kV	\$75,000	\$10,413	SH CIPCO LPC
Liberty-Dundee 161 kV #2	\$5,000,000	\$615,949	SH CIPCO LPC
Liberty-Hickory Crk 161 kV	\$500,000	\$37,966	SH CIPCO LPC
Jasper-Laurel 161 kV	\$600,000	\$45,779	SNU
Marshalltown-Laurel-Jasper 161 kV	\$37,500,000	\$37,066,316	NRIS
Marshalltown-Plaza-Timber Crk 161 kV	\$8,500,000	\$8,500,000	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$75,653,912	

1.4.18 J523 Summary

Network Upgrade	Cost	J523	NUs Type
J523 substation xfmr	IC Cost	IC Cost	SPK MISO
J523 161 kV interconnection line	IC Cost	IC Cost	SPK MISO
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$1,944,822	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,944,822	

1.4.19 J524 Summary

Network Upgrade	Cost	J524	NUs Type
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$6,178,730	SPP Affected System
Council Bluffs - SUB 3456 345KV CKT 1	\$5,500,000	\$53,105	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$6,231,835	

1.4.20 J525 Summary

Network Upgrade	Cost	J525	NUs Type
33 miles of 69 kV lines between Chandler tap and Lyon County	\$23,000,000	\$23,000,000	GRE LPC
Fenton 115-69 kV transformer	\$3,900,000	\$3,900,000	GRE LPC
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$2,978,516	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$35,244	SPP Affected System
Pahoja - Sioux Falls 230kV CKT 1	\$14,000,000	\$583,032	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$30,496,792	

1.4.21 J526 Summary

Network Upgrade	Cost	J526	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$72,694,379	Backbone NU Big Stone South-Alexandria 345 kV
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$4,622,240	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$102,391,430	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$29,720,925	Other NUs for MWEX Voltage Stability

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Network Upgrade	Cost	J526	NUs Type
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$25,930	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$1,417,031	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$444,972	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$216,262	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$121,467	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$108,903	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$169,910	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$599,879	Other Reactive Power NU
Brookings Co-White 345 kV #2	\$60,000	\$24,147	Affected System Upgrade
Panther-McLeod 230 kV	\$1,000,000	\$1,000,000	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$66,975	SH MISO
Benton-T-094 345 kV	\$19,600,000	\$1,658,000	PJM Affected System
Aurora - Brookings 115kV CKT 1	\$8,800,000	\$2,443,466	SPP Affected System
Aurora - Flandreau 115kV CKT 1	\$13,600,000	\$3,776,266	SPP Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$10,387,405	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$197,406	SPP Affected System
Meadow Grove - North Petersburg 115kV CKT 1	\$17,500,000	\$6,824,012	SPP Affected System
Meadow Grove 230/115/13kV Transformer CKT 1	\$12,500,000	\$4,874,294	SPP Affected System
Pahoja - Sioux Falls 230kV CKT 1	\$14,000,000	\$3,381,006	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$247,166,304	

1.4.22 J527 Summary

Network Upgrade	Cost	J527	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$19,205,450	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$15,190,934	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$4,285,039	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$60,662	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$1,842,591	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$245,089	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$57,353	Other Reactive Power NU

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Network Upgrade	Cost	J527	NUs Type
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$138,100	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$309,472	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$448,470	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$88,108	Other Reactive Power NU
Bondurant-Sycamore 345 kV	\$1,000,000	\$151,162	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$212,034	SH MISO
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$24,522,061	SPP Affected System
Council Bluffs - SUB 3456 345KV CKT 1	\$5,500,000	\$757,568	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$67,514,096	

1.4.23 J528 Summary

Network Upgrade	Cost	J528	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$15,209,024	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$12,116,470	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$3,413,733	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$48,570	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$1,475,715	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$195,638	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$45,742	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$110,089	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$247,920	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$359,348	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$69,930	Other Reactive Power NU
Bondurant-Sycamore 345 kV	\$1,000,000	\$116,082	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$169,844	SH MISO
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$19,650,093	SPP Affected System
Council Bluffs - SUB 3456 345KV CKT 1	\$5,500,000	\$605,738	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$53,833,936	

1.4.24 J529 Summary

Network Upgrade	Cost	J529	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$13,919,483	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$46,026,263	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$13,301,941	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$30,431	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$1,199,839	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$467,017	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$368,032	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$344,245	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$127,539	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$170,131	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$550,097	Other Reactive Power NU
J498 POI-Grimes 345 kV	\$2,500,000	\$260,051	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$81,420	SH MISO
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$15,978,543	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$92,825,032	

1.4.25 J530 Summary

Network Upgrade	Cost	J530	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$32,357,780	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$167,350	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$26,162	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$343,621	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$39,701	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$61,408	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$235,959	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$1,963,719	Other Reactive Power NU
J438 POI-Parnell 161 kV	\$200,000	\$200,000	SH MISO
J530 POI-Hills 345 kV	\$600,000	\$370,525	SH MISO

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Network Upgrade	Cost	J530	NUs Type
Henry Co-Denmark 161 kV	\$960,000	\$533,272	SH MISO
Henry Co-Jeff 161 kV	\$960,000	\$960,000	SH MISO
Wapello-Jeff 161 kV	\$27,000,000	\$14,998,422	SH MISO
Sub 56-Walcott 345 kV	\$4,000,000	\$4,000,000	SH MISO
Walcott-Sub 92 345 kV	\$300,000	\$300,000	SH MISO
Walto–Electric Junction 345 kV	\$350,000	\$62,796	PJM Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$7,367,433	SPP Affected System
Council Bluffs - SUB 3456 345KV CKT 1	\$5,500,000	\$253,766	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$64,241,913	

1.4.26 J531 Summary

Network Upgrade	Cost	J531	NUs Type
Big Stone South-Alexandria 345 kV	\$197,400,000	\$468,589	Backbone NU Big Stone South-Alexandria 345 kV
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$48,998,474	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$23,567,865	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$4,464,662	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$5,671	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$302,858	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$952,781	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$163,163	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$41,667	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$309,915	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$1,383,301	Other Reactive Power NU
J531 POI-Hazleton 345 kV	\$56,000,000	\$56,000,000	SH MISO
Adams S-Beaver Crk 161 kV	\$14,700,000	\$14,700,000	SH MISO
Seneca-Genoa-Gran Grae 161 kV	\$38,000,000	\$38,000,000	SH MISO
Rosehollow-Bertram 161 kV	\$75,000	\$64,587	SH CIPCO LPC
Hazleton S-Dundee 161 kV	\$100,000	\$63,296	SH CIPCO LPC
Liberty-Dundee 161 kV #2	\$5,000,000	\$4,384,051	SH CIPCO LPC

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Network Upgrade	Cost	J531	NUs Type
Liberty-Hickory Crk 161 kV	\$500,000	\$290,987	SH CIPCO LPC
Hazleton-Mitchell Co 345 kV	\$2,750,000	\$1,262,944	SNU
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$5,623,582	SPP Affected System
Grand Island - Grand Praire 345kV CKT 1	\$1,000,000	\$47,484	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$201,095,876	

1.4.27 J534 Summary

Network Upgrade	Cost	J534	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$19,159,783	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$42,519,500	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$12,229,565	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$33,158	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$1,178,807	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$469,419	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$364,938	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$358,115	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$140,465	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$175,201	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$538,620	Other Reactive Power NU
J498 POI-Grimes 345 kV	\$2,500,000	\$317,178	SH MISO
Raun-Tekamah MISO portion	\$5,000,000	\$879,554	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$90,056	SH MISO
Lee County-Byron 345 kV	\$6,000,000	\$6,000,000	PJM Affected System
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$16,266,931	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$100,721,288	

1.4.28 J535 Summary

Network Upgrade	Cost	J535	NUs Type
New Hartford-Morgan Valley-Beverly-Sub 92 345 kV	\$468,100,000	\$25,788,778	Backbone NU New Hartford-Morgan Valley-Beverly-Sub 92 345 kV
N Rochester-Rocky Run 345 kV	\$798,787,928	\$28,453,753	New 345 kV line North Roshester - Rocky Run
Other NUs for MWEX Voltage Stability	\$227,412,072	\$8,152,209	Other NUs for MWEX Voltage Stability
2*50 Mvar cap at Zachary 161 kV (344010)	\$1,000,000	\$34,898	Other Reactive Power NU
5*50 Mvar SVC at Overton 161 kV (345409)	\$25,000,000	\$1,264,247	Other Reactive Power NU
2*150 Mvar cap at Hickory 345 kV (631191)	\$8,000,000	\$308,162	Other Reactive Power NU
1*150 Mvar cap at New Hartford 345 kV (631240)	\$4,000,000	\$168,013	Other Reactive Power NU
1*150 Mvar cap at Blackhawk 345 kV (636199)	\$4,000,000	\$204,093	Other Reactive Power NU
1*150 Mvar cap at Montezuma 345 kV (635730)	\$4,000,000	\$164,400	Other Reactive Power NU
2*150 Mvar cap at Hills 345 kV (636400)	\$8,000,000	\$214,365	Other Reactive Power NU
2*150 Mvar cap at Morgan Valley 345 kV (631148)	\$8,000,000	\$240,146	Other Reactive Power NU
J498 POI-Grimes 345 kV	\$2,500,000	\$432,419	SH MISO
Raun-Tekamah MISO portion	\$5,000,000	\$992,295	SH MISO
Bondurant-Montezuma 345 kV	\$2,100,000	\$106,486	SH MISO
Franklin-Wall Lake 161 kV	\$500,000	\$134,355	SH MISO
Cooper - Stranger Creek 345kV CKT 1	\$301,000,000	\$14,833,819	SPP Affected System
Total Cost Per Project for Actual NRIS Elections for each Project		\$81,492,437	

1.5 Study Compliance with NERC FAC-002-2 Standard

This DPP 2016 February West Area study was completed in compliance with NERC FAC-002-2:

R1.1: The reliability impact of the new interconnection, or materially modified existing interconnection, on affected system(s).

Section 3 covers summer peak steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 4 covers summer shoulder steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 5.1 covers reliability impact of the new generating facilities in the CIPCO and MPC affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.2 covers reliability impact of the J525 generating facility per GRE Local planning Criteria (LPC). Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.3 covers reliability impact of the new generating facilities in the PJM affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 5.4 covers reliability impact of the new generating facilities in the SPP affected systems.

Section 6 covers transient stability analysis results.

Section 7 covers voltage stability (PV) analysis on the MWEX System Operating Limit (SOL). Network Upgrades required for MWEX voltage stability are identified.

Section 8 covers short circuit reliability impact of the new generating facilities.

Section 9 covers Deliverability reliability impact of the new NRIS generating facilities.

R1.2: Adherence to applicable NERC Reliability Standards; regional and Transmission Owner planning criteria; and Facility interconnection requirements.

Sections 2.2-2.4, Section 5, and Section 6 all cover NERC Reliability Standard TPL-001-4.

Section 5.1 covers CIPCO and MPC's Local Planning Criteria (LPC).

Section 5.2 covers GRE LPC.

Section 5.3 covers PJM system planning criteria.

Section 5.4 covers SPP system planning criteria.

Section 7 (voltage stability analysis) covers individual system planning criteria (ATC).

Section 9 covers MISO system planning criteria.

R1.3: Steady-state, short-circuit, and dynamics studies, as necessary, to evaluate system performance under both normal and contingency conditions.

Section 3 and Section 4 cover MISO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

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Section 5.1 covers CIPCO and MPC's LPC steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.2 covers GRE's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.3 covers PJM steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 5.4 covers SPP steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 6 covers transient stability studies under NERC category P0 to P7 contingencies (TPL-001-4).

Section 8 covers short-circuit assessment.

Section 9 covers MISO deliverability study (steady-state assessment) including NERC category P0 to P7 contingencies (TPL-001-4).

R1.4: Study assumptions, system performance, alternatives considered, and coordinated recommendations. While these studies may be performed independently, the results shall be evaluated and coordinated by the entities involved.

Section 2.1, Section 2.2, Section 2.3, Section 2.4, Section 6.2, Section 6.3, and Section 6.4 cover study assumptions and system performance criteria.

Jointly coordinated recommendations can be found in Sections 5.1 (MISO, CIPCO, and MPC), Section 5.2 (MISO and GRE), Section 5.3 (MISO and PJM), Section 5.4 (MISO and SPP), and Section 7 (MISO and ATC). Results in Section 3, 4, 5, 6, 8, and 9 have also been reviewed by PJM, SPP, CIPCO, and MPC.

Introduction

Twenty eight (28) generation projects, listed in Appendix A.1, Table A-1, have requested to interconnect to the MISO transmission network in the West Area and have advanced to the Definitive Planning Phase (DPP) 2016 February Phase 1 study (DPP West Area). Generating facilities J488 and J489 have requested Energy Resource Interconnection Service (ERIS), and other generating facilities have requested both ERIS and Network Resource Interconnection Service (NRIS).

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generating facilities in the DPP West Area Phase 1 study.

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

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Model Development and Study Criteria

2.1 Model Development

2.1.1 Benchmark Cases

DPP 2016 February West area power flow benchmark cases representing 2021 summer shoulder and summer peak conditions were developed from the corresponding models used in the DPP 2015 August West Area study, which were originally developed from MTEP15 models.

The benchmark cases for DPP 2016 February study were created as follows:

- MISO Prior queued generation projects and their associated Network Upgrades (NU) were modeled.
- DPP 2016 February generation projects in the West Area (DPP West Area, Table A-1) were modeled with offline status.
- DPP 2016 February generation projects in the Central Area (Table A-4), Michigan Area (Table A-5), Wisconsin Area (Table A-6), and South Area (Table A-7) were modeled and dispatched.
- For MISO generation projects in South area, their output was sunk to the MISO South (Appendix A.4, Table A-10), where generation was scaled in proportion to $P_{gen} - P_{min}$; For MISO generation projects in other areas, their output was sunk to the MISO Classic (Appendix A.3, Table A-9), where generation was scaled in proportion to $P_{gen} - P_{min}$;
- PJM generation projects were modeled and dispatched. The generation output was sunk to the PJM market (Appendix A.5, Table A-11), where generation was scaled in proportion to $P_{gen} - P_{min}$.
- SPP generation projects were modeled and dispatched. The generation output was sunk to the SPP market (Appendix A.6, Table A-12), where generation was scaled in proportion to $P_{gen} - P_{min}$.
- The Hickory Creek – Eden Prairie – Cardinal 345 kV project (MVP project 3127) was also included in the 2021 models; the Hickory Creek – Eden Prairie – Cardinal 345 kV project has an in-service date of 12/31/2023.
- Models were further reviewed by the Ad Hoc study members (transmission owners and customers). Model corrections and changes were made based on the comments and feedback. These modeling changes are listed in Appendix A.2.
- Adjusted Square Butte DC to match the total output of the Bison wind farms.
- Adjusted CU DC to match the total output of Coal Creek generation units #1 and #2.
- Set MHEX interface transfer level at 2431 MW for 2021 cases.

2.1.2 Study Cases

Summer peak study case was created by dispatching the DPP West Area generating facilities at the specified summer peak level (Table ES-1) from the benchmark cases.

Summer shoulder study case was created by dispatching the DPP West Area generating facilities at the specified summer shoulder level (Table ES-1) from the benchmark cases. Due to voltage collapse under system intact condition and King – Eau Claire – Arpin (most critical contingency) post contingency condition, eight (8) fictitious large size SVCs (Table 2-1) were added to the summer shoulder study case to achieve converged power flow thermal solutions.

Table 2-1: Fictitious SVCs Added Only in Summer Shoulder Case

Location	Bus #	SVC Mvar
Pershing 345 kV	694107	500
North Rochester 345 kV	601039	500
Montezuma 345 kV	635730	500
Killdeer 345 kV	631199	500
Hazleton 345 kV	631139	500
Hickory Creek 345 kV	631191	500
Eden 345 kV	693863	500
Arnold 161 kV	631088	500

The MISO Classic was used for power balance, where generation was scaled down in proportion to $P_{gen} - P_{min}$.

Both study and benchmark power flow cases were solved with transformer tap adjustment enabled, area interchange disabled, phase shifter adjustment enabled and switched shunt adjustment enabled.

The interface transfer levels in the 2020 study cases are summarized in Table 2-2.

Table 2-2: Interface Transfer Levels in 2021 Steady State Study Cases

Interface	2021 SH Case (MW)	2021 SPK Case (MW)
MHEX	2431	2431
MWEX	1928	1127
Arrowhead – Stone Lake 345 kV	840	455

2.2 Contingency Criteria

A variety of contingencies were considered for steady-state analysis:

- NERC Category P0 with system intact (no contingencies)
- NERC Category P1 contingencies
 - NERC Category P1 contingencies, at buses with a nominal voltage of 69 kV and above, in the following areas: CWLD (area 333), AMMO (area 356), AMIL (area 357), CWLP (area 360), SIPC (area 361), WEC (area 295), WEC MI (area 296), XCEL (area 600), MP (area 608), SMMPA (area 613), GRE (area 615), OTP (area 620), ITCM (area 627), MPW (area 633), MEC (area 635), MDU (area 661), MHEB (area 667), DPC (area 680), ALTE (area 694), WPS (area 696), MGE (area 697), UPPC (area 698), CE (area 222), NPPD (area 640), OPPD (area 645), LES (area 650), WAPA (area 652), AECI (area 330), MIPU (area 540), KCPL (area 541), KACY (area 542), INDN (area 545).
 - Multiple-element NERC Category P1 contingencies, in Dakotas, Illinois, Iowa, Manitoba, Minnesota, Missouri, and Wisconsin. These specified Category P1 contingencies are listed in Appendix A.7.
- NERC Category P2-P7 contingencies
 - Selected NERC Category P2-P7 contingencies provided by the Ad Hoc Study Group, in the study region of Dakotas, Illinois, Iowa, Manitoba, Minnesota, Missouri, and Wisconsin. These specified Category P2-P7 contingencies are listed in Appendix A.7.

For all contingency and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.3 Monitored Elements

The study area is defined in Table 2-3. Facilities in the study area were monitored for system intact and contingency conditions. Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal (PSS®E rate A) rating. Under NERC category P1-P7 conditions, branches were monitored for loading as shown in the column labeled "Post-Disturbance Thermal Limits".

Table 2-3: Monitored Elements

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
AECI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
AMIL	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.0	1.05/0.95
AMMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.0	1.05/0.95
ATCLLC	69 kV and above	95% of Rate A	95% of Rate B	1.05/0.95	1.1/0.9
CWLD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
CWLP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.0	1.1/0.95
CE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
DPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
GMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
GRE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.92

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
INDN	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
ITCM	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/1.07/0.93
KACY	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
KCPL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
LES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
MDU	57 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
MEC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.96/0.95	1.05/0.96/0.95 ³
MHEB	69 kV and above	100% of Rate A	100% of Rate B	1.12/1.1/1.07/1.05/1.04/0.99/0.97/0.96/0.95	1.15/1.1/0.94/0.9
MP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.0	1.1/0.95
MPW	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.95
NPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
OPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
OTP	40 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.97	1.1/0.92
PPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.95
SIPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.91
SMMPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
WAPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.1/0.9
XEL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.92

Notes

- 1: PSS®E Rate A, Rate B or Rate C
- 2: Limits dependent on nominal bus voltage
- 3: For facilities in Cedar Falls Utilities or Ames Municipal Utilities, post-contingency voltage limits are 1.05/0.94 for >200 kV, and 1.05/0.93 for others.

2.4 Performance Criteria

A branch is considered as a thermal injection constraint if the branch is loaded above its applicable normal or emergency rating for the post-change case, and any of the following conditions are met:

- 1) the generator (NR/ER) has a larger than 20% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, or
- 2) the megawatt impact due to the generator is greater than or equal to 20% of the applicable rating (normal or emergency) of the overloaded facility, or
- 3) the overloaded facility or the overload-causing contingency is at generator's outlet, or

- 4) for any other constrained facility, where none of the study generators meet one of the above criteria in 1), 2), or 3), however, the cumulative megawatt impact of the group of study generators (NR/ER) is greater than 20% of the applicable rating, then only those study generators whose individual MW impact is greater than 5% of the applicable rating and has DF greater than 5% (OTDF or PTDF) will be responsible for mitigating the cumulative MW impact constraint.

A bus is considered a voltage constraint if both of the following conditions are met. All voltage constraints must be resolved before a project can receive interconnection service.

- 1) the bus voltage is outside of applicable normal or emergency limits for the post-change case, and
- 2) the change in bus voltage is greater than 0.01 per unit.

All DPP 2016 February study generators must mitigate thermal injection constraints and voltage constraints in order to obtain unconditional Interconnection Service.

Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the deliverability algorithm, to meet the system performance criteria for NERC category P0-P1 events, if the constraint demonstrates an incremental flow caused by the generator equal to or greater than 5% of the generator's maximum dispatch level in each case.

2.5 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2016 February West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Except for generation projects J498, J499, J500, J501, and J529, all other non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

Collector system and shunt compensation of DPP West projects are modeled, which are listed in Appendix A.1, Table A-3. An analysis was performed to study the reactive power requirements (FERC Order 827) for the non-synchronous generation projects in the DPP 2016 February West study group. The analysis was performed as follows:

Step 1: Verify whether total dynamic reactive power (reactive power from generators and dynamic compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the generator terminal bus. The verification in Step 1 was performed when generator data was submitted and modeled.

Step 2: Verify whether total reactive power (reactive power from generators, dynamic compensation devices, and static compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the high-side of the generator substation. The testing procedure in Step 2 is described in the following:

- Lock the high-side of the generator substation at 1.0 pu voltage by adding a fictitious SVC. This is to ensure that the test result is not affected by the system condition.
- Lock the reactive power output of the generator to the maximum limit (Q_{max}). Make sure all shunt compensation devices within the substation are at the maximum capacitive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 lagging requirement.
- Lock the reactive power output of the generator to the minimum limit (Q_{min}). Make sure all shunt compensation devices within the substation are at the maximum reactive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 leading requirement.

Appendix C lists reactive power requirement analysis results for the DPP West generation projects. The results are summarized as following:

- J488, J489, and J493 do not meet the reactive power requirements per FERC Order 827.
- J498, J499, J500, J501, and J529 are exempted from the FERC Order 827 reactive power requirements.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

Summer Peak Steady-State Analysis

Summer peak steady-state analysis was performed in summer peak scenario to identify thermal and voltage upgrades required interconnecting the generating facilities in the DPP 2016 February West Area group to the transmission system.

3.1 Study Procedure

3.1.1 Computer Programs

Steady-state analyses were performed using PSS®E version 33.9 and PSS®MUST version 12.0.

3.1.2 Study Methodology

Summer peak power flow case was created in the procedure as described in Section 2.1. The summer peak case can converge under post-contingency conditions, so fictitious SVCs are not modeled. Nonlinear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any summer peak constraints.

3.2 Summer Peak Contingency Analysis Results

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP West Area projects) and study case (with DPP West Area projects). Analysis was performed in the 2021 summer peak scenario using PSS®E and PSS®MUST.

3.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table D-1, and no voltage constraints were identified (Table D-2).

3.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies in the 2021 summer peak scenario, thermal constraints are listed in Table D-3, and no voltage constraints were identified (Table D-4).

One category P2-P7 contingencies (Table D-7) were not converged, and their dc thermal results are listed in Table D-8. The contingency was not converged in the benchmark or study cases. No mitigation plan is required for the study projects for this contingency.

For P2-P7 contingencies in the 2021 summer peak scenario, thermal constraints are listed in Table D-5, and no voltage constraints were identified Table D-6.

3.3 Network Upgrades Identified in MISO ERIS Analysis for 2021 Summer Peak Scenario

Table 3-1 lists worst thermal constraints, Network Upgrades and their costs in the 2021 summer peak scenario.

Table 3-1: 2021 Summer Peak Thermal Constraints, Network Upgrades and Costs, Maximum Screened Loading

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Mitigation	Cost (\$)
			(MVA)	(%)				
J523 161-34.5-13.2 kV xfmr	60.0	J523	61.4	102.3	CEII Redacted	P0	J523 is responsible	NA
J523 - Adams North 161 kV	60.0	J523	60.1	100.2	CEII Redacted	P0	J523 is responsible	NA
Split Rock-White 345 kV	717.1	XEL WAPA	770.2	107.4	CEII Redacted	P1	Line is currently rated 1075 MVA for SN/SE no mitigation required	\$0
Split Rock-White 345 kV	717.1	XEL WAPA	743.8	103.7	CEII Redacted	P2-P7	Line is currently rated 1075 MVA for SN/SE no mitigation required	\$0

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Summer Shoulder Steady-State Analysis

Summer shoulder steady-state analysis was performed in summer shoulder scenario to identify thermal and voltage upgrades required interconnecting the generating facilities in the DPP 2016 February West Area group to the transmission system.

4.1 Study Procedure

Summer shoulder power flow case was created in the procedure as described in Section 2.1. Due to voltage collapse and severe thermal overloads in the initial power flow case, steady-state analysis was performed in the following four-step procedure:

- Step 1: With 8 fictitious large size SVCs (Table 2-1) added to the case, non-linear (AC) thermal analysis (1st round ACCC) was performed to identify thermal constraints.
- Step 2: Based on the identified thermal constraints in the 1st round ACCC, and based on voltage collapse under system intact condition and King – Eau Claire – Arpin (most critical contingency) post contingency condition, perform project justification analysis to determine Network Upgrades (NUs) required for interconnection of DPP West projects. These selected NUs are called Backbone NUs.
- Step 3: The fictitious SVCs were removed and the Backbone NUs were added to the summer shoulder case. 2nd round ACCC was performed to identify any remaining thermal and voltage constraints.
- Step 4: Based on the MWEX voltage stability analysis performed by ATC (Section 7), reactive power NUs were updated and verified.

4.2 Step 1 – 1st Round ACCC Analysis

Eight fictitious large size SVCs (Table 2-1) were added to the summer shoulder case. The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows between benchmark case (without DPP West Area projects or fictitious SVCs) and study case (with DPP West Area projects and fictitious SVCs). Analysis was performed in the 2021 summer shoulder scenario using PSS[®]E and PSS[®]MUST.

4.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-1.

4.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies.

One category P1 contingency was not converged, which is Big Stone South - J488&J489 POI 345 kV contingency. This contingency was solved in the benchmark case but not in the study case. Short Circuit Ratio (SCR) is a measure of system strength relative to the rating of the wind plant and is used by manufacturers to screen for weak grid risks. Under this N-1 contingency condition, Composite SCR (CSCR) of J488 and J489 is around 1.022 (Table 4-1). Further study is required to provide evidence that the J488 and J489 wind farms can operate at the CSCR of 1.022.

Table 4-1: Composite SC MVA and SCR of J488 and J489

System Condition	Composite SC MVA	Composite SCR
P0: System Intact	1825	1.022
P1: Big Stone South-POI of J488, J489	617	3.024

For P1 contingencies in the 2021 summer shoulder scenario, thermal constraints are listed in Table E-2.

One category P2-P7 contingency (Table E-4) was not converged. The contingency was not converged in the benchmark or study cases. No mitigation plan is required for the study projects for this non-converged contingency.

For P2-P7 contingencies in the 2021 summer shoulder scenario, thermal constraints are listed in Table E-3.

For non-converged contingencies, their dc thermal results are listed in Table E-5.

4.2.3 Worst Thermal Constraints in the 1st Round ACCC

Table 4-2 lists worst thermal constraints identified in the 1st round ACCC for 2021 summer shoulder scenario.

Table 4-2: 2021 Shoulder Thermal Constraints, Maximum Screened Loading, 1st Round ACCC

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J438 POI-Parnell 161 kV	282.0	ITCM MEC	346.7	122.9	CEII Redacted	P1	J530
J531 POI-Hazleton 345 kV	1000.0	ITCM	1199.5	120.0	CEII Redacted	P1	J531
J531 POI-Hazleton 345 kV	1000.0	ITCM	1276.3	127.6	CEII Redacted	P2-P7	J531
J506 POI-Raun 345 kV	864.0	MEC	946.1	109.5	CEII Redacted	P2-P7	J501,J506,J529,J534
J475 substation xfmr	212	J475	218.2	102.9	CEII Redacted	P0	J475
J475 substation xfmr	212.0	J475	215.5	101.6	CEII Redacted	P1	J475
J475 substation xfmr	212.0	J475	214.5	101.2	CEII Redacted	P2-P7	J475
J498 POI-Grimes 345 kV	956	MEC	1159.8	121.3	CEII Redacted	P0	J432,J460,J488,J489,J493, J498,J501,J506,J526,J529, J534,J535,J495

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Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J498 POI-Grimes 345 kV	1152.0	MEC	1355.4	117.7	CEII Redacted	P1	J498,J534,J535
J498 POI-Grimes 345 kV	1152.0	MEC	1484.0	128.8	CEII Redacted	P2-P7	J498,J534,J535
J530 POI-Montezuma 345 kV	1010.0	MEC	1099.1	108.8	CEII Redacted	P1	J475
J530 POI-Montezuma 345 kV	1010.0	MEC	1075.4	106.5	CEII Redacted	P2-P7	J475
J530 POI-Hills 345 kV	956	MEC	974.5	101.9	CEII Redacted	P0	J475,J498,J499,J500,J506, J527,J528,J530,J534,J535
J530 POI-Hills 345 kV	1010.0	MEC	1354.8	134.1	CEII Redacted	P1	J475,J530
J530 POI-Hills 345 kV	1010.0	MEC	1333.9	132.1	CEII Redacted	P2-P7	J475,J530
Split Rock-White 345 kV	717.1	XEL WAPA	984.5	137.3	CEII Redacted	P1	J432,J460,J488,J489,J493, J526
Split Rock-White 345 kV	717.1	XEL WAPA	959.0	133.7	CEII Redacted	P2-P7	J432,J460,J488,J489,J493, J526
King-Eau Claire 345 kV	1188.5	XEL	1198.7	100.9	CEII Redacted	P1	J531
Brookings Co-White 345 kV #1	788.8	XEL WAPA	816.1	103.5	CEII Redacted	P1	J432,J460,J488,J489,J493, J526

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Brookings Co-White 345 kV #1	788.8	XEL WAPA	805.9	102.2	CEII Redacted	P2-P7	J432,J460,J488,J489,J493, J526
Brookings Co-White 345 kV #2	788.8	XEL WAPA	816.1	103.5	CEII Redacted	P1	J432,J460,J488,J489,J493, J526
Brookings Co-White 345 kV #2	788.8	XEL WAPA	1169.6	148.3	CEII Redacted	P2-P7	J432,J488,J489,J493,J526
Helena-Scott Co 345 kV	1195.1	XEL WAPA	1545.5	129.3	CEII Redacted	P1	J460
Helena-Scott Co 345 kV	1195.1	XEL WAPA	1543.1	129.1	CEII Redacted	P2-P7	J460
Hazel Crk-Minn Valley 230 kV	338.0	XEL	349.4	103.4	CEII Redacted	P2-P7	J460,J526
Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	273.8	105.7	CEII Redacted	P1	J526
Panther-McLeod 230 kV	321.9	GRE	383.2	119.0	CEII Redacted	P2-P7	J526
Johnson Jct-Ortonville 115 kV	107.8	MRES GRE OTP	171.3	158.9	CEII Redacted	P1	J526

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Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Johnson Jct-Ortonville 115 kV	107.8	MRES GRE OTP	134.3	124.6	CEII Redacted	P2-P7	J526
Johnson Jct-Morris 115 kV	92.2	GRE WAPA	158.7	172.1	CEII Redacted	P1	J526
Johnson Jct-Morris 115 kV	92.2	GRE WAPA	121.7	132.0	CEII Redacted	P2-P7	J526
Canby-Granite Falls 115 kV	113.0	OTP WAPA	130.9	115.9	CEII Redacted	P1	J488,J489,J493,J526
Big Stone-Highway 12 115 kV	230.0	OTP	263.3	114.5	CEII Redacted	P1	J488,J489,J493,J526
Big Stone 115-230-13.8 kV xfmr	262.9	OTP	332.4	126.4	CEII Redacted	P1	J488,J489,J493,J526
Highway 12-Ortonville 115 kV	230.0	OTP	254.7	110.8	CEII Redacted	P1	J488,J489,J493,J526
Big Stone 230-115-13.8 kV xfmr	291.2	OTP	345.0	118.5	CEII Redacted	P1	J488,J489,J493,J526
Big Stone-Browns Valley 230 kV	285.0	OTP	368.2	129.2	CEII Redacted	P1	J488,J489,J493,J526

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Big Stone-Blair 230 kV	526.0	OTP BEPC	959.4	182.4	CEII Redacted	P1	J488,J489,J493,J526
Big Stone-Blair 230 kV	526.0	OTPBE PC	710.0	135.0	CEII Redacted	P2-P7	J488,J489,J493,J526
Browns Valley-Hankinson 230 kV	353.0	OTP	353.0	100.0	CEII Redacted	P1	J488,J489,J493,J526
Hankinson-Wahpeton 230 kV	351.0	OTP	461.7	131.5	CEII Redacted	P1	J488,J489,J493,J526
Hankinson-Wahpeton 230 kV	351.0	OTP	359.7	102.5	CEII Redacted	P2-P7	J488,J489,J493,J526
Marshalltown-Blairtown 115 kV	90	ITCM	94.8	105.3	CEII Redacted	P0	J514
Marshalltown-Blairtown 115 kV	90.0	ITCM	125.0	138.9	CEII Redacted	P1	J514
Blairtown-Prairie Crk 115 kV	88.0	CIPCO ITCM	117.6	133.6	CEII Redacted	P1	J514

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Hayward-Glenworth 161 kV	325.0	ITCM	332.6	102.4	CEII Redacted	P2-P7	J495
Killdeer 345-161 kV xfmr	450.0	ITCM	667.5	148.3	CEII Redacted	P1	J501
Emery-Colby 161 kV	327.0	ITCM	376.7	115.2	CEII Redacted	P1	J501,J495
Emery-Floyd 161 kV	272.0	ITCM MEC	351.9	129.4	CEII Redacted	P2-P7	J495
Boone Jct-Sub T Fort Dodge 161 kV	199.0	ITCM MEC CIPCO	292.3	146.9	CEII Redacted	P1	J498
Boone Jct-Sub T Fort Dodge 161 kV	199.0	ITCM MEC CIPCO	312.5	157.1	CEII Redacted	P2-P7	J498
Parnell-Hills 161 kV	325.0	MEC ITCM	336.6	103.6	CEII Redacted	P1	J530
Henry Co-Denmark 161 kV	169.0	ITCM	201.9	119.5	CEII Redacted	P1	J530

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Henry Co-Jeff 161 kV	197.0	ITCM	221.2	112.3	CEII Redacted	P1	J530
Jasper-Laurel 161 kV	361.0	ITCM	380.2	105.3	CEII Redacted	P2-P7	J514
Wapello-Jeff 161 kV	223.0	ITCM	269.3	120.8	CEII Redacted	P1	J530
Poweshiek-Reasnor 161 kV	311.0	ITCM	419.6	134.9	CEII Redacted	P1	J498,J499,J500
Poweshiek-Reasnor 161 kV	311.0	ITCM	449.2	144.5	CEII Redacted	P2-P7	J498,J499,J500
Adams S-Beaver Crk 161 kV	264.0	ITCM DPC	324.4	122.9	CEII Redacted	P1	J531
Adams S-Beaver Crk 161 kV	264.0	ITCM DPC	317.1	120.1	CEII Redacted	P2-P7	J531
Hazleton-Arnold 345 kV	1036.0	ITCM MEC	1129.6	109.0	CEII Redacted	P1	J531
Hazleton-Arnold 345 kV	1036.0	ITCM MEC	1330.3	128.4	CEII Redacted	P2-P7	J531

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Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Hazleton-Hickory Crk 345 kV	1195	ITCM	1312.2	109.8	CEII Redacted	P0	J514,J460,J501,J506,J529,J531,J534,J535,J495
Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1532.4	128.2	CEII Redacted	P1	J531
Hazleton-Hickory Crk 345 kV	1195.0	ITCM	1568.7	131.3	CEII Redacted	P2-P7	J531
Hazleton-Blackhawk 345 kV	1195.0	ITCM MEC	1322.9	110.7	CEII Redacted	P1	J531
Arnold-Morgan Valley 345 kV	1183.0	ITCM MEC	1283.3	108.5	CEII Redacted	P1	J531
Arnold-Morgan Valley 345 kV	1183.0	ITCM MEC	1397.5	118.1	CEII Redacted	P2-P7	J531
Ottumwa-Montezuma 345 kV	1139.0	ITCM MEC	1144.0	100.4	CEII Redacted	P1	J475,J530
Ottumwa-Montezuma 345 kV	1139.0	ITCM MEC	1183.9	103.9	CEII Redacted	P2-P7	J475,J530

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Morgan Valley-Tiffin 345 kV	1175.0	ITCM MEC	1266.9	107.8	CEII Redacted	P1	J531
Morgan Valley-Tiffin 345 kV	1175.0	ITCM MEC	1401.0	119.2	CEII Redacted	P2-P7	J531
Council Bluffs-S3456 345 KV	956.0	MEC OPPD	1010.9	105.7	CEII Redacted	P2-P7	J499
Raun-S3451 345 kV	956.0	MEC OPPD	1219.3	127.5	CEII Redacted	P1	J498
Raun-S3451 345 kV	956.0	MEC OPPD	1216.8	127.3	CEII Redacted	P2-P7	J501
Raun-Tekamah 161 kV	217.0	MEC OPPD	376.1	173.3	CEII Redacted	P2-P7	J506
Grimes-Sycamore 345 kV #1	1152.0	MEC	1239.6	107.6	CEII Redacted	P1	J498,J499
Grimes-Sycamore 345 kV #1	1152.0	MEC	1179.4	102.4	CEII Redacted	P2-P7	J498,J499
Grimes-Sycamore 345 kV #2	1152.0	MEC	1232.2	107.0	CEII Redacted	P1	J498,J499

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Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Grimes-Sycamore 345 kV #2	1152.0	MEC	1418.0	123.1	CEII Redacted	P2-P7	J498,J499
Grimes-Granger Tap 161 kV	395.0	MEC	434.6	110.0	CEII Redacted	P2-P7	J498,J499
Granger Tap-108th & 54th 161 kV	395.0	MEC	443.2	112.2	CEII Redacted	P2-P7	J498,J499
Bondurant-Sycamore 345 kV	1152.0	MEC	1212.1	105.2	CEII Redacted	P1	J499
Bondurant-Sycamore 345 kV	1152.0	MEC	1303.2	113.1	CEII Redacted	P2-P7	J498,J499,J500,J527,J528
Bondurant-Montezuma 345 kV	926	MEC	1245.2	134.5	CEII Redacted	P0	J514,J432,J460,J488,J489,J490,J493,J498,J499,J500,J501,J506,J526,J527,J528,J529,J534,J535,J495
Bondurant-Montezuma 345 kV	1010.0	MEC	1394.1	138.0	CEII Redacted	P1	J499

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Bondurant-Montezuma 345 kV	1010.0	MEC	1405.1	139.1	CEII Redacted	P2-P7	J499,J500,J527,J528
Franklin-Wall Lake 161 kV	201.0	MEC	214.9	106.9	CEII Redacted	P1	J498
Franklin-Wall Lake 161 kV	201.0	MEC	221.9	110.4	CEII Redacted	P2-P7	J498,J535
Hills-Tiffin 345 kV	1079.0	MEC	1246.8	115.6	CEII Redacted	P2-P7	J531
Hills-Sub 92 345 kV	1304.0	MEC	1417.4	108.7	CEII Redacted	P1	J475,J530
Hills-Sub 92 345 kV	1304.0	MEC	1445.1	110.8	CEII Redacted	P2-P7	J475,J530

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Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Sub 56-Walcott 345 kV	1152.0	MEC	1243.7	108.0	CEII Redacted	P2-P7	J530
Walcott-Sub 92 345 kV	1189.0	MEC	1410.4	118.6	CEII Redacted	P2-P7	J530
Blair-Watertown 230 kV	410.0	WAPA	480.1	117.1	CEII Redacted	P1	J488,J489,J493,J526
Ellendale-Aberdeen Jct 115 kV	128.0	MDU NWPS	131.4	102.7	CEII Redacted	P1	J488,J489
Glenham 230-115-41.6 kV xfmr	31.0	MDU	32.7	105.6	CEII Redacted	P1	J490,J490
Seneca-Genoa 161 kV	332.0	DPC	333.7	100.5	CEII Redacted	P1	J531
Wabaco-Alma 161 kV	291.0	DPC	305.6	105.0	CEII Redacted	P1	J531

4.3 Step 2 – Project Justification Analysis

Based on the identified thermal constraints in the 1st round ACCC, and based on voltage collapse under system intact condition and King – Eau Claire – Arpin (most critical contingency) post contingency condition, the Ad Hoc group proposed tens of potential transmission Network Upgrades (NUs) and their combinations. To justify the needs of the NUs only required for interconnection of the DPP West projects, project justification analysis was performed considering the following aspects:

- 1) Evaluate the 1st round ACCC identified constraints to be relieved by the NU.
- 2) Evaluate the 1st round ACCC identified constraints to be aggravated by the NU
- 3) Evaluate the new constraints caused by the NU.
- 4) Evaluate effectiveness of reactive power loss reduction by the NU.
- 5) Select the transmission NUs required and justified for interconnection of DPP West projects.
- 6) Model the justified transmission NUs in the summer shoulder case; Remove the 8 fictitious large size SVCs (Table 2-1) added in the initial case. Justify the needs of reactive power NUs to resolve the voltage collapse issue under system intact condition and King – Eau Claire – Arpin post contingency condition.

Project justification analysis was performed in 4 study areas: Big Stone area, Central-East Iowa, West-South-North Iowa, and Xcel area.

4.3.1 Project Justification in Big Stone Area

Five transmission NUs (Table 4-3) in Big Stone area were proposed and evaluated. Within these 5 transmission NUs, the Big Stone South-Alexandria 345 kV line was justified and selected as required NU based on cost and effectiveness. Big Stone area project justification details are in Appendix E.2.1.

Table 4-3: Transmission NUs Evaluated in Big Stone Area

Transmission NUs	Miles
Big Stone South-Alexandria 345 kV	95
Big Stone South-Hazel Creek 345 kV	80
Big Stone South-Hazel Crk-Lyon County 345 kV	105
Big Stone South-Brookings 2nd 345 kV	NA
Lakefield Jct-Lyon County 345 kV	100

4.3.2 Project Justification in Central-East Iowa Area

Seventeen transmission NUs (Table 4-4) in Central-East Iowa area were proposed and evaluated. Within these 17 transmission NUs, the Blackhawk-Morgan Valley-Beverly-Sub 92

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345 kV line was justified and selected as required NU. Central-East Iowa area project justification details are in Appendix E.2.2.

Table 4-4: Transmission NUs Evaluated in Central-East Iowa Area

Transmission NUs	Miles
Blkhwk-Morgan & Beverly-Sub 92 1 - New Hartford-Morgan Valley Section; 2 - Morgan Valley Sub-Beverly Sub 3 - Beverly-Sub 92 345 kV	115
Blkhwk-Mntzuma_Tama-Morgan_Beverly-Sub 92 1 - New north-south 345 kV connecting Blackhawk, or new 345 kV switching station due north of Montezuma (New Hartford), to Montezuma 345 kV; 2 - New 345 kV "Tama County" switching station to Morgan Valley 345 kV 3 - Beverly-Sub 92 345 kV	155
Blkhwk-Mntzuma_Tama-Morgan_Beverly-E Calamus 1 - New north-south 345 kV connecting Blackhawk, or new 345 kV switching station due north of Montezuma (New Hartford), to Montezuma 345 kV; 2 - New 345 kV "Tama County" switching station to Morgan Valley 345 kV 5 - Beverly to East Calamus 345 kV	168
Franklin-Marshalltown-Montezuma 345 kV, Marshalltown-Morgan Valley-Sub 92 345 kV 2 - Franklin-Marshalltown 345 kV 4 - Marshalltown-Montezuma 345 kV 3 - Marshalltown-Morgan Valley 345 kV 5 - Morgan Valley-Sub 92 345 kV	264
Blkhwk-Mntzuma 1- New north-south 345 kV connecting Blackhawk, or new 345 kV switching station due north of Montezuma (New Hartford), to Montezuma 345 kV	70
Blkhwk-Mntzuma & Tama-Morgan 1-New north-south 345 kV connecting Blackhawk, or new 345 kV switching station due north of Montezuma (New Hartford), to Montezuma 345 kV 2-New 345 kV "Tama County" switching station to Morgan Valley 345 kV	110
Beverly-Sub 92	45
Beverly-East Calamus	58
Beverly-E Calamus-Rock Crk - 5-Beverly to East Calamus 345 kV - 6-East Calamus-Rock Creek 345 kV	80
Morgan-Sub 92 - 5-Morgan Valley-Sub 92 345 kV	65
Franklin-Marshalltown-Mntzuma - 2-Franklin-Marshalltown 345 kV - 4-Marshalltown-Montezuma 345 kV	123
Franklin-Marshalltown-Morgan - 2-Franklin-Marshalltown 345 kV - 3-Marshalltown-Morgan Valley 345 kV	145

Transmission NUs	Miles
Franklin-Marshalltown-Morgan-Sub 92 - 2-Franklin-Marshalltown 345 kV - 3-Marshalltown-Morgan Valley 345 kV - 5-Morgan Valley-Sub 92 345 kV	210
Webster-Franklin - 1-Webster-Franklin 345 kV	50
Beverly-Bertram - 1-Beverly-Bertram 345 kV	9
Beverly-Bertram-Sub 92 - 1-Beverly-Bertram 345 kV - 2-Bertram-Sub 92 345 kV line	47
Beverly-Bertram-E Calamus - 1-Beverly-Bertram 345 kV - 4-Bertram to East Calamus 345 kV	60

4.3.3 Project Justification in West-South-North Iowa Area

Eight transmission NUs (Table 4-5) in West-South-North Iowa area were proposed and evaluated. Within these 8 transmission NUs, none of the NUs was justified as required NU. West-South-North Iowa area project justification details are in Appendix E.2.3.

Table 4-5: Transmission NUs Evaluated in West-South-North Iowa Area

Transmission NUs	Miles
Ida Co-Avoca 345 kV	76
Madison-Ottumwa-Sub T Haskins 345 kV	187
A new Winterset area 345 kV to Ottumwa to Sub T	165
Madison-Ottumwa 345 kV	105
Ottumwa-Sub T Haskins 345 kV	82
Colby-Mitchell County 345 kV	55
Colby-Mitchell County-Rochester 345 kV - 1-Colby-Mitchell County 345 kV - 2-Mitchell County-Rochester 345 kV - 3-Convert Mitchell County to breaker and ½ configuration and configure as 2 separate Mitchell buses with normally open breakers between the two buses	95
Mitchell County-Rochester 345 kV	45

4.3.4 Project Justification in Xcel Area

Three transmission NUs in Xcel area were proposed and evaluated. The Huntley-Wilmarth 345 kV line is MTEP Appendix A project with in service data of 2022. This Huntley-Wilmarth

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345 kV project will be included in the model. The Briggs Road 60% series compensation was justified and selected as required NU based on cost and effectiveness. Xcel area project justification details are in Appendix E.2.4.

Table 4-6: Transmission NUs to be Evaluated in Xcel Area

Transmission NUs	Miles
Briggs Road Series Comp	
Huntley-Wilmarth 345 kV	NA
Helena - Chub Lake 2nd CKT	NA

4.3.5 Project Justification of Reactive Power NUs

The transmission NUs required and justified for interconnection of DPP West projects are listed in Table 4-7.

Table 4-7: Justified and Required Transmission NUs

Area	NUs	Miles	Cost (\$)
MN-WI	Hickory Creek-Eden Prairie-Cardinal 345 kV	NA	NA
Minnesota Area	Huntley-Wilmarth 345 kV	NA	NA
Big Stone Area	Big Stone South-Alexandria 345 kV	95	\$197,400,000
Central-East Iowa	New Hartford-Morgan Valley Section	117	\$309,150,000
	Morgan Valley Sub-Beverly Sub	NA	NA
	Beverly-Sub 92 345 kV	65	\$158,950,000
Xcel	Briggs Road Series Comp		\$18,000,000

These transmission NUs were modeled in the summer shoulder case as backbone NUs. The 8 fictitious SVCs were removed. Summer shoulder power flow case under system intact condition was solved without voltage collapse. Since the King – Eau Claire – Arpin contingency still has voltage collapse issue, reactive power NUs were justified as follows:

- Based on voltage constraints identified in system intact conditions, 2×150 Mvar switched capacitor bank at Hickory Creek 345 kV bus and 2×150 Mvar switched capacitor bank at New Hartford 345 kV bus were required and justified.
- With the addition of 2×150 Mvar switched capacitor banks at Hickory Creek and New Hartford 345 kV buses, a ±200 Mvar STATCOM at Arrowhead 230 kV was required and justified for solving the King – Eau Claire – Arpin contingency.

4.4 Backbone Network Upgrades

Based on the project justifications in Section 4.3, a minimal set of transmission and reactive power NUs as Backbone NUs was required and justified for interconnecting of the DPP West projects. The Backbone NUs are listed in Table 4-8.

Table 4-8: Backbone NUs Required and Justified

Area	NUs
MN-WI	Hickory Creek-Eden Prairie-Cardinal 345 kV
Minnesota Area	Huntley-Wilmarth 345 kV
Big Stone Area	Big Stone South-Alexandria 345 kV
Central-East Iowa	New Hartford-Morgan Valley Section
	Morgan Valley Sub-Beverly Sub
	Beverly-Sub 92 345 kV
Xcel	Briggs Road Series Comp
Arrowhead	±200 Mvar STATCOM at Arrowhead 230 kV
Hickory Creek	2×150 Mvar switched shunt capacitor at Hickory 345 kV (631191)
New Hartford	2×150 Mvar switched shunt capacitor at New Hartford 345 kV (631240)

4.5 Step 3 – 2nd Round ACCC Analysis

With the Backbone NUs added and the fictitious SVCs removed, 2nd round AC contingency analysis was performed to identify any remaining thermal and voltage constraints not mitigated by the Backbone NUs.

4.5.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-16, and voltage constraints are listed in Table E-17.

4.5.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies.

All category P1 contingencies were converged. For P1 contingencies in the 2021 summer shoulder scenario, thermal constraints are listed in Table E-18, and voltage constraints are listed in Table E-19.

As noted in Section 4.2, under the Big Stone South - J488&J489 POI 345 kV post-contingency condition, Composite SCR (CSCR) of J488 and J489 is around 1.022. Further study is required to provide evidence that the J488 and J489 wind farms can operate at the CSCR of 1.022.

All category P2-P7 contingencies (Table E-22) were converged. For P2-P7 contingencies in the 2021 summer shoulder scenario, thermal constraints are listed in Table E-20, and voltage constraints are listed in Table E-21.

4.5.3 Summer Shoulder Worst Thermal Constraints in the 2nd Round ACCC

Table 4-9 lists worst thermal constraints identified in the 2nd round ACCC for 2021 summer shoulder scenario.

4.5.4 Summer Shoulder Worst Voltage Constraints in the 2nd Round ACCC

For P0 system intact conditions, worst voltage constraints are listed in Appendix E.4, Table E-24.

For P1 contingencies, worst voltage constraints are listed in Appendix E.4, Table E-25.

For P2-P7 contingencies, worst voltage constraints are listed in Appendix E.4, Table E-26.

Table 4-9: 2021 Summer Shoulder Thermal Constraints, Maximum Screened Loading, 2nd Round ACCC

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
J438 POI-Parnell 161 kV	282.0	ITCM MEC	300.3	106.5	CEII Redacted	P1	J530
J531 POI-Hazleton 345 kV	1000.0	ITCM MEC	1150.5	115.1	CEII Redacted	P1	J531
J531 POI-Hazleton 345 kV	1000.0	ITCM MEC	1224.2	122.4	CEII Redacted	P2-P7	J531
J475 substation xfmr	212.0	J475	222.1	104.8	CEII Redacted	P0	J475
J475 substation xfmr	212.0	J475	224.4	105.9	CEII Redacted	P1	J475
J475 substation xfmr	212.0	J475	225.0	106.1	CEII Redacted	P2-P7	J475
J498 POI-Grimes 345 kV	1152.0	MEC	1242.8	107.9	CEII Redacted	P1	J498, J535
J498 POI-Grimes 345 kV	1152.0	MEC	1361.3	118.2	CEII Redacted	P2-P7	J498, J534, J535
J530 POI-Hills 345 kV	1010.0	MEC	1254.5	124.2	CEII Redacted	P1	J475, J530
J530 POI-Hills 345 kV	1010.0	MEC	1223.4	121.1	CEII Redacted	P2-P7	J475, J530

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Split Rock-White 345 kV	717.1	XEL WAPA	796.6	111.1	CEII Redacted	P1	J432, J460, J488, J489, J493, J526
Split Rock-White 345 kV	717.1	XEL WAPA	772.6	107.7	CEII Redacted	P2-P7	J432, J460, J488, J489, J493, J526
Brookings Co-White 345 kV #2	788.8	XEL WAPA	914.7	116.0	CEII Redacted	P2-P7	J432, J488, J489, J493, J526
Helena-Scott Co 345 kV	1195.1	XEL WAPA	1431.1	119.7	CEII Redacted	P1	J501
Helena-Scott Co 345 kV	1195.1	XEL WAPA	1429.3	119.6	CEII Redacted	P2-P7	J501
Panther-McLeod 230 kV	321.9	GRE	340.0	105.6	CEII Redacted	P2-P7	J526
Big Stone-Blair 230 kV	526.0	OTP BEPC	558.4	106.2	CEII Redacted	P1	J488, J489, J493, J526
Hankinson-Forman 230 kV	257.0	OTP	318.4	123.9	CEII Redacted	P1	J488, J489
Oakes-Forman 230 kV	305.0	OTP	437.8	143.5	CEII Redacted	P1	J488, J489
Oakes-Ellendale 230 kV	351.0	OTP MDU	441.5	125.8	CEII Redacted	P1	J488, J489

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Marshalltown-Blairstown 115 kV	90.0	ITCM	90.0	100.0	CEII Redacted	P0	J514
Marshalltown-Blairstown 115 kV	90.0	ITCM	118.5	131.7	CEII Redacted	P1	J514
Blairstown-Prairie Crk 115 kV	80.0	CIPCO ITCM	82.3	102.9	CEII Redacted	P0	J514
Blairstown-Prairie Crk 115 kV	80.0	CIPCO ITCM	111.0	138.7	CEII Redacted	P1	J514
Killdeer 345-161 kV xfmr	450.0	ITCM	628.8	139.7	CEII Redacted	P1	J501
Emery-Colby 161 kV	327.0	ITCM	372.7	114.0	CEII Redacted	P1	J495, J501
Boone Jct-Sub T Fort Dodge 161 kV	199.0	MEC CIPCO	268.9	135.1	CEII Redacted	P1	J498
Boone Jct-Sub T Fort Dodge 161 kV	199.0	MEC CIPCO	288.8	145.1	CEII Redacted	P2-P7	J498
Traer-Dysart 161 kV	258.0	ITCM	264.9	102.7	CEII Redacted	P1	J514
Henry Co-Denmark 161 kV	169.0	ITCM	188.1	111.3	CEII Redacted	P1	J530
Henry Co-Jeff 161 kV	197.0	ITCM	207.6	105.4	CEII Redacted	P1	J530
Wapello-Jeff 161 kV	223.0	ITCM	254.1	113.9	CEII Redacted	P1	J530

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Poweshiek-Reasnor 161 kV	311.0	ITCM	387.9	124.7	CEII Redacted	P1	J498, J499, J500
Poweshiek-Reasnor 161 kV	311.0	ITCM	420.4	135.2	CEII Redacted	P2-P7	J498, J499, J500
Adams S-Beaver Crk 161 kV	264.0	ITCM DPC	269.7	102.2	CEII Redacted	P1	J531
Council Bluffs-S3456 345 kV	956.0	MEC OPPD	1022.8	107.0	CEII Redacted	P2-P7	J499, J500
Raun-S3451 345 kV	956.0	MEC OPPD	1130.8	118.3	CEII Redacted	P1	J498
Raun-Tekamah 161 kV	217.0	MEC OPPD	352.8	162.6	CEII Redacted	P2-P7	J506
Grimes-Sycamore 345 kV #1	1152.0	MEC	1161.5	100.8	CEII Redacted	P1	J498, J499
Grimes-Sycamore 345 kV #2	1152.0	MEC	1154.5	100.2	CEII Redacted	P1	J498, J499
Grimes-Sycamore 345 kV #2	1152.0	MEC	1338.9	116.2	CEII Redacted	P2-P7	J498, J499

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Grimes-Granger Tap 161 kV	395.0	MEC	415.8	105.3	CEII Redacted	P2-P7	J498, J499
Granger Tap-108th & 54th 161 kV	395.0	MEC	418.0	105.8	CEII Redacted	P2-P7	J498, J499
Bondurant-Sycamore 345 kV	1152.0	MEC	1184.9	102.9	CEII Redacted	P2-P7	J498, J499, J500, J527, J528
Bondurant-Montezuma 345 kV	926.0	MEC	1119.3	120.9	CEII Redacted	P0	J432, J460, J488, J489, J490, J493, J498, J499, J500, J501, J506, J514, J526, J527, J528, J529, J534, J535,
Bondurant-Montezuma 345 kV	1010.0	MEC	1245.7	123.3	CEII Redacted	P1	J499, J527, J528
Bondurant-Montezuma 345 kV	1010.0	MEC	1294.2	128.1	CEII Redacted	P2-P7	J499, J500, J527, J528
Franklin-Wall Lake 161 kV	201.0	MEC	216.3	107.6	CEII Redacted	P1	J498
Franklin-Wall Lake 161 kV	201.0	MEC	222.8	110.8	CEII Redacted	P2-P7	J498, J535

Summer Shoulder Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
Sub 56-Walcott 345 kV	1152.0	MEC	1349.1	117.1	CEII Redacted	P2-P7	J530
Walcott-Sub 92 345 kV	1189.0	MEC	1518.2	127.7	CEII Redacted	P2-P7	J530
Ellendale-Aberdean Jct 115 kV	128.0	MDU NWPS	143.6	112.2	CEII Redacted	P1	J488, J489
Glenham 230-115-41.6 kV xfmr	31.0	MDU	32.7	105.5	CEII Redacted	P1	J490
Seneca-Genoa 161 kV	332.0	DPC	343.7	103.5	CEII Redacted	P1	J531
Seneca-Gran Grae 161 kV	221.0	DPC	224.0	101.4	CEII Redacted	P1	J531

4.6 Step 4 – Update Reactive Power NUs Based on the MWEX Voltage Stability Study Results

ATC performed steady state voltage stability analysis for the MWEX interface (Section 7). The following Network Upgrades were identified and required for the MWEX voltage stability.

Table 4-10: NUs Identified and Required for the MWEX Voltage Stability

Area	NUs	Cost (\$)
MWEX	North Rochester Substation Rework	\$8,000,000
MWEX	161/69 kV Xcel/DPC substation in Minnesota	\$5,500,000
MWEX	New 345 kV line North Rochester - Rocky Run (170 miles)	\$793,287,928
MWEX	Install a 300 Mvar SVC at North Rochester 345 kV	\$32,000,000
MWEX	Rock Run Substation Rework; 300 MVar SVC	\$169,412,072
MWEX	60% series compensation at the end of Briggs Road - North Madison 345 kV line	\$18,000,000

Starting from the MWEX reactive power NUs (Table 4-10), additional reactive power NUs were updated based on the 2nd round ACCC voltage constraint results and new voltage constraints under new contingencies (North Rochester-Rocky Run, Briggs Road-North Madison) and some critical contingencies (Stone Lake-Gardner Park, Gardner Park-Rocky Run, etc.). The effectiveness of reactive power NUs were also verified in AC contingency analysis.

In addition to the MWEX reactive power NUs, other reactive power NUs required for the DPP West projects are listed in Table 4-11.

Table 4-11: Additional Reactive Power NUs Required in MISO Analysis

Network Upgrades	Owner	Cost (\$)
2*50 Mvar switched shunt capacitor at Zachary 161 kV (344010)	Ameren	\$1,000,000
5*50 Mvar SVC at Overton 161 kV (345409)	Ameren	\$25,000,000
2*150 Mvar switched shunt capacitor at Hickory 345 kV (631191)	ITCM	\$8,000,000
1*150 Mvar switched shunt capacitor at New Hartford 345 kV (631240)	MEC	\$4,000,000
1*150 Mvar switched shunt capacitor at Blackhawk 345 kV (636199)	MEC	\$4,000,000
1*150 Mvar switched shunt capacitor at Montezuma 345 kV (635730)	MEC	\$4,000,000
2*150 Mvar switched shunt capacitor at Hills 345 kV (636400)	MEC	\$8,000,000
2*150 Mvar switched shunt capacitor at Morgan Valley 345 kV (631148)	MEC	\$8,000,000

4.7 Network Upgrades Identified in MISO ERS Analysis for 2021 Summer Shoulder Scenario

Based on the MISO 2021 summer shoulder steady state analyses, the Backbone thermal NUs and cost are listed in Table 4-12. Additional thermal NUs and cost are listed in Table 4-13.

Table 4-12: Backbone Thermal NUs

Area	NUs	Miles	Cost (\$)
MN-WI	Hickory Creek-Eden Prairie-Cardinal 345 kV	NA	NA
Minnesota Area	Huntley-Wilmarth 345 kV	NA	NA
Big Stone Area	Big Stone South-Alexandria 345 kV	95	\$197,400,000
Central-East Iowa	New Hartford-Morgan Valley Section	117	\$309,150,000
	Morgan Valley Sub-Beverly Sub	NA	NA
	Beverly-Sub 92 345 kV	65	\$158,950,000

Table 4-13: Additional Thermal NUs

Constraint	Owner	Mitigation	Cost (\$)
J438 POI-Parnell 161 kV	ITCM MEC	ITCM: Not an ITCM constraint; ITCM terminal limit is 335 MVA. \$0 MEC: Structure replacements. \$200,000	\$200,000
J531 POI-Hazleton 345 kV	ITCM MEC	ITCM: ITCM owns 28.1 miles and MEC owns 16.4 miles of that line segment. The 1224.2 MVA loading indicated below is beyond the conductor limit and rebuild ITCM's 28.1 with 2-T2 795 ACSR would be required. Rating of 1781 MVA for rebuild ITCM section. \$53M. MEC: Structure replacements. \$3,000,000	\$56,000,000
J475 substation xfmr	J475	J475 is responsible.	IC Cost
J498 POI-Grimes 345 kV	MEC	MEC: Reconfigure Raun Sub to eliminate Raun 0270 CON and reduce loading below maximum conductor rating (1333 MVA) for next worst CON (\$5 M). Structure replacements on J498-Grimes 345 kV (\$2.5 M).	\$2,500,000
J530 POI-Hills 345 kV	MEC	MEC: Structure replacements and terminal equipment upgrades	\$600,000
Split Rock-White 345 kV	XEL WAPA	Line is currently rated 1075 MVA for SN/SE no mitigation required	\$0
Brookings Co-White 345 kV #2	XEL WAPA	XEL: Line is limited by WAPA terminal limits. NSP's equipment able to handle the required loading. \$0 WAPA: Uprate the White terminal. \$60,000. Affected system upgrades. M3 calculations will not reflect these costs	\$60,000
Helena-Scott Co 345 kV	XEL WAPA	This line has an emergency rating of 1515 MVA.	\$0
Panther-McLeod 230 kV	GRE	Remediate 4 Structures	\$1,000,000

Constraint	Owner	Mitigation	Cost (\$)
Big Stone-Blair 230 kV	OTP BEPC	Mitigation required for DPP 2015 Aug will increase the rating to 601.8 MVA. \$0	\$0
Hankinson-Forman 230 kV	OTP	Line clearance mitigations and terminal equipment upgrades.	\$500,000
Oakes-Forman 230 kV	OTP	Line clearance mitigations and terminal equipment upgrades.	\$750,000
Oakes-Ellendale 230 kV	OTP MDU	OTP: Line clearance mitigations and terminal equipment upgrades. \$1,550,000 MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$1,550,000
Marshalltown-Blairtown 115 kV	ITCM	Raise/replace structures to reach 123 MVA	\$2,700,000
Blairtown-Prairie Crk 115 kV	CIPCO ITCM	CIPCO: No Mitigation Needed. \$0 ITCM: Raise structures to reach a minimum rating of 114 MVA	\$2,700,000
Killdeer 345-161 kV xfmr	ITCM	Add second transformer	\$6,500,000
Emery-Colby 161 kV	ITCM	Upgrade approx .5 miles of 161 kV - new rating at 418 MVA	\$500,000
Boone Jct-Sub T Fort Dodge 161 kV	MEC CIPCO	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor line and terminal equipment upgrades. \$7,500,000	\$7,500,000
Traer-Dysart 161 kV	ITCM	Not a constraint, the line is currently rated at 276 MVA	\$0
Henry Co-Denmark 161 kV	ITCM	Raise/replace 4 spans to 223 MVA	\$960,000
Henry Co-Jeff 161 kV	ITCM	Raise/replace 4 spans to 223 MVA	\$960,000
Wapello-Jeff 161 kV	ITCM	Line is conductor limited at 240 MVA. Rebuild 22.6 miles of 161 kV with T2-795ACSR	\$27,000,000
Poweshiek-Reasnor 161 kV	ITCM	Line is conductor limited at 327 MVA. Rebuild 20.7 miles of 161 kV with T2-795 ACSR	\$25,000,000
Adams S-Beaver Crk 161 kV	ITCM DPC	ITCM: Not an ITCM constraint - ITCM terminal limit is 335 MVA. \$0 DPC: Rebuild 15 miles 636 ACSR 161 kV line. \$14.7 M	\$14,700,000
Council Bluffs-S3456 345 KV	MEC OPPD	MEC: No mitigation required. Existing MEC emergency rating is 1439 MVA. \$0 OPPD: Upgrade terminal equipment; Replace structure. \$2 M. Affected system upgrades. M3 calculations will not reflect these costs	\$2,000,000
Raun-S3451 345 kV	MEC OPPD	MEC: No mitigation required. Existing MEC emergency rating is 1152 MVA. \$0 OPPD: Upgrade terminal equipment; Rebuild 63 miles line. \$72 M. Affected system upgrades. M3 calculations will not reflect these costs	\$72,000,000
Raun-Tekamah 161 kV	MEC OPPD	MEC: Reconfigure Raun Sub to eliminate Raun 0270 CON. \$5,000,000 OPPD: \$0	\$5,000,000

Constraint	Owner	Mitigation	Cost (\$)
Grimes-Sycamore 345 kV #1	MEC	MEC: No mitigation required. Current MEC ring open emergency rating is 1195 MVA	\$0
Grimes-Sycamore 345 kV #2	MEC	Current MEC ring open emergency rating is 1018 MVA. Terminal equipment upgrades; Add breaker between Grimes 345-161 kV Transformer 9T1 and Grimes-Sycamore 345 kV Ckt 1 to eliminate CON	\$3,100,000
Grimes-Granger Tap 161 kV	MEC	Reconductor line	\$2,000,000
Granger Tap-108th & 54th 161 kV	MEC	Current rating is 360 MVA. Reconductor line and terminal equipment upgrades	\$1,250,000
Bondurant-Sycamore 345 kV	MEC	Structure replacements and terminal equipment upgrades	\$1,000,000
Bondurant-Montezuma 345 kV	MEC	Structure replacements and terminal equipment upgrades	\$2,100,000
Franklin-Wall Lake 161 kV	MEC	Structure replacements	\$500,000
Sub 56-Walcott 345 kV	MEC	Current emergency rating is 1195 MVA. Add breakers at Sub T Haskins to change CON by removing loss of Hills-Sub T 345 kV. Reduces loading to 1155 MVA	\$4,000,000
Walcott-Sub 92 345 kV	MEC	Add breakers at Sub T Haskins to change CON by removing loss of Hills-Sub T 345 kV. Reduces loading to 1319 MVA (\$4 M). Replace structures and terminal equipment to achieve this rating (\$300,000).	\$300,000
Ellendale-Aberdeen Jct 115 kV	MDU NWPS	MDU: The Ellendale 115 kV bus will be rebuilt to 318 MVA as part of MTEP #9120. \$0 NWPS: A switching substation at A-Tap (Aberdeen Jct) will be built in 2019. \$0	\$0
Glenham 230-115-41.6 kV xfmr	MDU	New Glenham 230/115/41.6 kV sub & transformer	\$6,500,000
Seneca-Genoa 161 kV	DPC	The DPC 161 kV lines from Genoa to Seneca and Seneca to Gran Grae are limited by conductor	\$38,000,000
Seneca-Gran Grae 161 kV	DPC	The DPC 161 kV lines from Genoa to Seneca and Seneca to Gran Grae are limited by conductor	Incl.

To mitigate MWEX voltage stability constraint, the required Network Upgrades are listed in Table 4-14.

Table 4-14: Required Reactive Power NUs for MWEX Voltage Stability

Area	NUs	Cost (\$)
MWEX	North Rochester Substation Rework	\$8,000,000
MWEX	161/69 kV Xcel/DPC substation in Minnesota	\$5,500,000
MWEX	New 345 kV line North Rochester - Rocky Run (170 miles)	\$793,287,928
MWEX	Install a 300 Mvar SVC at North Rochester 345 kV	\$32,000,000

Area	NUs	Cost (\$)
MWEX	Rock Run Substation Rework; 300 MVar SVC	\$169,412,072
MWEX	60% series compensation at the end of Briggs Road - North Madison 345 kV line	\$18,000,000

In addition to the MWEX reactive power NUs, Table 4-15 lists additional reactive power NUs required for mitigating the voltage constraints identified in AC contingency analysis.

Table 4-15: Additional Reactive Power NUs Required for Voltage Constraints

Network Upgrades	Owner	Cost (\$)
2*50 Mvar switched shunt capacitor at Zachary 161 kV (344010)	Ameren	\$1,000,000
5*50 Mvar SVC at Overton 161 kV (345409)	Ameren	\$25,000,000
2*150 Mvar switched shunt capacitor at Hickory 345 kV (631191)	ITCM	\$8,000,000
1*150 Mvar switched shunt capacitor at New Hartford 345 kV (631240)	MEC	\$4,000,000
1*150 Mvar switched shunt capacitor at Blackhawk 345 kV (636199)	MEC	\$4,000,000
1*150 Mvar switched shunt capacitor at Montezuma 345 kV (635730)	MEC	\$4,000,000
2*150 Mvar switched shunt capacitor at Hills 345 kV (636400)	MEC	\$8,000,000
2*150 Mvar switched shunt capacitor at Morgan Valley 345 kV (631148)	MEC	\$8,000,000

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Affected System Steady-State Analysis

Steady state analyses were performed to identify constraints on affected systems.

5.1 Local Planning Criteria Analysis for CIPCO and MPC Companies

Per CIPCO and MPC's Local Planning Criteria (LPC), transmission facilities which meet all of the following three (3) conditions are considered as constraints:

- 1) the branch is loaded above its applicable normal or emergency rating for the post-change case, and
- 2) the generator has a larger than 3% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, and
- 3) the loading increase of the overloaded facility is greater than 1 MVA compared with that in the pre-change case under system intact or contingency conditions.

AC contingency analysis was performed for this LPC analysis, using the following benchmark and study cases:

- Summer peak benchmark and study cases (without any network upgrades)
- Summer shoulder benchmark and study cases (with Backbone NUs)

All NERC category P0-P7 contingencies described in Section 2.2 were simulated. The CIPCO and MPC affected systems were monitored.

The additional thermal constraints in the affected systems identified in this LPC analysis are listed in Appendix F.1. These include the thermal constraints for the 2021 summer peak and shoulder scenarios. Whether these constraints will be mitigated or not will be determined by the affected transmission owning company's local planning criteria.

The highest loading of CIPCO and MPC local planning criteria thermal constraints and potential network upgrades are listed in Table 5-1 for the 2021 summer shoulder scenario. No CIPCO or MPC's thermal constraints were identified in the 2021 summer peak scenario.

Table 5-1: Additional Thermal Constraints Identified in LPC Analysis and their Highest Loadings in 2021 SH Scenario

Generator	Constraint	Rating	Owner	Worst Loading		Cont Type	Mitigation	Cost (\$)
				(MVA)	(%)			
IR-20, J514 ¹	CIPCO IR20-Dundee 161 kV	240.0	CIPCO ITCM	310.8	129.5	P2-P7	IR-20 is queued prior to the Feb 16 DPP group with impact of 48.643 MW. J514 has impact of 1.156 MW	\$6,000,000
J514	Blairstown-Prairie Crk 115 kV	80.0	CIPCO ITCM	110.1	137.7	P1	CIPCO: No Mitigation Needed to achieve 116.7 MVA. \$0 ITCM: CIPCO LPC. \$0	\$0
J514	Blairstown-Prairie Crk 115 kV	80.0	CIPCO ITCM	139.4	174.3	P2-P7	CIPCO: No Mitigation Needed to achieve 116.7 MVA. \$0 ITCM: CIPCO LPC. \$0	
J514	Prairie Crk-Marion 115 kV	197.0	CIPCO ITCM	199.0	101.0	P2-P7	CIPCO: Replace Jumpers at Prairie Creek \$50,000 ITCM: CIPCO LPC. \$0	\$50,000
J514	Prairie Crk-Bertram 115 kV	268.0	CIPCO ITCM	295.6	110.3	P2-P7	CIPCO: Rebuild of Double Circuit from Prairie Creek to Bertram. Would require new structures and conductor on one circuit. Old conductor can be re-used on Prairie Creek - Marion. \$16,000,000 ITCM: CIPCO LPC. \$0	\$16,000,000
J514	Prairie Crk-Squared 115 kV	164.0	CIPCO ITCM	168.5	102.7	P1	CIPCO: Replace Jumpers at Prairie Creek. \$50,000 ITCM: CIPCO LPC. \$0	\$50,000
J514	Prairie Crk-Squared 115 kV	164.0	CIPCO ITCM	208.0	126.9	P2-P7	CIPCO: Replace Jumpers at Prairie Creek. \$50,000 ITCM: CIPCO LPC. \$0	
J514	Bertram 161-115 kV xfmr	300.0	CIPCO	309.4	103.1	P2-P7	Replace Bertram 161-115 Transformer	\$4,000,000
J514, J531	Rosehollow-Bertram 161 kV	261.0	CIPCO ITCM	280.7	107.6	P2-P7	CIPCO: Change 2 CT ratios and replace jumpers. \$75,000 ITCM: CIPCO LPC. \$0	\$75,000
J495, J531	Hazleton S-Dundee 161 kV	327.0	CIPCO ITCM	356.5	109.0	P1	CIPCO: Replace 4 CT's at Dundee \$100,000. ITCM: CIPCO LPC. \$0	\$100,000

Affected System Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Cont Type	Mitigation	Cost (\$)
				(MVA)	(%)			
J531	Hazleton S-Dundee 161 kV	327.0	CIPCO ITCM	437.3	133.7	P2-P7	CIPCO: Replace 4 CT's at Dundee \$100,000. ITCM: CIPCO LPC. \$0	
J501, J529, J534, J535	Boone Jct-Sub T Fort Dodge 161 kV	199.0	ITCM MEC CIPCO	235.1	118.1	P1	CIPCO: No Mitigation Needed. ITCM/MEC: CIPCO LPC. \$0	\$0
J501, J529, J534, J535	Boone Jct-Sub T Fort Dodge 161 kV	199.0	ITCM MEC CIPCO	246.4	123.8	P2-P7	CIPCO: No Mitigation Needed. ITCM/MEC: CIPCO LPC. \$0	
J514, J531	Liberty-Dundee 161 kV #2	327.0	CIPCO ITCM	405.8	124.1	P2-P7	CIPCO: \$25,000,000 if a new line is required or \$5,000,000 if reconductor option is feasible. ITCM: CIPCO LPC. \$0	\$5,000,000
J514, J531	Liberty-Hickory Crk 161 kV	327.0	CIPCO ITCM	452.9	138.5	P1	CIPCO: Upgrade Liberty terminal to minimum of 2000 amps. ITCM: CIPCO LPC. \$0	\$500,000
J514, J531	Liberty-Hickory Crk 161 kV	327.0	CIPCO ITCM	510.4	156.1	P2-P7	CIPCO: Upgrade Liberty terminal to minimum of 2000 amps. ITCM: CIPCO LPC. \$0	

Note 1: IR-20 is CIPCO project queued prior to the DPP 2016 Feb 2016 group but system impact study has not been completed. IR-20 has impact of 48.643 MW, and J514 has impact of 1.156 MW. Based on the MW impact, IR-20 has cost share of \$5,860,720 and J514 has cost share of \$139,280.

5.2 J525 GRE Local Planning Criteria Analysis

GRE performed the local planning criteria analysis for J525 based on GRE's LPC. The GRE local planning criteria analysis details can be found in Appendix F.2.

The analysis indicates that the following Network Upgrades will be required prior to putting the J525 project into service:

5.2.1 Transmission Line Upgrades Required

Constraints	Owner	Length (miles)
Chandler - Lake Wilson 69 kV	Xcel	1.0
Hadley - Lake Wilson 69 kV	Xcel	2.5
Hadley - Slayton 69 kV	Xcel	4.3
Slayton (GRE) - Slayton (Xcel) 69 kV	Xcel	0.5
Currie - Slayton 69 kV	Xcel	9.7
Currie - Lake Sarah Tap 69 kV	Xcel	8.6
Lake Sarah Tap - Tracy Switch 69 kV	Xcel	6.0
Tracy Switch - Walnut Grove 69 kV	Xcel	1.0
Total		33.6

The total cost of the transmission line upgrades is approximately \$23 million. This cost is based on rebuilding the existing transmission with new 477 ACSR transmission line.

5.2.2 Fenton Transformer Replacement

The overload of the Fenton 115-69 kV transformer can be mitigated by the addition of a second 115-69 kV transformer at Fenton. The planning estimated cost for the transformer addition is \$3.9 million.

5.2.3 Low Short Circuit Ratio

The customer provides evidence that the inverter-based generator can operate at an SCR of 1.96.

5.2.4 Possible Instability of the Inverter Controls in Low SCR Conditions

GRE has performed a transient stability analysis in the local area around the proposed interconnection of a 50 MW Solar PV project at the Lake Wilson substation (Appendix I.18). The results indicate that there is a possible instability of the inverter controls due to the low SCR at the point of interconnection during a system fault on the Chandler tap-Lake Wilson 69 kV line circuit that results in the tripping of the 69 kV line from Lake Wilson to Fenton. This leaves the 50 MW inverter-based generation on a long radial circuit from Lake Wilson to Lyon County through Tracy and results in low short-circuit ratio (approximately 1.96) at the point of interconnection.

Before the interconnection can be allowed to operate, the developer and/or system manufacturer must provide assurance that the generation can operate during the low SCR conditions.

5.3 PJM Affected System AC Contingency Analysis

The PJM affected system analysis details can be found in Appendix F.3.

5.3.1 Study Results

5.3.1.1 Overload on Twin Branch–Argenta 345 kV line

To relieve the Twin Branch–Argenta 345 kV line overload:

- a. A sag check will be required for the ACSR ~ 954 ~ 45/7 ~ RAIL - Conductor Section 1 to determine if the line section can be operated above its emergency rating of 1409 MVA. The result could prove that no additional upgrades are necessary, that some upgrades on the circuit are necessary, or that the entire 52.0 mile section of line would need to be rebuilt. PJM Network Upgrade N5240.
- b. MISO end rating (1691 MVA) is sufficient through 2016 February MISO DPP.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$208 K)
J488	6.4096	23.47%	48.8
J501	20.9	76.53%	159.2

5.3.1.2 Overload on Benton–T-094 345 kV line

To relieve the Benton–T-094 345 kV overload:

- a) Reconductor or rebuild (depending on the existing structures) the portions of 345 kV lines between the Benton Harbor and Sagreto 345 kV substations. Replace the Benton H wavetrap. Replace the Benton H Line Riser. PJM Network Upgrade N5106.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$19.6 M)
J411	10.8334	6.17%	1.210
J515	16.672	9.50%	1.863
AB1-121	52.506	29.93%	5.866
J498	16.3798	9.34%	1.830
J499	16.1677	9.22%	1.806
J500	23.516	13.40%	2.627
J501	24.532	13.98%	2.741
J526	14.8392	8.46%	1.658

5.3.1.3 Overload on Stillwell–Dumont 345 kV line

To relieve the Stillwell-Dumont 345 kV line overload: There are 3 scopes of Network Upgrades.

Per PJM cost allocation rules, the DPP 2016 February West projects presently do not receive any cost allocation for this upgrade.

5.3.1.4 Overload on Casey–Breed 345 kV line

To relieve the Casey–Breed 345 kV overload:

- a) MISO-end: The MISO-end SE rating is 1466 MVA and is sufficient. No upgrade needed on the MISO-end.
- b) AEP-end: The AEP-end SN rating is 1443 MVA and not sufficient. AEP Upgrade: 0.8 miles AEP line will be re-conducted which will increase the ratings to SN – 1930MVA and SE – 1930MVA. Cost – \$700K. PJM Network Upgrade N5251.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$700 K)
J475	10.8	19.9%	139.483
J499	17.4	32.1%	224.723
J500	26	48.0%	335.793

5.3.1.5 Overload on Wilton–Dumont 765 kV line

To relieve the Wilton–Dumont 765 kV overload:

- a) AEP-end: The AEP-end SE rating is 4801 MVA. This rating is not sufficient due to a prior queue. The upgrade is to replace the Dumont wavetrapp and relay compliance thermal limit to obtain new AEP-end ratings of 6198/7362 MVA (SN/SE) at a cost of \$1.5M. PJM Network Upgrade N4789.
- b) ComEd-end: The ComEd-end ALDR rating is 5522 MVA. This rating is not sufficient due to a prior queue. The upgrade is to mitigate the sag on the Wilton - Dumont 765 kV line L11215 to achieve an ALDR that exceeds 6166 MVA. A preliminary estimate for this work is \$9M with an estimated construction timeline of 30 months. Upon completion of this work the new line ratings will be 4494/4494/5646/6493 MVA (SN/SE/SLD/ALDR). PJM Network Upgrade N5252.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$9 M)
AB1-122	404.2	88%	7.891
J501	17.4	12%	1.109

5.3.1.6 Overload on Crete–St John 345 kV line

To relieve the Crete–St John 345 kV overload:

- a) ComEd-end: ALDR rating on Crete end is 1925MVA and not sufficient. ComEd upgrade is to reconductor the line to achieve ratings of 1837 SSTE, SLD of 2360 MVA (ALDR of 2714 MVA). Preliminary estimate for the upgrade is \$18M with an estimated construction timeline of 24-30 months. PJM Network Upgrade N5253.
- b) MISO-end: SE rating of St John end line is 1508 MVA and not sufficient. MISO end upgrade is to upgrade St John substation conductor and switches. \$1M cost estimate. New MISO end SE rating will be 1900 MVA. This upgrade is driven and needed for a prior queue. Projects in DPP 2016 February West area do not receive any cost allocation for this upgrade.
- c) MISO-end: Additional MISO-end upgrades: Upgrade to 4000 A switch(es) and to bundled 1590 AL substation conductor. \$3M. New MISO-end ratings to be 1961/2390 MVA SN/SE. This upgrade is driven and needed for a prior queue. Projects in DPP 2016 February West area do not receive any cost allocation for this upgrade.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$18 M)
AB1-122	158.5	86.38%	15.5477
J501	25.0	13.62%	2.4523

5.3.1.7 Overload on Wilton 765-345 kV transformers

To relieve the Wilton–Dumont 765 kV overload:

- a) ALDR rating for each transformer is 1841 MVA. Build out the Wilton 765kV bus thereby allowing for 765kV L11216 (currently on Bus 6) to be relocated to Bus 8. Along with this line relocation, installation of 2-765kV BT CB's (6-8 & 8-2). Cost estimate is \$11M.
- b) ComEd-end: The ComEd-end ALDR rating is 5522 MVA. This rating is not sufficient due to a prior queue. The upgrade is to mitigate the sag on the Wilton - Dumont 765 kV line L11215 to achieve an ALDR that exceeds 6166 MVA. A preliminary estimate for this work is \$9M with an estimated construction timeline of 30 months. Upon completion of this work the new line ratings will be 4494/4494/5646/6493 MVA (SN/SE/SLD/ALDR). PJM Network Upgrade N5252.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$11 M)
AA2-035	102.4	10.79%	1.1873
AA2-039	20.4	2.15%	0.2365
AB1-086	97.0	10.22%	1.1247
AB1-089	80.5	8.49%	0.9334
AB1-090	80.5	8.49%	0.9334
AB1-091	82.8	8.73%	0.9601
AB1-121 MTX	279.4	29.45%	3.2396
AB1-122	180.3	19.00%	2.0905
J501	25.4	2.68%	0.2945

5.3.1.8 Overload on Walto–Electric Junction 345 kV line

To relieve the Walto–Electric Junction 345 kV overload:

- a) SSTE rating of the lines is 1837 MVA and not sufficient. This overload is caused by the MISO 2016 February DPP projects. Upgrade 345kV switches at TSS 155 for L15502, upgrade station conductor at the two station terminals and adjust the CT ratios for the line at the station terminals. A preliminary estimate is \$350K with an estimated construction timeline of 20 months. The post construction SSTE will be 2107 MVA. Network Upgrade N4734.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$350 K)
J498	19.0	17.30%	60.5647
J500	25.1	22.86%	80.0091
J501	29.1	26.50%	92.7596
J510	16.9	15.39%	53.8707
J530	19.7	17.94%	62.7960

5.3.1.9 Overload on Lee County–Byron 345 kV line

To relieve the Lee County–Byron 345 kV overload:

- a) The SSTE is 1837 MVA and not sufficient. Re-conductor a portion of line L0626. A preliminary estimate for this work is \$6M. Upon completion the new line ratings will be 1679/2058/2107/2280 MVA (SN/SLTE/SSTE/SLD). PJM Network Upgrade N5254.

The cost allocation is as follows:

Queue	MW contribution	Percentage of Cost	\$ cost (\$6 M)
J534		100.0%	6.0

5.3.1.10 Overload on Pleasant Prairie–Zion Station R- Zion EC 345 kV line

To relieve the Pleasant Prairie–Zion Station R- Zion EC 345 kV overload:

- a) The MISO-end of the Pleasant Prairie–Zion Station R 345 kV line SE rating is 1528 MVA and not sufficient. The planned upgrade is to use MTEP project 8065 which builds a new 345 kV sub by looping in the Zion-Libertyville Blue 345kV line and the Pleasant Prairie-Arcadian 345kV line. The Projected IS date for MTEP 8065 is 12-31-2020.
- b) The ComEd SSTE rating of the Zion EC–Zion Station R 345 kV line is 1251 MVA and not sufficient. The planned upgrade is to use MTEP project 8065 which builds a new 345 kV sub by looping in the Zion-Libertyville Blue 345kV line and the Pleasant Prairie-Arcadian 345kV line. The Projected IS date for MTEP 8065 is 12-31-2020.

Per PJM cost allocation rules, the DPP 2016 February West projects presently do not receive any cost allocation for this upgrade.

5.3.2 Study Summary

Multiple projects in the MISO DPP 2016 February West Area group contribute loading to the overloads in the PJM system. These projects are responsible for the cost of Network Upgrades per PJM cost allocation rules.

5.4 SPP Affected System AC Contingency Analysis

Southwest Power Pool (SPP) conducted an Affected System Interconnection System Impact Study (ASISIS) to evaluate potential impacts to the SPP Transmission System related to the interconnection of generators on the Mid-Continent Independent System Operation (MISO) Transmission System.

Power flow analysis has indicated that for the power flow cases studied, the SPP Transmission System is impacted by the interconnection of the MISO DPP 2016 February cycle West Area generation. These impacts are described in Section 8 and Section 9 of Appendix F.4. Steady state thermal and voltage impacts can be mitigated by determinations listed in Table 9 of Appendix F.4. Within the scope of this ASISIS, SPP coordinated with associated SPP Transmission Owners and MISO Transmission Owners for planning level estimates for mitigation of constraints requiring transmission reinforcements and will require Affected System Facilities Study (ASFS) if the MISO Interconnection Request(s) impacting the SPP transmission system proceed forward.

Certain constraints identified can be mitigated with either previously approved SPP or MISO upgrades. These include the Gentleman Generation Station (GGS) – Thedford - Holt 345 kV (R-Plan) transmission line and MISO MVP projects. The R-plan and MISO MVP projects are scheduled to be placed in service by January 1, 2019. Full Interconnection Service for these requests may need to be delayed until that time period.

If the MISO Interconnection Requests proceed, additional analysis will be required through an ASFS. The purpose of the ASFS is to refine the scope, cost estimates, and expected time of construction of Network Upgrades.

Potential mitigations for constraints identified in the SPP affected system are listed in Table 5-2.

MISO Generation Interconnection projects that come into service prior to completion of required Network Upgrades on the SPP transmission system will be subject to SPP's study process for Limited Operation. SPP updates the output limits on all Generator Interconnection Agreements when events occur on the Transmission System that are listed in the Limited Operation Impact Study for that generator to account for changing transmission and generation assumptions. SPP will coordinate project output limitations with MISO on an annual basis or more often as events are needed, and SPP will provide MISO the list of conditions that will be added to MISO Generator Interconnection Agreement.

The SPP affected system analysis results for this study are in Appendix F.4.

Table 5-2: SPP Suggested Mitigations for DPP 2016 February West Area Study

Mitigation	Area	Purpose	ERIS and / or NRIS Affecting Requests	Estimated Cost Cost
Rebuild approximately 11 miles of 115kV	WAPA	Required to mitigate AURORA - BROOKINGS 115KV CKT 1 NRIS thermal constraints for contingency conditions	NRIS: J432, J460, J493, J510, J526	\$8,800,000
Rebuild approximately 17 miles of 115kV	WAPA	Required to mitigate AURORA - FLANDRU 115KV CKT 1 NRIS thermal constraints for contingency conditions	NRIS: J432, J460, J493, J510, J526	\$13,600,000
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	NPPD/AECI	Required to mitigate COOPER - FAIRPORT 345KV CKT 1 NRIS thermal constraints for contingency conditions	NRIS: J432, J460, J475, J485, J490, J493, J495, J498, J499, J500, J501, J504, J506, J510, J514, J523, J524, J525, J526, J527, J528, J529, J530, J531, J534, J535	\$301,000,000
	NPPD/KCPL	Required to mitigate COOPER - ST JOE 345KV CKT 1 NRIS thermal constraints for system intact and contingency conditions	NRIS: J233/J514, J432, J460, J475, J485, J490, J493, J495, J498, J499, J500, J501, J504, J506, J510, J523, J524, J525, J526, J527, J528, J529, J530, J531, J534, J535	
	NPPD/OPPD/KCPL	Required to mitigate COOPER_S flowgate for system intact conditions	ERIS and NRIS: J233/J514, J432, J460, J475, J485, J488, J489, J490, J493, J495, J498, J499, J500, J501, J504, J506, J510, J523, J524, J525, J526, J527, J528, J529, J530, J531, J534, J535	
	NPPD/OPPD/KCPL	Required to mitigate various steady state voltage violations observed in NPPD, OPPD, KCPL areas after SPP previously allocated, MISO MVP, and MISO current study baseline “backbone” upgrades are inservice. Please refer to Appendix C-V for the incremental voltage constraints.	ERIS and NRIS: J233/J514, J432, J460, J475, J485, J488, J489, J490, J493, J495, J498, J499, J500, J501, J504, J506, J510, J523, J524, J525, J526, J527, J528, J529, J530, J531, J534, J535	

Affected System Steady-State Analysis

Mitigation	Area	Purpose	ERIS and / or NRIS Affecting Requests	Estimated Cost Cost
Rebuild OPPD's portion of the Council Bluffs – Sub 3456 345kV transmission circuit and upgrade Sub 3456 terminal equipment upgrade	MEC/OPPD	Required to mitigate COUNCIL BLUFFS - SUB 3456 345KV CKT 1 ERIS thermal constraints for system intact and contingency conditions	ERIS: J475, J498, J499, J500, J524, J527, J528, J530	\$5,500,000
Upgrade breakers, CT, and increase clearance	WAPA	Required to mitigate FT RANDAL - FT THOMPSON 230KV CKT 1 NRIS thermal constraint for contingency conditions	NRIS: J490	\$550,000
CT uprate and clearance increase	WAPA/NPPD	Required to mitigate GR ISLD-LNX3345.00 - HOLT.CO3 345.00 345KV CKT 1 NRIS thermal constraint for system intact and contingency conditions	NRIS: J432, J460, J485, J490, J493, J510, J525, J526, J531	\$1,000,000
Build Meadow Grove 230/115 Transformer and Meadow Grove - Petersburg 115kV line	NPPD	Required to mitigate KELLY - MEADOWGROVE4230.00 230KV CKT 1 NRIS thermal constraint for system intact and contingency conditions	NRIS: J432, J460, J490, J526	\$30,000,000
Rebuild Pahoja – Sioux Falls	WAPA	Required to mitigate PAHOJA - SIOUX FALLS 230KV CKT 1 NRIS thermal constraint for contingency conditions	NRIS: J432, J460, J490, J493, J510, J525, J526	\$14,000,000

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Stability Analysis

Stability analysis was performed to evaluate the transient stability and impact on the region of the generating facilities in the DPP 2016 February West Area group.

6.1 Procedure

6.1.1 Computer Programs

Stability analysis was performed using PSS®E revision 33.9.

6.1.2 Study Methodology

Study stability package representing 2021 summer shoulder (SH) conditions with generating facilities in the DPP 2016 February West Area group was created from the stability package used for the DPP 2015 August West area study, which was originally developed from MTEP15 stability package. Benchmark case was created by removing the DPP West Area generating facilities from the study case. Disturbances were simulated to evaluate the transient stability and impact on the region of the generating facilities. If a simulation for the study case violates MISO transient stability criteria or local TOs' planning criteria, the simulation was repeated on the benchmark case to assess the impact of the generating facilities on the violation.

6.2 Case Development

6.2.1 Study Case

Study case representing 2021 shoulder (SH) high transfer conditions was developed from the DPP 2015 August West area stability package, which was originally developed from MTEP15 stability package.

The stability study case for DPP 2016 February study was created as described in Section 2.1.1. Required reactive power NUs for MWEX voltage stability (Table 4-14) and additional reactive power NUs required for voltage constraints (Table 4-15) were also added in the stability case.

The interface transfer levels are summarized in Table 6-1.

Table 6-1: Interface Transfer Levels in 2021 Stability Study Cases

Interface	2021 SH Case (MW)
MHEX	2431
MWEX	2001
Arrowhead – Stone Lake 345 kV	893

6.2.2 Benchmark Case

The DPP West Area generating facilities as described in Table A-1 (Appendix A.1) and reactive power NUs (Table 4-14, Table 4-15) were removed from the study case. MISO Classic was used for power balance, where generation was scaled in proportion to Pmax - Pgen.

6.3 Disturbance Criteria

The stability simulations performed as part of this study considered all the regional and local contingencies listed in Table 6-2. Regional contingencies with the pre-defined switching sequences were selected from the MISO MTEP15 study; switching sequences for local contingencies were developed based on the generic clearing times shown in Table 6-3. The admittance for local single line-to-ground (SLG) faults were estimated by assuming that the Thevenin impedance of the positive, negative and zero sequence networks at the fault point are equal.

Table 6-2: Regional and Local Disturbance Descriptions

CEII Redacted

Table 6-3: Generic Clearing Time Assumption

Voltage Level (kV)	Primary Clearing Time (cycle)	Backup Clearing Time (cycle)
345 kV	4	11
230 kV	5	13
161/138 kV	6	18
115 kV	6	20
69 kV	8	24

6.4 Performance Criteria

All generators must mitigate the stability constraints listed below in order to obtain any type of Interconnection Service:

- System instability
- Transient voltage constraint

- Damping violation

6.4.1 MISO Criteria

Stability simulation results are evaluated based on the following MISO criteria:

- All on-line generating units are stable
- No unexpected generator tripping
- Post-fault transient voltage limits: 1.2 per unit maximum, 0.7 per unit minimum.
- Per local TOs' planning criteria, specific transient voltage limits are applied to specific buses, areas or companies that have different requirements.
- All machine rotor angle oscillations must be positively damped with a minimum damping ratio of 0.81633% for disturbances with a fault or 1.6766% for line trips without a fault.

A bus is considered a transient voltage constraint if both of the following conditions are met. All transient voltage constraints must be resolved before a project can receive interconnection service.

1. the bus transient voltage is outside of specified transient voltage limits during transient period, and
2. the bus voltage is at least 0.01 per unit worse than the benchmark case voltage for the same contingency.

6.4.2 Local Planning Criteria

6.4.2.1 ATC Local Planning Criteria

ATC has the following local transient voltage recovery criteria. For facilities in the ATC footprint, transient voltage recovery is evaluated based on ATC's local planning criteria.

- Voltage recovery within 80 percent and 120 percent of nominal for between 2 and 20 seconds following the clearing of a disturbance.

6.4.2.2 ITCM Local Planning Criteria

ITCM has the following local transient voltage and damping criteria. For facilities in the ITCM footprint, transient voltages and dampings are evaluated based on ITCM's local planning criteria.

- Voltages at all busses on the Transmission Systems should not drop below 0.70 per unit after the first swing for more than 5 cycles. The duration for the minimum voltage dip starts after the first swing post clearing of fault.
- Voltage at all Transmission System buses should recover to the applicable post-contingency steady-state voltage level (ITCM post-disturbance limits in Table 2-3), within 1.0 second of the clearing of the fault.
- Rotor angle oscillation damping ratios are not to be less than 0.03.

6.4.2.3 MEC Local Planning Criteria

MEC has the following local transient voltage and damping criteria. For facilities in the MEC footprint, transient voltages and dampings are evaluated based on MEC's local planning criteria.

- Generator bus transient voltage limits shall adhere to the high voltage duration and low voltage duration curve in Attachment 2 of NERC PRC-024, which is:
 - Generator bus transient over voltage limits (after fault clearing): 1.2 pu voltage from 0.0 to and including 0.2 s; 1.175 pu voltage from 0.2 to and including 0.5 s; 1.15 pu voltage from 0.5 to and including 1.0 s; 1.1 pu voltage for greater than 1.0 s.
 - Generator bus transient low voltage limits (after fault clearing): may be less than 0.45 pu voltage from 0 to 0.15 seconds; Voltage shall remain above 0.45 pu from 0.15 to 0.3 s; Voltage shall remain above 0.65 pu from 0.3 to 2.0 s; Voltage shall remain above 0.75 pu from 2.0 to 3.0 s; Voltage shall recover to 0.9 pu after 3 s.
- Load bus transient voltage limits:
 - Load bus transient over voltage limits (after fault clearing): 1.6 pu voltage from 0.01 to and including 0.04 s; 1.2 pu voltage from 0.04 to and including 0.5 s; 1.1 pu voltage from 0.5 to and including 5 s; and 1.05 pu voltage for greater than 5 s. These voltage limits also apply to buses without loads or generators.
 - Load bus transient low voltage limits (after fault clearing): may be less than 0.7 pu voltage from 0 to 2 s; Voltage shall remain above 0.7 pu from 2 to 20 s; Voltage shall recover to 0.9 pu after 20 s.
- Angular transient stability minimum damping ratio (ζ) should not be less than 0.03.

6.5 Stability Results

The contingencies listed in Table 6-2 were simulated using the 2021 summer shoulder study case. If a transient stability criteria violation was identified, the same disturbance was repeated in the benchmark case.

Appendix G.2 contains plots of generator rotor angles, generator power output, generator terminal voltages, bus voltages, and branch flows for each simulation. Simulations were performed with a 2.0 seconds steady-state run followed by the appropriate disturbance. Simulations were run for a 12-second duration.

6.5.1 2021 Stability Analysis Results

Stability study results summary is in Appendix G, Table G-1. All simulations are transiently stable, transient period voltage criteria are met, oscillations are well damped.

6.6 Network Upgrades Identified in Stability Analysis

With the required reactive power NUs for MWEX voltage stability (Table 4-14) and additional reactive power NUs required for voltage constraints (Table 4-15) modeled in the stability case, no additional Network Upgrades are required in stability analysis.

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MWEX Voltage Stability Study

ATC performed steady state voltage stability analysis. Voltage stability analysis is required to determine if the initial conditions of the DPP system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for loss of the worst contingency.

The voltage stability analysis used 2021 shoulder load cases to compare the Pre-DPP and Post-DPP + MISO Backbone Network Upgrades. The MISO Backbone Network Upgrades are the following projects.

1. Construction of a New Hartford – Tama County – Morganville 345 kV line,
2. Construction of a Big Stone South – Alexandria 345 kV line,
3. Construction of a Beverly – Sub 92 345 kV line,
4. Installation of 2x150 MVAR capacitors at New Hartford 345 kV,
5. Installation of 2x150 MVAR capacitors at Hickory Creek 345 kV,
6. Installation of a +/- 200 MVAR STATCOM at Arrowhead 230 kV,
7. Installation of a 60% series capacitor on North Rochester – Briggs Road 345 kV

As shown in Table 7-1, the Post-DPP + MISO Backbone Network Upgrades scenario in the 2021SH cases is not voltage stable.

Table 7-1: MWEX Margins to Collapse in the 2021SH Cases

	Real Power Flows (MW)						
	AHD-SLK ¹	MWEX			Margin to Nose ^{2,3}		
Case	N-0 Initial Condition	N-0 ⁴ I.C.	N-1 ⁴ I.C.	N-1 Nose	(MW)	(%)	Notes
Pre-DPP	755	1715	1490	1582	92	5.8	Voltage Stable Margin < 10%
Post-DPP + MISO Backbone NUs	849	1915	N/A				Not Voltage Stable
Post-DPP + MWEX Solution	758	1734	1491	1584	93	5.9	Voltage Stable Margin < 10% ⁵

Notes:

1. As described in the active MWEX Operating Guide, the AHD-SLK interface is a single element PTDF interface measured at the Minnesota Power 230 kV side of the Arrowhead 230 kV phase shifter.

2. Margin to Nose (MW) is defined as the MWEX N-1 Nose minus the N-1 Initial Condition.
3. Margin to Nose (%) is defined as the Margin to Nose (MW) divided by the MWEX N-1 Nose.
4. Initial Condition flows were measured in the base cases with the worst contingency plus operation of various control systems as needed with all transformer taps, switched shunts, and PARs locked.
5. ATC Planning Criteria requires a 10% voltage stability margin but the Interconnection Customers are only responsible for alleviating their impact on the system.

The Post-DPP + MISO Backbone Network Upgrades scenario does not converge with the worst contingency and therefore is in a violation of ATC Planning Criteria such that Network Upgrades will be assigned to the Interconnection Customers. Based on the assumptions used in this analysis, including existing generation dispatch, ATC recommends the following Network Upgrades in addition to the MISO Backbone Network Upgrades, which will result in the Post-DPP case with a voltage stability margin equivalent to the Pre-DPP case.

1. Construction of a North Rochester – Rocky Run 345 kV line to be constructed such that there are no valid P2, P4, P5, or P7 contingencies with other critical 345 kV lines.
2. A rebuild of the Rocky Run 345 kV Substation as a breaker-and-a-half configuration replacing the three existing 345/115 kV transformers with two new 345/115 kV transformers.
3. Installation of a 300 MVAR SVC at North Rochester 345 kV.
4. Installation of a 300 MVAR SVC at Rocky Run 345 kV.
5. Installation of a 60% series capacitor on Briggs Road – North Madison 345 kV.
6. Eliminating the Arrowhead 230 kV STATCOM from the MISO Backbone Network Upgrades.
7. Eliminating the North Rochester – Briggs Road series capacitor from the MISO Backbone Network Upgrades.

The combination of the MISO Backbone Upgrades and these Network Upgrades is hereafter referred to as the MWEX Solution.

The results of this analysis show that the Post-DPP scenario in the 2021 shoulder load cases is not voltage stable because the post-contingency initial condition doesn't converge. With the proposed MWEX Solution, the Post-DPP case is voltage stable and has a voltage stability margin equivalent to the Pre-DPP case.

The MWEX voltage stability study details can be found in Appendix H.

Short Circuit Analysis

8.1 Introduction

Siemens PTI and several transmission owning companies performed short circuit analysis for the DPP 2016 February West Area generation projects.

8.2 J432 Short Circuit Study Performed by Siemens PTI

The J432 short circuit study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J432 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.1.

8.3 J460 Short Circuit Study Performed by Siemens PTI

The J460 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 11577 A (increased by 715.2 A) and SLG fault current is 9751 A (increased by 1399.6 A) at the Hawks Nest 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J460 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.2.

8.4 J475 Short Circuit Study Performed by MEC

The J475 short circuit study was performed by MEC. The study results show that the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project.

Study details can be found in Appendix I.3.

8.5 J485 Short Circuit Study Performed by Siemens PTI

The J485 short circuit facilities study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J485 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.4.

8.6 J488 & J489 Short Circuit Study Performed by OTP

The J488 and J489 short circuit study was performed by OTP. With the proposed J488/J489 project additions, the fault currents are estimated to be roughly 8.6 kA at the Twin Brooks 345 kV bus, 11.2 kA at the Big Stone South 345 kV bus, 19.7 kA at the Big Stone South 230 kV bus, and 33.6 kA at the Big Stone South 34.5 kV bus. Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required.

Study details can be found in Appendix I.5.

8.7 J490 Short Circuit Study Performed by Siemens PTI

The J490 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 1370.8 A (increased by 357.5 A) and SLG fault current is 954.6 A (increased by 60.5 A) at the MCLGHLN5 115 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J490 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.6.

8.8 J493 and J510 Short Circuit Study Performed by OTP

The J493 and J510 short circuit study was performed by OTP.

With the proposed J493 project additions, the fault currents are estimated to be roughly 16 kA at the Big Stone South 230 kV bus and the faults on the 345 kV bus were below these values.

With the proposed J493 and J510 project additions, the fault currents are estimated to be roughly 16.6 kA at the Big Stone South 230 kV bus.

Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required.

Study details can be found in Appendix I.7.

8.9 J495 Short Circuit Study Performed by ITCM

The J495 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J495. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. New circuit breakers at the POI switch station were assumed to have 50 kA interrupting capability.

Study details can be found in Appendix I.8.

8.10 J498 Short Circuit Facilities Study Performed by MEC

The J498 short circuit facilities study was performed by MEC. The study results show that the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no

MidAmerican breakers become overdutied as a result of the Interconnection Customer's project.

Study details can be found in Appendix I.9.

8.11 J499 Short Circuit Facilities Study Performed by MEC

The J499 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J499 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.10.

8.12 J500 Short Circuit Facilities Study Performed by MEC

The J500 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J500 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.11.

8.13 J501 Short Circuit Facilities Study Performed by MEC

The J501 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J501 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.12.

8.14 J504 Short Circuit Facilities Study Performed by ITCM

The J504 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J504. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. There are several existing breakers at DAEC 161 kV substation that are expected to be replaced as part of planned work at this location, as well as new breakers being installed with the bus expansion. 40 kA interrupting capability was assumed for the new and replaced breakers.

Study details can be found in Appendix I.13.

8.15 J506 Short Circuit Facilities Study Performed by Siemens PTI

The J506 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 12205.7 A (increased by 806.3 A) and SLG fault current is 9955.5 A (increased by 1698.8 A) at the J506 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J506 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.14.

8.16 J514 Short Circuit Facilities Study Performed by ITCM

The J514 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J514. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. Circuit breaker 2680 at Marshalltown 115 kV was previously identified as a constraint for J233, but has since been removed.

Study details can be found in Appendix I.15.

8.17 J523 Short Circuit Facilities Study Performed by ITCM

The J523 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J523. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix I.16.

8.18 J524 Short Circuit Facilities Study Performed by Siemens PTI

The J524 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 19095 A (increased by 580.4 A) and SLG fault current is 19487 A (increased by 1244.2 A) at the Webster 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J524 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.17.

8.19 J525 Short Circuit Facilities Study Performed by GRE

The J525 short circuit facilities study was performed by GRE. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J525 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.18.

8.20 J526 Short Circuit Facilities Study Performed by OTP

The J526 short circuit facilities study was performed by OTP.

With the proposed J526 project additions, the fault currents are estimated to be roughly 11.7 kA at the Big Stone South 345 kV bus, 20.6 kA at the Big Stone South 230 kV bus, and 37.1 kA at the Big Stone South 34.5 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J526 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.19.

8.21 J527 Short Circuit Facilities Study Performed by Siemens PTI

The J527 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 17037 A (increased by 1119.2 A) and SLG fault current is 14696 A (increased by 2339.9 A) at the J527 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J527 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.20.

8.22 J528 Short Circuit Facilities Study Performed by Siemens PTI

The J528 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 16673 A (increased by 868.9 A) and SLG fault current is 14134 A (increased by 1757.2 A) at the J528 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J528 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.21.

8.23 J529 Short Circuit Facilities Study Performed by MEC

The J529 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J529 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.22.

8.24 J530 Short Circuit Facilities Study Performed by Siemens PTI

The J530 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 13118 A (increased by 1118.4 A) and SLG fault current is 10678 A (increased by 2793.4 A) at the J530 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J530 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.23.

8.25 J531 Short Circuit Facilities Study Performed by ITCM

The J531 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J531. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. New circuit breakers at the POI switch station were assumed to have 50kA interrupting capability.

Study details can be found in Appendix I.24.

8.26 J534 Short Circuit Facilities Study Performed by Siemens PTI

The J534 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 13335 A (increased by 1119.5 A) and SLG fault current is 11425

A (increased by 2413.0 A) at the J534 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J534 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.25.

8.27 J535 Short Circuit Facilities Study Performed by Siemens PTI

The J535 short circuit facilities study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J535 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix I.26.

Deliverability Study

9.1 Project Description

Interconnection requests requesting Network Resource Interconnection Services (NRIS) were considered for deliverability analysis.

9.2 Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS).

If the generator is determined as not fully deliverable, the customer can choose either to change his project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable.

Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottlenecked. The wind generators are tested at 100 % of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

9.3 Study Methodology

MISO Generator Deliverability Study whitepaper describing the algorithm can be found at ["https://www.misoenergy.org/Library/Repository/Study/Generator%20Interconnection/Generator%20Deliverability%20Study%20Methodology.pdf"](https://www.misoenergy.org/Library/Repository/Study/Generator%20Interconnection/Generator%20Deliverability%20Study%20Methodology.pdf).

9.4 Determining the MW restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study is in the “Top 30 DF List” (see white paper for detail), part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

The generator is also responsible for any NEW base case (pre-shift) overload or NEW “severe yet credible dispatch overload” where the generator is not in the “Top 30 DF List”, if the generator’s DF is greater than 5%. Please see white paper for detail. The formula above also applies to these situations.

9.5 2021 Deliverability Study Result

9.5.1 J432

J432 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS upgrades and case assumptions)	98 MW (100%)
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9.5.2 J460

J460 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J460 34.5/345 kV Transformer (IC Facility)	200.00	99.99%	No		J460	NA	NA

9.5.3 J475

J475 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS upgrades and case assumptions)	200 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV ¹	200.00	5.45%	Yes	J499,J500, J527,J528	J475,J499, J500,J527, J528		

Note 1: J475 has impact on this constraint only in the NRIS analysis. If all ERIS impacted generation projects are withdrawn, J475 will be responsible for this NU.

9.5.4 J485

J485 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS upgrades and case assumptions)	46.85 MW (100%)
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9.5.5 J490

J490 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIIS upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated With ERIIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Fargo - Sheynne 230 kV	0.00	8.59%	No		J490	\$750,000	\$750,000
Merricourt - Ellendale MVP 230 kV	60.00	5.57%	No		J490	\$5,000,000	\$5,000,000

9.5.6 J493

J493 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIIS upgrades and case assumptions)	150 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated With ERIIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Big Stone - Blair 345 kV	150.00	28.67%	Yes	J488,J489, J493,J510, J526	J493,J510, J526		

9.5.7 J495

J495 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIIS and IC upgrades and case assumptions)	200 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated With ERIIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J495 34.5/345 kV Transformer (IC Facility)	180.81	99.99%	No		J495		
Killdeer 345/Emery 161 kV Transformer ¹	200.00	10.24%	Yes	J501	J495,J501, J524,J529, J534		

Note 1: J495 has impact on this constraint only in the NRIS analysis. If all ERIIS impacted generation projects are withdrawn, J495 will be responsible for this NU.

9.5.8 J498

J498 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIIS and IC upgrades and case assumptions)	340 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated With ERIIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J498 34.5/345 kV Transformer (IC Facility)	331.40	99.98%	No		J498		
Fort Dodge - Boone Junction 161 kV	340.00	6.78%	Yes	J498,J534, J535	J498,J524, J534		

9.5.9 J499

J499 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIIS and IC upgrades and case assumptions)	340 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIIS Analysis?	Projects Associated With ERIIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	180.07	57.97%	Yes	J499,J500, J527,J528	J475,J499, J500,J527, J528		
J499 34.5/345 kV Transformer (IC Facility)	340.00	99.98%	No		J499		

9.5.10 J500

J500 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS and IC upgrades and case assumptions)	500 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	180.07	17.53%	Yes	J499, J500, J527, J528	J475, J499, J500, J527, J528		
J500 34.5/345 kV Transformer (IC Facility)	500.00	99.98%	No		J500		

9.5.11 J501

J501 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS and IC upgrades and case assumptions)	500 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J501 34.5/345 kV Transformer (IC Facility)	452.03	99.98%	No		J501		
Killdeer 345/Emery 161 kV Transformer	500.00	6.99%	Yes	J501	J495, J501, J524, J529, J534		

9.5.12 J504

J504 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS upgrades and case assumptions)	21.34 MW (42.68%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Marshalltown - Jasper County 161 kV	50.00	5.42%	No		J504, J514	\$433,684	\$37,500,000

9.5.13 J506

J506 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Dickinson County - Triboji 161KV	200.00	5.13%	No		J506	\$3,000,000	\$3,000,000

9.5.14 J510

J510 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS upgrades and case assumptions)	321.9 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Big Stone - Blair 345 kV	321.90	28.67%	Yes	J488,J489, J493,J510, J526	J493,J510, J526		

9.5.15 J514

J514 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Marshalltown - Jasper County 161 kV	0.00	34.83%	No		J504, J514	\$37,066,316	\$37,500,000
Marshalltown - Timber Creek 161 kV	30.00	25.26%	No		J514	\$8,500,000	\$8,500,000

9.5.16 J523

J523 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERS upgrades and case assumptions)	50 MW (100%)
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9.5.17 J524

J524 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERS upgrades and case assumptions)	100 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated With ERS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Killdeer 345/Emery 161 kV Transformer ¹	97.47	6.77%	Yes	J501	J495,J501, J524,J529, J534		
Fort Dodge - Boone Junction 161 kV ²	100.00	11.58%	Yes	J498,J534, J535	J498,J524, J534		

Note 1: J524 has impact on this constraint only in the NRIS analysis. If all ERS impacted generation projects are withdrawn, J524 will be responsible for this NU.

Note 2: J524 has impact on this constraint only in the NRIS analysis. If all ERS impacted generation projects are withdrawn, J524 will be responsible for this NU.

9.5.18 J525

J525 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERS upgrades and case assumptions)	50 MW (100%)
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9.5.19 J526

J526 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS and IC upgrades and case assumptions)	300 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J526 34.5/345 kV Transformer (IC Facility)	219.36	99.98%	No		J526		
Big Stone - Blair 345 kV	300.00	28.67%	Yes	J488,J489, J493,J510, J526	J493,J510, J526		

9.5.20 J527

J527 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS and IC upgrades and case assumptions)	250 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	150.06	16.89%	Yes	J499,J500, J527,J528	J475,J499, J500,J527, J528		
J527 34.5/345 kV Transformer (IC Facility)	250.00	99.98%	No		J527		

9.5.21 J528

J528 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS and IC upgrades and case assumptions)	200 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Council Bluffs - Sub 3456 345 kV	110.04	16.96%	Yes	J499,J500, J527,J528	J475,J499, J500,J527, J528		
J528 34.5/345 kV Transformer (IC Facility)	200.00	99.99%	No		J528		

9.5.22 J529

J529 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS upgrades and case assumptions)	250 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Killdeer 345/Emery 161 kV Transformer ¹	250.00	6.67%	Yes	J501	J495,J501, J524,J529, J534		

Note 1: J529 has impact on this constraint only in the NRIS analysis. If all ERIS impacted generation projects are withdrawn, J529 will be responsible for this NU.

9.5.23 J530

J530 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERIS and IC upgrades and case assumptions)	250 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERIS Analysis?	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J530 34.5/345 kV Transformer (IC Facility)	250.00	99.98%	No		J530		

9.5.24 J531

J531 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERS and IC upgrades and case assumptions)	200 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated With ERS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J531 34.5/345 kV Transformer (IC Facility)	200.00	99.99%	No		J531		

9.5.25 J534

J534 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERS and IC upgrades and case assumptions)	250 MW (100%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraint in ERS Analysis?	Projects Associated With ERS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
J534 34.5/345 kV Transformer (IC Facility)	226.01	99.98%	No		J534		
Killdeer 345/Emery 161 kV Transformer ¹	243.68	6.74%	Yes	J501	J495,J501, J524,J529, J534		
Fort Dodge - Boone Junction 161 kV	250.00	5.17%	Yes	J498,J534, J535	J498,J524, J534		

Note 1: J534 has impact on this constraint only in the NRIS analysis. If all ERS impacted generation projects are withdrawn, J534 will be responsible for this NU.

9.5.26 J535

J535 Deliverable (NRIS) Amount in 2021 Case: (Conditional on ERS upgrades and case assumptions)	210 MW (100%)
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Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study.

Generation projects J504, J514, and J531 are the only three projects to have the SNUs. These Network Upgrades were originally identified in the MISO DPP 2013 August and 2015 August West Area study cycles which are listed below:

- Hazleton-Mitchell Co 345 kV (\$2,750,000, DPP 2013 August)
- Jasper-Laurel 161 kV (\$600,000, DPP 2013 August)
- Arnold-Morgan Valley 161 kV (\$2,245,000, DPP 2015 August)

The maximum MW impacts and Shared Network Upgrade (SNU) cost allocations are listed in Table 10-1.

Table 10-1: Maximum MW Impact and SNU Cost Allocations

Network Upgrades	Original Driver Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
Hazleton-Mitchell Co 345 kV	DPP-2013-AUG	J278	52.93	\$2,750,000	\$606,955
	DPP-2015-AUG	J449	76.7499		\$880,101
	DPP-2016-FEB	J531	110.136		\$1,262,944
Jasper-Laurel 161 kV	DPP-2013-AUG	J233	202.31	\$600,000	\$479,075
	DPP-2015-AUG	J298	31.734		\$75,147
	DPP-2016-FEB	J514	19.332		\$45,779
Arnold-Morgan Valley 161 kV	DPP-2015-AUG	J298	45.219	\$2,245,000	\$1,752,629
	DPP-2016-FEB	J504	12.7035		\$492,371

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Section

11

Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

11.1 Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

11.2 ERIS Network Upgrades Proposed for DPP West Area Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the MISO ERIS analysis and the affected system analysis. The ERIS network upgrades include voltage support network upgrades and thermal network upgrades identified in the MISO steady-state analysis, thermal network upgrades identified in the Local Planning Criteria analysis and affected system analysis, voltage support network upgrades identified in the MWEX voltage stability analysis, stability network upgrades identified in the MISO transient stability analysis, and short circuit network upgrades identified in the MISO short circuit analysis. The total costs of ERIS network upgrades for the 2021 scenario are summarized in Table 11-1.

Table 11-1: Summary of ERIS Network Upgrades

Category of Network Upgrades	2021 Cost (\$)
Backbone Network Upgrades	\$665,500,000
Additional Thermal Network Upgrades Identified in MISO Steady-State Analysis	\$289,430,000
Network Upgrades Identified in MWEX Voltage Stability analysis	\$1,026,200,000
Additional Reactive Power Network Upgrades for Voltage Constraints	\$62,000,000
Network Upgrades Identified in Transient Stability Analysis	\$0
Network Upgrades Identified in Short Circuit Analysis	\$0
Network Upgrades Identified in GRE LPC Analysis	\$26,900,000
Network Upgrades Identified in CIPCO/MPC Affected Systems	\$25,914,280
Network Upgrades Identified in PJM Affected Systems	\$21,775,800
Network Upgrades Identified in SPP Affected Systems	\$374,450,000
Shared Network Upgrades	\$1,801,093

Cost Allocation

Category of Network Upgrades	2021 Cost (\$)
Total	\$2,493,971,173

ERIS network upgrades are listed below.

Table 11-2: Backbone Network Upgrades

Area	NUs	Miles	Cost (\$)
MN-WI	Hickory Creek-Eden Prairie-Cardinal 345 kV	NA	NA
Minnesota Area	Huntley-Wilmarth 345 kV	NA	NA
Big Stone Area	Big Stone South-Alexandria 345 kV	95	\$197,400,000
Central-East Iowa	New Hartford-Morgan Valley Section	117	\$309,150,000
	Morgan Valley Sub-Beverly Sub	NA	NA
	Beverly-Sub 92 345 kV	65	\$158,950,000

Table 11-3: Additional Thermal Network Upgrades in MISO Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
J438 POI-Parnell 161 kV	ITCM MEC	ITCM: Not an ITCM constraint; ITCM terminal limit is 335 MVA. \$0 MEC: Structure replacements. \$200,000	\$200,000
J531 POI-Hazleton 345 kV	ITCM MEC	ITCM: ITCM owns 28.1 miles and MEC owns 16.4 miles of that line segment. The 1224.2 MVA loading indicated below is beyond the conductor limit and rebuild ITCM's 28.1 with 2-T2 795 ACSR would be required. Rating of 1781 MVA for rebuild ITCM section. \$53M. MEC: Structure replacements. \$3,000,000	\$56,000,000
J475 substation xfmr	J475	J475 is responsible.	IC Cost
J498 POI-Grimes 345 kV	MEC	MEC: Reconfigure Raun Sub to eliminate Raun 0270 CON and reduce loading below maximum conductor rating (1333 MVA) for next worst CON (\$5 M). Structure replacements on J498-Grimes 345 kV (\$2.5 M).	\$2,500,000
J530 POI-Hills 345 kV	MEC	MEC: Structure replacements and terminal equipment upgrades	\$600,000
J523 substation xfmr	J523	J523 is responsible.	IC Cost
J523 161 kV interconnection line	J523	J523 is responsible.	IC Cost
Brookings Co-White 345 kV #2	XEL WAPA	XEL: Line is limited by WAPA terminal limits. NSP's equipment able to handle the required loading. \$0 WAPA: Upgrade the White terminal. \$60,000. Affected system	\$60,000

Cost Allocation

Constraint	Owner	Mitigation	Cost (\$)
		upgrades. M3 calculations will not reflect these costs	
Panther-McLeod 230 kV	GRE	Remediate 4 Structures	\$1,000,000
Hankinson-Forman 230 kV	OTP	Line clearance mitigations and terminal equipment upgrades.	\$500,000
Oakes-Forman 230 kV	OTP	Line clearance mitigations and terminal equipment upgrades.	\$750,000
Oakes-Ellendale 230 kV	OTP MDU	OTP: Line clearance mitigations and terminal equipment upgrades. \$1,550,000 MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$1,550,000
Marshalltown-Blairstown 115 kV	ITCM	Raise/replace structures to reach 123 MVA	\$2,700,000
Blairstown-Prairie Crk 115 kV	CIPCO ITCM	CIPCO: No Mitigation Needed. \$0 ITCM: Raise structures to reach a minimum rating of 114 MVA	\$2,700,000
Killdeer 345-161 kV xfmr	ITCM	Add second transformer	\$6,500,000
Emery-Colby 161 kV	ITCM	Upgrade approx .5 miles of 161 kV - new rating at 418 MVA	\$500,000
Boone Jct-Sub T Fort Dodge 161 kV	MEC CIPCO	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor line and terminal equipment upgrades. \$7,500,000	\$7,500,000
Henry Co-Denmark 161 kV	ITCM	Raise/replace 4 spans to 223 MVA	\$960,000
Henry Co-Jeff 161 kV	ITCM	Raise/replace 4 spans to 223 MVA	\$960,000
Wapello-Jeff 161 kV	ITCM	Line is conductor limited at 240 MVA. Rebuild 22.6 miles of 161 kV with T2-795ACSR	\$27,000,000
Poweshiek-Reasnor 161 kV	ITCM	Line is conductor limited at 327 MVA. Rebuild 20.7 miles of 161 kV with T2-795 ACSR	\$25,000,000
Adams S-Beaver Crk 161 kV	ITCM DPC	ITCM: Not an ITCM constraint - ITCM terminal limit is 335 MVA. \$0 DPC: Rebuild 15 miles 636 ACSR 161 kV line. \$14.7 M	\$14,700,000
Council Bluffs-S3456 345 KV	MEC OPPD	MEC: No mitigation required. Existing MEC emergency rating is 1439 MVA. \$0 OPPD: Upgrade terminal equipment; Replace structure. \$2 M. Affected system upgrades. M3 calculations will not reflect these costs	\$2,000,000
Raun-S3451 345 kV	MEC OPPD	MEC: No mitigation required. Existing MEC emergency rating is 1152 MVA. \$0 OPPD: Upgrade terminal equipment; Rebuild 63 miles line. \$72 M. Affected system upgrades. M3 calculations will not reflect these costs	\$72,000,000

Cost Allocation

Constraint	Owner	Mitigation	Cost (\$)
Raun-Tekamah 161 kV	MEC OPPD	MEC: Reconfigure Raun Sub to eliminate Raun 0270 CON. \$5,000,000 OPPD: \$0	\$5,000,000
Grimes-Sycamore 345 kV #2	MEC	Current MEC ring open emergency rating is 1018 MVA. Terminal equipment upgrades; Add breaker between Grimes 345-161 kV Transformer 9T1 and Grimes-Sycamore 345 kV Ckt 1 to eliminate CON	\$3,100,000
Grimes-Granger Tap 161 kV	MEC	Reconductor line	\$2,000,000
Granger Tap-108th & 54th 161 kV	MEC	Current rating is 360 MVA. Reconductor line and terminal equipment upgrades	\$1,250,000
Bondurant-Sycamore 345 kV	MEC	Structure replacements and terminal equipment upgrades	\$1,000,000
Bondurant-Montezuma 345 kV	MEC	Structure replacements and terminal equipment upgrades	\$2,100,000
Franklin-Wall Lake 161 kV	MEC	Structure replacements	\$500,000
Sub 56-Walcott 345 kV	MEC	Current emergency rating is 1195 MVA. Add breakers at Sub T Haskins to change CON by removing loss of Hills-Sub T 345 kV. Reduces loading to 1155 MVA	\$4,000,000
Walcott-Sub 92 345 kV	MEC	Add breakers at Sub T Haskins to change CON by removing loss of Hills-Sub T 345 kV. Reduces loading to 1319 MVA (\$4 M). Replace structures and terminal equipment to achieve this rating (\$300,000).	\$300,000
Glenham 230-115-41.6 kV xfmr	MDU	New Glenham 230/115/41.6 kV sub & transformer	\$6,500,000
Seneca-Genoa 161 kV	DPC	The DPC 161 kV lines from Genoa to Seneca and Seneca to Gran Grae are limited by conductor	\$38,000,000
Seneca-Gran Grae 161 kV	DPC	The DPC 161 kV lines from Genoa to Seneca and Seneca to Gran Grae are limited by conductor	Incl.

Table 11-4: Network Upgrades Required for MWEX Voltage Stability

Area	NUs	Cost (\$)
MWEX	North Rochester Substation Rework	\$8,000,000
MWEX	161/69 kV Xcel/DPC substation in Minnesota	\$5,500,000
MWEX	New 345 kV line North Rochester - Rocky Run (170 miles)	\$793,287,928
MWEX	Install a 300 Mvar SVC at North Rochester 345 kV	\$32,000,000
MWEX	Rock Run Substation Rework; 300 MVar SVC	\$169,412,072
MWEX	60% series compensation at the end of Briggs Road - North Madison 345 kV line	\$18,000,000

Table 11-5: Additional Reactive Power NUs Required for Voltage Constraints

Network Upgrades	Owner	Cost (\$)
2*50 Mvar switched shunt capacitor at Zachary 161 kV (344010)	Ameren	\$1,000,000
5*50 Mvar SVC at Overton 161 kV (345409)	Ameren	\$25,000,000
2*150 Mvar switched shunt capacitor at Hickory 345 kV (631191)	ITCM	\$8,000,000
1*150 Mvar switched shunt capacitor at New Hartford 345 kV (631240)	MEC	\$4,000,000
1*150 Mvar switched shunt capacitor at Blackhawk 345 kV (636199)	MEC	\$4,000,000
1*150 Mvar switched shunt capacitor at Montezuma 345 kV (635730)	MEC	\$4,000,000
2*150 Mvar switched shunt capacitor at Hills 345 kV (636400)	MEC	\$8,000,000
2*150 Mvar switched shunt capacitor at Morgan Valley 345 kV (631148)	MEC	\$8,000,000

Table 11-6: GRE Local Planning Criteria Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	J525 Cost (\$)
33 miles of 69 kV lines between Chandler tap and Lyon County	Xcel	rebuilding the existing transmission with new 477 ACSR	\$23,000,000	\$23,000,000
Fenton 115-69 kV transformer	Xcel	Install 2nd 115-69 kV transformer at Fenton	\$3,900,000	\$3,900,000

Table 11-7: CIPCO Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
CIPCO IR20-Dundee 161 kV ¹	CIPCO ITCM	IR-20 is queued prior to the Feb 16 DPP group with impact of 48.643 MW. J514 has impact of 1.156 MW	\$6,000,000
Prairie Crk-Marion 115 kV	CIPCO ITCM	CIPCO: Replace Jumpers at Prairie Creek \$50,000 ITCM: CIPCO LPC. \$0	\$50,000
Prairie Crk-Bertram 115 kV	CIPCO ITCM	CIPCO: Rebuild of Double Circuit from Prairie Creek to Bertram. Would require new structures and conductor on one circuit. Old conductor can be re-used on Prairie Creek - Marion. \$16,000,000 ITCM: CIPCO LPC. \$0	\$16,000,000
Prairie Crk-Squared 115 kV	CIPCO ITCM	CIPCO: Replace Jumpers at Prairie Creek. \$50,000 ITCM: CIPCO LPC. \$0	\$50,000
Bertram 161-115 kV xfmr	CIPCO	Replace Bertram 161-115 Transformer	\$4,000,000
Rosehollow-Bertram 161 kV	CIPCO ITCM	CIPCO: Change 2 CT ratios and replace jumpers. \$75,000 ITCM: CIPCO LPC. \$0	\$75,000

Cost Allocation

Constraint	Owner	Mitigation	Cost (\$)
Hazleton S-Dundee 161 kV	CIPCO ITCM	CIPCO: Replace 4 CT's at Dundee \$100,000. ITCM: CIPCO LPC. \$0	\$100,000
Liberty-Dundee 161 kV #2	CIPCO ITCM	CIPCO: \$25,000,000 if a new line is required or \$5,000,000 if reconductor option is feasible. ITCM: CIPCO LPC. \$0	\$5,000,000
Liberty-Hickory Crk 161 kV	CIPCO ITCM	CIPCO: Upgrade Liberty terminal to minimum of 2000 amps. ITCM: CIPCO LPC. \$0	\$500,000

Note 1: IR-20 is CIPCO project queued prior to the DPP 2016 Feb 2016 group but system impact study has not been completed. IR-20 has impact of 48.643 MW, and J514 has impact of 1.156 MW. Based on the MW impact, IR-20 has cost share of \$5,860,720 and J514 has cost share of \$139,280.

Table 11-8: PJM Affected System Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	DPP Projects
Twin Branch–Argenta 345 kV	PJM	a. A sag check will be required for the ACSR ~ 954 ~ 45/7 ~ RAIL - Conductor Section 1 to determine if the line section can be operated above its emergency rating of 1409 MVA.	\$208,000	J488, J501
Benton–T-094 345 kV	PJM	Reconductor or rebuild (depending on the existing structures) the portions of 345 kV lines between the Benton Harbor and Sagreto 345 kV substations. Replace the Benton H wavetrapp. Replace the Benton H Line Riser.	\$19,600,000	J498, J499, J500, J501, J526
Stillwell–Dumont 345 kV	PJM	0.8 miles AEP line will be re-conducted which will increase the ratings to SN – 1930MVA and SE – 1930MVA.	\$700,000	J475, J499, J500
Wilton–Dumont 765 kV	PJM	1) Replace the Dumont wavetrapp and relay compliance thermal limit to obtain new AEP-end ratings of 6198/7362 MVA (SN/SE); 2) Mitigate the sag on the Wilton - Dumont 765 kV line L11215 to achieve an ALDR that exceeds 6166 MVA.	\$9,000,000	J501
Crete–St John 345 kV	PJM	1) Reconductor the line to achieve ratings of 1837 SSTE, SLD of 2360 MVA (ALDR of 2714 MVA); 2) Upgrade St John substation conductor and switches; 3) Upgrade to 4000 A switch(es) and to bundled 1590 AL substation conductor	\$18,000,000	J501
Wilton 765-345 kV transformers	PJM	1) Build out the Wilton 765kV bus thereby allowing for 765kV L11216 (currently on Bus 6) to be relocated to Bus 8. Along with this line relocation, installation of 2-765kV BT CB's (6-8 & 8-2); 2) Mitigate the sag on the Wilton - Dumont 765 kV line L11215 to achieve an ALDR that exceeds 6166 MVA	\$11,000,000	J501
Walto–Electric Junction 345 kV	PJM	Upgrade 345kV switches at TSS 155 for L15502, upgrade station conductor at the two station terminals and adjust the CT ratios for the line at the station terminals.	\$350,000	J498, J500, J501, J510, J530
Lee County–Byron 345 kV	PJM	Re-conductor a portion of line L0626	\$6,000,000	J534

Table 11-9: SPP Affected System Network Upgrades

Mitigation	Area	Purpose	ERIS and / or NRIS Affecting Requests	Estimated Cost
Rebuild approximately 11 miles of 115kV	WAPA	Required to mitigate AURORA - BROOKINGS 115KV CKT 1 NRIS thermal constraints for contingency conditions	NRIS: J432, J460, J493, J510, J526	\$8,800,000
Rebuild approximately 17 miles of 115kV	WAPA	Required to mitigate AURORA - FLANDRU 115KV CKT 1 NRIS thermal constraints for contingency conditions	NRIS: J432, J460, J493, J510, J526	\$13,600,000
Build approximately 120 miles of new 345kV from Cooper – Stranger Creek 345kV, Cooper substation build and retermination, expand Stranger Creek, and install network reactor.	NPPD/ AECI	Required to mitigate COOPER - FAIRPORT 345KV CKT 1 NRIS thermal constraints for contingency conditions	NRIS: J432, J460, J475, J485, J490, J493, J495, J498, J499, J500, J501, J504, J506, J510, J514, J523, J524, J525, J526, J527, J528, J529, J530, J531, J534, J535	\$301,000,000
	NPPD/ KCPL	Required to mitigate COOPER - ST JOE 345KV CKT 1 NRIS thermal constraints for system intact and contingency conditions	NRIS: J233/J514, J432, J460, J475, J485, J490, J493, J495, J498, J499, J500, J501, J504, J506, J510, J523, J524, J525, J526, J527, J528, J529, J530, J531, J534, J535	
	NPPD/ OPPD/ KCPL	Required to mitigate COOPER_S flowgate for system intact conditions	ERIS and NRIS: J233/J514, J432, J460, J475, J485, J488, J489, J490, J493, J495, J498, J499, J500, J501, J504, J506, J510, J523, J524, J525, J526, J527, J528, J529, J530, J531, J534, J535	
	NPPD/ OPPD/ KCPL	Required to mitigate various steady state voltage violations observed in NPPD, OPPD, KCPL areas after SPP previously allocated, MISO MVP, and MISO current study baseline "backbone" upgrades are inservice. Please refer to Appendix C-V for the incremental voltage constraints.	ERIS and NRIS: J233/J514, J432, J460, J475, J485, J488, J489, J490, J493, J495, J498, J499, J500, J501, J504, J506, J510, J523, J524, J525, J526, J527, J528, J529, J530, J531, J534, J535	
Rebuild OPPD's portion of the Council Bluffs – Sub 3456 345kV transmission circuit and upgrade Sub 3456 terminal equipment upgrade	MEC/O PPD	Required to mitigate COUNCIL BLUFFS - SUB 3456 345KV CKT 1 ERIS thermal constraints for system intact and contingency conditions	ERIS: J475, J498, J499, J500, J524, J527, J528, J530	\$5,500,000
Upgrade breakers, CT, and increase clearance	WAPA	Required to mitigate FT RANDAL - FT THOMPSON 230KV CKT 1 NRIS thermal constraint for contingency conditions	NRIS: J490	\$550,000
CT uprate and clearance increase	WAPA/ NPPD	Required to mitigate GR ISLD-LNX3345.00 - HOLT.CO3 345.00 345KV CKT 1 NRIS thermal constraint for system intact and contingency conditions	NRIS: J432, J460, J485, J490, J493, J510, J525, J526, J531	\$1,000,000
Build Meadow Grove 230/115 Transformer and Meadow Grove - Petersburg 115kV line	NPPD	Required to mitigate KELLY - MEADOWGROVE4230.00 230KV CKT 1 NRIS thermal constraint for system intact	NRIS: J432, J460, J490, J526	\$30,000,000

Cost Allocation

Mitigation	Area	Purpose	ERIS and / or NRIS Affecting Requests	Estimated Cost
		and contingency conditions		
Rebuild Pahoja – Sioux Falls	WAPA	Required to mitigate PAHOJA - SIOUX FALLS 230KV CKT 1 NRIS thermal constraint for contingency conditions	NRIS: J432, J460, J490, J493, J510, J525, J526	\$14,000,000

Table 11-10: Additional Network Upgrades in Transient Stability Analysis

Constraint	Owner	Mitigation	Cost (\$)
No additional NUs			\$0

Table 11-11: Network Upgrades in Short Circuit Analysis

Constraint	Owner	Mitigation	Cost (\$)
No Constraints			\$0

Table 11-12: Shared Network Upgrades

Network Upgrades	Original Driver Study Cycle	Projects sharing cost	MW Contribution	Total NU Cost (\$)	Cost Responsibility
Hazleton-Mitchell Co 345 kV	DPP-2013-AUG	J278	52.93	\$2,750,000	\$606,955
	DPP-2015-AUG	J449	76.7499		\$880,101
	DPP-2016-FEB	J531	110.136		\$1,262,944
Jasper-Laurel 161 kV	DPP-2013-AUG	J233	202.31	\$600,000	\$479,075
	DPP-2015-AUG	J298	31.734		\$75,147
	DPP-2016-FEB	J514	19.332		\$45,779
Arnold-Morgan Valley 161 kV	DPP-2015-AUG	J298	45.219	\$2,245,000	\$1,752,629
	DPP-2016-FEB	J504	12.7035		\$492,371

11.3 Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Post Case. For constraints identified in the shoulder peak scenario, the MW impact is calculated using the shoulder peak post-DPP case. The MW impact on constraints identified in the summer peak scenario is calculated using the summer peak post-DPP case. With all Group Study generation projects dispatched in the

Post Case, all thermal and voltage constraints will be identified and a distribution factor from each project on each constraint will be obtained.

Constraints which are mitigated by one or a subset of NU are identified. The MW contribution on these constraints from each generating facility is calculated in the Post Case without any network upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj. A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj. i MW contribution on constraint})} \right)$$

11.4 Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

The Distribution Factor (DF) from each generating facility is calculated on the constraints identified in the steady-state analysis in the Post Case without any network upgrades. For a voltage support network upgrade required for mitigating voltage constraints identified in the steady-state AC contingency analysis, DFs are calculated under the most critical contingency on all branches (proxy branches for voltage support network upgrade) connecting to the high voltage side of the transformer, where the voltage support network upgrade is located. For a voltage support network upgrade required for mitigating MWEX voltage stability constraints identified in the voltage stability analysis, DFs are calculated under the most critical contingency on all branches (proxy branches) connecting to the high voltage side of the transformer, where the voltage collapse occurs.

For each thermal constraint, the maximum MW contribution (increasing flow) from each generating facility is calculated. MW contribution from one generating facility is set as zero if the MW contribution is less than 1 MW, or the constraint is not categorized as MISO ERIS constraint or affected system constraint for that specific generating facility.

For a voltage support network upgrade, the generators in same local area with positive MW impact will be responsible for mitigating the voltage or voltage stability constraints. No cutoff is applied to the MW contribution from each generating facility.

The calculated DF results and the MW contribution on each constraint are in Appendix J.1 for the 2021 scenario.

Finally, the cost allocation for each NU is calculated based on the MW contribution of each generating facility, as detailed in Appendix J.2 for the 2021 scenario.

Assuming all generating facilities in the DPP 2016 February West Area group advance, a summary of the costs for total NUs (NUs for ERIS, NRIS, and Interconnection Facilities) allocated to each generating facility is listed in Table 11-13.

Table 11-13: Summary of Total NU Costs Allocated to Each Generation Project

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J432	98	\$70,143,354	\$715,749	2.54%
J460	200	\$152,653,541	\$763,268	5.53%
J475	200	\$59,562,055	\$297,810	2.16%
J485	46.85	\$4,122,023	\$87,983	0.15%
J488	151.8	\$108,432,742	\$714,313	3.93%
J489	151.8	\$108,383,942	\$713,992	3.93%
J490	60	\$52,835,429	\$880,590	1.91%
J493	150	\$116,443,512	\$776,290	4.22%
J495	200	\$77,119,563	\$385,598	2.80%
J498	340	\$259,022,969	\$761,832	9.39%
J499	340	\$140,913,502	\$414,451	5.11%
J500	500	\$170,902,120	\$341,804	6.19%
J501	500	\$193,444,815	\$386,890	7.01%
J504	50	\$5,861,342	\$117,227	0.21%
J506	200	\$83,759,596	\$418,798	3.04%
J510	326.9	\$35,423,255	\$108,361	1.28%
J514	65	\$75,668,912	\$1,164,137	2.74%
J523	50	\$3,144,822	\$62,896	0.11%
J524	100	\$7,231,835	\$72,318	0.26%
J525	50	\$33,496,792	\$669,936	1.21%
J526	300	\$258,416,304	\$861,388	9.37%
J527	250	\$79,014,096	\$316,056	2.86%
J528	200	\$65,333,936	\$326,670	2.37%
J529	250	\$104,788,332	\$419,153	3.80%
J530	250	\$75,741,913	\$302,968	2.75%
J531	200	\$212,095,876	\$1,060,479	7.69%
J534	250	\$112,221,288	\$448,885	4.07%
J535	210	\$92,992,437	\$442,821	3.37%
Total/Average	5690.35	\$2,759,170,304	\$501,167	100.00%

Model Development for Steady-State and Stability Analysis

A.1 DPP 2016 February Generation Projects

Table A-1: DPP 2016 February West Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J432	SD	Brookings	XEL	Brookings 345 kV	98	Wind	NRIS
J460	MN	Lincoln	GRE	Brookings-H081 345 kV line (10 mi west of proposed sub)	200	Wind	NRIS
J475	IA	Poweshiek	MEC	Montezuma 345 kV	200	Wind	NRIS
J485	MN	Olmsted	RPU	West Side 161 kV	46.85	CT	NRIS
J488	SD	Grant, Roberts	OTP	Big Stone-Ellendale 345 kV	151.8	Wind	ERIS
J489	SD	Grant, Roberts	OTP	Big Stone-Ellendale 345 kV	151.8	Wind	ERIS
J490	ND, SD	Corson, Sioux	MDU	McIntosh Substation	60	Wind	NRIS
J493	MN	Yellow Medicine	OTP	Big Stone-Brookings 345 kV	150	Wind	NRIS
J495	IA	Winnebago	ITCM	Ledyard-Colby 345 kV line	200	Wind	NRIS
J498	IA	Boone, Greene	MEC	Grimes-Lehigh 345 kV (18 mi south of Lehigh sub)	340	Wind	NRIS
J499	IA	Adair, Madison	MEC	Fallow-Grimes 345 kV (18 mi east of Fallow sub)	340	Wind	NRIS
J500	IA	Adair	MEC	New substation at intersection of Boone-Atchison 345 kV and Rolling Hills-Madison Co 345 kV	500	Wind	NRIS
J501	IA	Hancock, Kossuth	MEC	Kossuth Co 345 kV	500	Wind	NRIS
J504	IA	Linn	ITCM	Bertram-Duane Arnold 161 kV (0.5 mi from Duane Arnold)	50	Solar	NRIS
J506	IA	Plymouth	MEC	Raun-Lakefield Jct 345 kV	200	Wind	NRIS
J510	SD	Deuel	OTP	Scandinavia Township	321.9 sum / 326.9 win	CT	NRIS
J514	IA	Marshall	ITCM	Marshalltown substation	30 sum / 65 win	CC	NRIS
J523	MN	Mower	ITCM	Adams 161 kV	50	Solar	NRIS
J524	IA	Webster	MEC	Webster 161 kV	100	Solar	NRIS
J525	MN	Murray	GRE	Lake Wilson-Hadley 69 kV	50	Solar	NRIS
J526	SD	Deuel	OTP	Brooking Co-Big Stone South 345 kV	300	Wind	NRIS
J527	IA	Adair	MEC	Booneville-Cooper 345 kV	250	Wind	NRIS

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J528	IA	Adair	MEC	Rolling Hills-Madison 345 kV	200	Wind	NRIS
J529	IA	Palo Alto	MEC	Obrien-Kossuth 345 kV	250	Wind	NRIS
J530	IA	Iowa	MEC	Montezuma-Hills 345 kV	250	Wind	NRIS
J531	IA	Chickasaw	ITCM	Hazleton-Adams 345 kV	200	Wind	NRIS
J534	IA	Humboldt, Kossuth	MEC	Kossuth-Webster 345 kV	250	Wind	NRIS
J535	IA	Sac	MEC	J411-Lehigh 345 kV	210	Wind	NRIS

Table A-2: Dynamic Modeling for DPP West Area Projects

MISO Project #	Dynamic Modeling	Generator Reactive Power Capability (power factor)
J432	49 Gamesa G9X 2.0 MW WTG	± 0.95
J460	100 Vestas V110 2.0 MW WTG	± 0.95
J475	100 Vestas V110 2.0 MW WTG	± 0.95
J485	5 9.37 MW natural gas fired reciprocating engines	0.8 lagging -0.95 leading
J488	61 GE 2.5 MW WTG	± 0.9
J489	61 GE 2.5 MW WTG	± 0.9
J490	24 Siemens 120 2.5 MW WTG	± 0.9
J493	45 V136 GS 3.45 MW WTG	0.902 lagging 0.918 leading
J495	100 Vestas V110 2.0 MW WTG	± 0.95
J498	100 Vestas V110 2.0 MW Mk 10D WTG	± 0.95
J499	100 Vestas V110 2.0 MW Mk 10D WTG	± 0.95
J500	250 Vestas V110 2.0 MW Mk 10D WTG	± 0.95
J501	250 Vestas V110 2.0 MW Mk 10D WTG	± 0.95
J504	14 GE 1500V 4 MW Solar	± 0.9
J506	100 Vestas V110 2.0 MW WTG	± 0.95
J510	GE CT generator	0.85 lagging 0.95 leading
J514	J233 CC expansion, 2 CTGs, 1 STG	NA
J523	20 SMA SC 2500 ev 2.5 MVA inverters	± 0.9
J524	40 SMA SC 2500 ev 2.5 MVA inverters	± 0.9
J525	14 GE 1500V 4 MW Solar	± 0.9

MISO Project #	Dynamic Modeling	Generator Reactive Power Capability (power factor)
J526	120 GE 2.5 MW WTG	± 0.9
J527	100 GE 2.5 MW WTG	± 0.9
J528	80 GE 2.5 MW WTG	± 0.9
J529	140 GE 1.79 MW WTG	± 0.9
J530	100 GE 2.5 MW WTG	± 0.9
J531	80 GE 2.5 MW WTG	± 0.9
J534	100 GE 2.5 MW WTG	± 0.9
J535	84 GE 2.5 MW WTG	± 0.9

Table A-3: Collector System and Shunt Compensation Modeling for DPP West Area Non-Synchronous Projects

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J432	One 98 MW unit	Req = 0.010510 pu Xeq = 0.069124 pu Beq = 0.018199 pu	30 Mvar capacitor bank on 34.5 kV system
J460	Two 100 MW units	CKT #1: R=0.008357 pu X=0.0101 pu B=0.057923 pu CKT #2: R=0.019103 pu X=0.025036 pu B=0.047283 pu	2*15 Mvar capacitor bank on 34.5 kV system
J475	One 200 MW unit	R=0.00168 pu X=0.003781 pu B=0.12064 pu	45 Mvar capacitor bank on 34.5 kV system
J485	Five 9.37 MW units	None	None
J488	One 151.8 MW unit	R=0.01448 pu X=0.02156 pu B=0.0 pu	None
J489	One 151.8 MW unit	R=0.010728 pu X=0.01406 pu B=0.0 pu	None
J490	One 60 MW unit	Req = 0.005265 pu Xeq = 0.006547 pu Beq = 0.013785 pu	7.2 Mvar capacitor bank on 34.5 kV system
J493	One 150 MW unit	Req = 0.008928 pu Xeq = 0.039978 pu	None

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
		Beq = 0.061625 pu	
J495	Two 100 MW units	Two equivalent collector systems, each has: Req = 0.02395 pu Xeq = 0.026499 pu Beq = 0.103059 pu	2*27 Mvar capacitor bank on 34.5 kV system
J498	Two 170 MW units	None	None
J499	Two 170 MW units	None	None
J500	Two 166 MW units One 168 MW unit	None	None
J501	Two 166 MW units One 168 MW unit	None	None
J504	One 50 MW unit	Req = 0.001848 pu Xeq = 0.001092 pu Beq = 0.002321 pu	None
J506	Two 100 MW units	Two equivalent collector systems, each has: Req = 0.026793 pu Xeq = 0.030726 pu Beq = 0.08777 pu	2*27.5 Mvar capacitor bank on 34.5 kV system
J510	One 326.9 MW unit	None	None
J514	Two 232 MW units One 257 MW unit	None	None
J523	One 50 MW unit	Req = 0.007079 pu Xeq = 0.006495 pu Beq = 0.00475 pu	None
J524	One 100 MW unit	Req = 0.003379 pu Xeq = 0.003045 pu Beq = 0.00907 pu	None
J525	One 50 MW unit	R: 0.0045 pu X: 0.0055 pu B: 0.0046 pu	4 Mvar capacitor bank on 34.5 kV system
J526	Two 150 MW units	Two equivalent collector systems, each has: R: 0.0086 pu X: 0.0110 pu B: 0.1065 pu	2*10 Mvar capacitor bank on 34.5 kV bus
J527	Two 125 MW units	Two equivalent collector systems, each has: R: 0.0104 pu X: 0.0132 pu B: 0.0884 pu	2*7 Mvar capacitor bank on 34.5 kV bus

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J528	Two 100 MW units	Two equivalent collector systems, each has: R: 0.0130 pu X: 0.0166 pu B: 0.0703 pu	2*7 Mvar capacitor bank on 34.5 kV bus
J529	One 250 MW unit	None	None
J530	Two 125 MW units	Two equivalent collector systems, each has: R: 0.0104 pu X: 0.0132 pu B: 0.0884 pu	2*7 Mvar capacitor bank on 34.5 kV bus
J531	Two 100 MW units	Two equivalent collector systems, each has: R: 0.0130 pu X: 0.0166 pu B: 0.0703 pu	2*7 Mvar capacitor bank on 34.5 kV bus
J534	Two 125 MW units	Two equivalent collector systems, each has: R: 0.0104pu X: 0.0132pu B: 0.0884pu	2*7 Mvar capacitor bank on 34.5 kV bus
J535	One 210 MW unit	R: 0.0043pu X: 0.0055pu B: 0.2131pu	12 Mvar capacitor bank on 34.5 kV bus

Table A-4: DPP 2016 February Central Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J351	IN	St. Joseph	NIPS	Stillwell 345 kV	705 sum / 715 win	CC	NRIS
J468	IL	Douglas	AMIL	Sidney-Kansas West 345 kV line	202	Wind	NRIS
J515	IN	Benton, Warren	DEI	Cayuga 345 kV	400	Wind	ERIS

Table A-5: DPP 2016 February Michigan Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J533	MI	Gratiot	METC	Slate 345 kV	200	Wind	NRIS
J537	MI	Sanilac	ITCT	Banner 345 kV	200	Wind	NRIS
J538	MI	Hillsdale	METC	Moore Rd-Dowling 138 kV	150	Wind	NRIS

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J540	MI	Kalkaska	WPSC	Kalkaska County	48.6	CT	NRIS Only

Table A-6: DPP 2016 February ATC Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J505	WI	Manitowoc	ATC	Mishicott-Kewaunee 138 kV line	100	Solar	NRIS

Table A-7: DPP 2016 February South Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J473	MS	Lamar	SMEPA	Sumrall II substation	52	Solar	ERIS
J482	LA	St. Charles Parish	Entergy	Little Gypsy Power Station	19.8 sum / 26.2 win	CC	NRIS
J486	TX	Grimes	Entergy	Grimes 345 kV	51.5	CC	NRIS Only
J518	LA	Calcasieu Parish, Jefferson Davis Parish	Entergy	Lacassine 69 kV	49.5	Solar	NRIS

A.2 Model Review Comments

Table A-8: Model Review Comments

Company	Python/ Idev File Name	2021 SH	2021 SPK	2021 Stability
MISO	ITCM_3127_MVP5_HCK-EDEN_345-kV.idv	x	x	x
MISO	ATC_(3127)_Z3_SB_EEN_345-138kV.idv	x	x	x
MISO	ATC_(3127)_Z3_SB_EEN-CDL_345kV.idv	x	x	x
MISO	RMV_J375-NU.py	x	x	x
ITCM	Aug_15_DPP_Model_Corrections_ITCM_11-28-16.idv	x	x	x
ITCM	MTEP_4116_and_4459_corrections.idv	x	x	x
MISO	RMV_J497.idv	x	x	x
MISO	DPP-2015Aug_Set-Wind-PF_095.py	x	x	
MISO	DPP-2016Feb_Set-Wind-PF_095_SH.py	x		
MISO	DPP-2016Feb_Set-Wind-PF_095_SPK.py		x	
MISO	Ortonville-Johnson_Jct_Rating-Update.idv	x	x	x
ATC	Correct_Arrowhead_345-230kV_TR_LTC_Reg_Bus.idv	x	x	x
MISO	TurnOff_Saratog-J454-Withdrawl_SH.py	x		x
MEC	Parnell-Hills 161 kV.py	x	x	x
MISO	J488-489_Chng.py	x	x	x
MISO	J436-437_Chng.py	x	x	x
MISO	J436-437_Gentie.py	x	x	x
MISO	J436-437_D-VAR System Solution.idv	x	x	x
MISO	Add 2016 Feb DPP Power Factor upgrades_SH.idv	x		x
MISO	Add 2016 Feb DPP Power Factor upgrades_SPK.idv		x	
MISO	J493_Chng.py	x	x	x
MISO	J495_Chng.py	x	x	x
DPC	i-chng-dpc.idv	x	x	x
MISO	RMV_J034.py	x	x	x
MISO	RMV_SPP_GEN-2015-097.py	x	x	x
CIPCO	CIPCO_IR20_SH.idv	x		x
CIPCO	CIPCO_IR20_SPK.idv		x	
AMRN	Cronus Off peak.idv	x		x
AMRN	Cronus summer peak.idv		x	
AMRN	AMRN_rating_impedance_changes_per_email.py	x	x	x
AMRN	Turnoff_AMRN_units_dueTo_AttachmentY.py	x	x	x

Company	Python/ Idev File Name	2021 SH	2021 SPK	2021 Stability
ComEd	RMV_Will County#4.py	x	x	x
SPP	RMV_Ft Cal.py	x	x	x
ATC	2021SH_ATC_20210715_DPP-Feb16_Dynamics.idv			x
ATC	2021SH_ATC_20210715_DPP-Feb16_Minimal.idv	x		
ATC	2021SP_ATC_20210715_DPP-Feb16_Study.idv		x	
ATC	turnoff_J394_withdrawn.py	x	x	x
BEPC	659_BEPC_MISO-DPP-2016-FEB_00_Extraneous_FtThompson-Holt_20170125.idv	x	x	x
BEPC	659_BEPC_MISO-DPP-2016-FEB_01_Extraneous_Hanlon-Fodness115_20170110.idv	x	x	x
BEPC	659_BEPC_MISO-DPP-2016-FEB_02_Mallard-Logan_Minotfix.idv	x	x	x
WAPA	GRPrairieHoltAreaCorrectionDPPFeb16.idv	x	x	x
WAPA	RMV_Summit-Watertn_Duplicate.py	x	x	x
DPC	DPC_ModelUpdates.idv	x	x	x
MDU	MDU_Updates-2021SH-Dynamics.idv			x
MDU	MDU_Updates-2021SH-Steady.idv	x		
MDU	MDU_Updates-2021SPK-Steady.idv		x	
MEC	DPP FEB 16 Shoulder MidAmerican Updates.py	x		x
MEC	DPP FEB 16 Summer Peak MidAmerican Updates.py		x	
MPC	MPC-fix-ratings-DPP-FEB16.idv	x	x	x
MPC	MPC-Queue-fix-ratings-DPP-FEB16.idv	x	x	x
MPC	OTP-fix-ratings-DPP-FEB16.idv	x	x	x
MPC	WAPA-fix-ratings-DPP-FEB16.idv	x	x	x
MPC	MPC-MapleR-Frontier-115kV-DPP-FEB16-SH.idv	x		x
MPC	MPC-MapleR-Frontier-115kV-DPP-FEB16-SPK.idv		x	
OTP	2021SH_Load_Correction_2-8-17.idv	x		x
OTP	OTPSummerRatingUpdates_2-8-17.idv	x	x	x
OTP	OTPCapacitorRemoval_2-8-17.idv	x	x	x
Xcel	xcel_turnon_2_more_fixedCaps_at_LakeYankton_bus603010_SH.py	x		x
SPP	Correct-FtThompson-GrandPrairie-Holt-GrandIsland.IDV	x	x	x
SPP	Thedford_Correction_SPK.idv		x	
SPP	THEDFORD-TRANSFORMER-345-115kV.IDV	x	x	x
SPP	RMV_SPP_GEN-2015-097.py	x	x	x
ITCM	Montezuma-Ottumwa_Uprating.py	x	x	x

Company	Python/ Idev File Name	2021 SH	2021 SPK	2021 Stability
J432	J432_IC_comments_SH.py	x		x
J432	J432_IC_comments_SPK.py		x	
J460	J460_IC_Comments_SH.py	x		x
J460	J460_IC_Comments_SPK.py		x	
J475	J475_IC_Comments.py	x	x	x
J490	J490_IC_Comments.py	x	x	x
J493	J493_IC_Comments_SH.py	x		
J493	J493_IC_Comments_ST.py			x
J493	J493_IC_Comments_SPK.py		x	
J495	J495_IC_Comments_SH.py	x		
J495	J495_IC_Comments_ST.py			x
J495	J495_IC_Comments_SPK.py		x	
J498	J498_IC_Comments_SH.py	x		
J498	J498_IC_Comments_ST.py			x
J498	J498_IC_Comments_SPK.py		x	
J499	J499_IC_Comments_SH.py	x		
J499	J499_IC_Comments_ST.py			x
J499	J499_IC_Comments_SPK.py		x	
J500	J500_IC_Comments_SH.py	x		
J500	J500_IC_Comments_ST.py			x
J500	J500_IC_Comments_SPK.py		x	
J501	J501_IC_Comments_SH.py	x		
J501	J501_IC_Comments_ST.py			x
J501	J501_IC_Comments_SPK.py		x	
J506	J506_IC_Comments_SH.py	x		
J506	J506_IC_Comments_ST.py			x
J506	J506_IC_Comments_SPK.py		x	
J510	update_J510-Pmax.py	x	x	x
J514	J514_IC_Comments_SH.py	x		
J514	J514_IC_Comments_ST.py			x
J514	J514_IC_Comments_SPK.py		x	
J523	J523_IC_Comments_SH.py	x		
J523	J523_IC_Comments_ST.py			x

Company	Python/ Idev File Name	2021 SH	2021 SPK	2021 Stability
J523	J523_IC_Comments_SPK.py		x	
J524	J524_IC_Comments_SH.py	x		
J524	J524_IC_Comments_ST.py			x
J524	J524_IC_Comments_SPK.py		x	
J526	J526_IC_Comments_SH.py	x		
J526	J526_IC_Comments_ST.py			x
J526	J526_IC_Comments_SPK.py		x	
J527	J527_IC_Comments_SH.py	x		
J527	J527_IC_Comments_ST.py			x
J527	J527_IC_Comments_SPK.py		x	
J528	J528_IC_Comments_SH.py	x		
J528	J528_IC_Comments_ST.py			x
J528	J528_IC_Comments_SPK.py		x	
J530	J530_IC_Comments_SH.py	x		
J530	J530_IC_Comments_ST.py			x
J530	J530_IC_Comments_SPK.py		x	
J531	J531_IC_Comments_SH.py	x		
J531	J531_IC_Comments_ST.py			x
J531	J531_IC_Comments_SPK.py		x	
J534	J534_IC_Comments_SH.py	x		
J534	J534_IC_Comments_ST.py			x
J534	J534_IC_Comments_SPK.py		x	
J535	J535_IC_Comments_SH.py	x		
J535	J535_IC_Comments_ST.py			x
J535	J535_IC_Comments_SPK.py		x	
ICs	Misc-Corrections.py	x	x	x
J493	J493_45-Turbines.py	x	x	x
SPP	RMV_SPP_GEN-2015-042.py	x	x	x
MEC	MEC Voltage Schedule Update.py	x	x	x
ITCM	ITCM_DPP FEB 2016 Ratings Corrections.idv	x	x	x
ITCM	ITCM DPP Feb16 voltage schedules.idv	x	x	x
MISO	RMV_Stanton-1.py	x	x	x
SPTI	TurnOff disconnected_gen.py	x	x	x

Company	Python/ Idev File Name	2021 SH	2021 SPK	2021 Stability
CIPCO	Coggon to IR 20 Sub 161 kV Rating Update.idv	x	x	x
GRE	GRE-add-G667-at-RoundLakeTap_SH.py	x		x
GRE	GRE-add-G667-at-RoundLakeTap_SPK.py		x	
J498	J498_changes_Qlimits_PPC_adj_SH.py	x		x
J498	J498_changes_Qlimits_PPC_adj_SPK.py		x	
J499	J499_changes_Qlimits_GSU_updates_SH.py	x		x
J499	J499_changes_Qlimits_GSU_updates_SPK.py		x	
J500	J500_Qlimit_GSU_Adjustments_SH.py	x		x
J500	J500_Qlimit_GSU_Adjustments_SPK.py		x	
J501	J501_Qlimit_GSU_Adjustments_SH.py	x		x
J501	J501_Qlimit_GSU_Adjustments_SPK.py		x	
MISO	Huntley - Wilmarth.py	x		x
CIPCO	CR-North Liberty Area Upgrades Phase 1.idv	x	x	x
MISO	dispatch-dpp-2015AUG_SSH.py	x		
MISO	dispatch-dpp-2015AUG_Stability.py			x
MISO	Update J488 J489.idv	x		x
MISO	Update J488 J489_SPK.idv		x	
Ameren	Adair.idv	x	x	x
Ameren	Hall Street.idv	x	x	x
Ameren	Hannibal.idv	x	x	x
Ameren	Hunter.idv	x	x	x
Ameren	Huster.idv	x	x	x
Ameren	HWY-N.idv	x	x	x
Ameren	JC Zion.idv	x	x	x
Ameren	Lakeshire.idv	x	x	x
Ameren	Lakeside.idv	x	x	x
Ameren	McClay.idv	x	x	x
Ameren	Ofallon.idv	x	x	x
Ameren	Pt Prairie.idv	x	x	x
Ameren	Troy.idv	x	x	x
Ameren	Warrenton.idv	x	x	x

A.3 MISO Classic as the Study Sink

Table A-9: MISO Classic as the Study Sink

Area #	Area Name	Area #	Area Name
207	HE	600	Xcel
208	DEI	608	MP
210	SIGE	613	SMMPA
216	IPL	615	GRE
217	NIPS	620	OTP
218	METC	627	ALTW
219	ITC	633	MPW
295	WEC	635	MEC
296	MIUP	661	MDU
314	BREC	680	DPC
333	CWLD	694	ALTE
356	AMMO	696	WPS
357	AMIL	697	MGE
360	CWLP	698	UPPC
361	SIPC		

A.4 MISO South as the Study Sink

Table A-10: MISO South as the Study Sink

Area #	Area Name	Area #	Area Name
327	EES-EAI	502	CLEC
332	LAGN	503	LAFA
349	SMEPA	504	LEPA
351	EES		

A.5 PJM Market as PJM Projects Sink

Table A-11: PJM Market as PJM Projects Sink

Area #	Area Name	Area #	Area Name
201	AP	229	PPL
202	ATSI	230	PECO
205	AEP	231	PSE&G
209	DAY	232	BGE
212	DEO&K	233	PEPCO
215	DLCO	234	AE
222	CE	235	DP&L
225	PJM	236	UGI
226	PENELEC	237	RECO
227	METED	320	EKPC
228	JCP&L	345	DVP

A.6 SPP Market as SPP Projects Sink

Table A-12: SPP Market as SPP Projects Sink

Area #	Area Name	Area #	Area Name
515	SWPA	540	GMO
520	AEPW	541	KCPL
523	GRDA	542	KACY
524	OKGE	544	EMDE
525	WFEC	545	INDN
526	SPS	546	SPRM
527	OMPA	640	NPPD
531	MIDW	645	OPPD
534	SUNC	650	LES
536	WERE	652	WAPA

A.7 ContingencyFiles used in Steady-State Analysis

Table A-13: List of Contingencies used in Steady-State Analysis

Contingency File Name	Description	2020
Automatic single element contingencies	Single element outages at buses 60 kV and above in the study region	x
P1-1_ATC_20210715_DPP-Feb16.con	Specified category P1 contingencies in ATC	x
P1-2_ATC_20210715_DPP-Feb16.con	Specified category P1 contingencies in ATC	x
P1-3_ATC_20210715_DPP-Feb16.con	Specified category P1 contingencies in ATC	x
P1-4_ATC_20210715_DPP-Feb16.con	Specified category P1 contingencies in ATC	x
P1-4_SVC_Portfolio_DPP-Feb16.con	Specified category P1 contingencies in ATC	x
P2-1_ATC_20210715_DPP-Feb16.con	Specified category P2 contingencies in ATC	x
P2-2_ATC_20210715_DPP-Feb16.con	Specified category P2 contingencies in ATC	x
P2-3_ATC_20210715_DPP-Feb16.con	Specified category P2 contingencies in ATC	x
P2-4_ATC_20210715_DPP-Feb16.con	Specified category P2 contingencies in ATC	x
P5-5_ATC_20210715_DPP-Feb16.con	Specified category P5 contingencies in ATC	x
P7-1_ATC_20210715_DPP-Feb16.con	Specified category P7 contingencies in ATC	x
2021_MRO-MN Region-P1-Single Contingency.con	Specified category P1 contingencies in Minnesota	x
2021_MRO-MN Region-P2-Single Contingency.con	Specified category P2 contingencies in Minnesota	x
2021_MRO-MN Region-P4-P5-P7 Multiple Contingency.con	Specified category P4, P5, P7 contingencies in Minnesota	x
2021_MDU-MTEP15-2021-P1.con	Specified category P1 contingencies in MDU	x
2021_MDU-MTEP15-2021-P2_b.con	Specified category P2 contingencies in MDU	x
2021_MDU-MTEP15-2021-P4_b.con	Specified category P4 contingencies in MDU	x
2021_MDU-MTEP15-2021-P7.con	Specified category P7 contingencies in MDU	x
CIPCO DPP-2016-FEB Additional P6.con	Specified category P6 contingencies in CIPCO	x
MEC DPP2016 FEB 2021 Cat P1 05.30.2017.con	Specified category P1 contingencies in MEC	x
2021_MTEP15_Ames_P1.con	Specified category P1 contingencies in Ames	x
2021_MTEP15_CFU_P1.con	Specified category P1 contingencies in CFU	x
MEC DPP2016 FEB 2021 Cat P2 05.30.2017.con	Specified category P2 contingencies in MEC	x
MEC DPP2016 FEB 2021 Cat P4.con	Specified category P4 contingencies in MEC	x

Contingency File Name	Description	2020
MEC DPP2016 FEB 2021 Cat P5 05.09.2017.con	Specified category P5 contingencies in MEC	x
MEC DPP2016 FEB 2021 Cat P7 05.09.2017.con	Specified category P7 contingencies in MEC	x
2021_MTEP15_Ames_P2.con	Specified category P2 contingencies in Ames	x
2021_MTEP15_Ames_P457.con	Specified category P4, P5, P7 contingencies in Ames	x
2021_MTEP15_CFU_P2.con	Specified category P2 contingencies in CFU	x
2021_MTEP15_2021_CATP1_ITCM.con	Specified category P1 contingencies in ITCM	x
2021_MTEP15_2021_CATP2_ITCM.CON	Specified category P2 contingencies in ITCM	x
2021_MTEP15_2021_CATP4_ITCM.con	Specified category P4 contingencies in ITCM	x
2021_MTEP15_2021_CATP5_ITCM.CON	Specified category P5 contingencies in ITCM	x
2021_MTEP15_2021_CATP7_ITCM.CON	Specified category P7 contingencies in ITCM	x
2021_MPW_MTEP15_Cat_P1_2015.con	Specified category P1 contingencies in MPW	x
2021_MPW_MTEP15_Cat_P2_2015.con	Specified category P2 contingencies in MPW	x
2021_MPW_MTEP15_Cat_P5_2015.con	Specified category P5 contingencies in MPW	x
2021_MPW_MTEP15_Cat_P7_2015.con	Specified category P7 contingencies in MPW	x
HVDC_Red_2021SH.con	Specified category P1 contingencies with HVDC reduction in 2020 shoulder	x
HVDC_Red_2021SPK.con	Specified category P1 contingencies with HVDC reduction in 2020 peak	x
Without_SPS_MN_2021.con	Specified category P1 contingencies without LGS and Mankato Energy Center SPS	x
MISO_Feb_2016_DPP_Alex_MRES_CONs.con	Specified category P2 contingencies in backbone	x
2021_AMRN-P1-1 2021.con	Specified category P1 contingencies in AMRN	x
2021_AMRN-P1-2 2021.con	Specified category P1 contingencies in AMRN	x
2021_AMRN-P1-3 2021.con	Specified category P1 contingencies in AMRN	x
2021_AMRN-P1-4 2021.con	Specified category P1 contingencies in AMRN	x
2021_AMRN-P2-1 2021.con	Specified category P2 contingencies in AMRN	x
2021_AMRN-P2-2 2021.con	Specified category P2 contingencies in AMRN	x
2021_AMRN-P2-3 2021.con	Specified category P2 contingencies in AMRN	x
2021_AMRN-P2-4 2021.con	Specified category P2 contingencies in AMRN	x

Contingency File Name	Description	2020
2021_AMRN-P7-1 2021.con	Specified category P7 contingencies in AMRN	x
2021_CWLD_P1.con	Specified category P1 contingencies in CWLD	x
2021_P2_Explicit_CWLD.con	Specified category P2 contingencies in CWLD	x
2021_P4_Explicit_CWLD.con	Specified category P4 contingencies in CWLD	x
2021_P7_Explicit_CWLD.con	Specified category P7 contingencies in CWLD	x
2021_CWLP_MTEP15_CatP1.con	Specified category P1 contingencies in CWLP	x
2021_CWLP_MTEP15_CatP2.con	Specified category P2 contingencies in CWLP	x
2021_CWLP_MTEP15_CatP7.con	Specified category P7 contingencies in CWLP	x
2021_P1_Explicit_SIPC_MISO.con	Specified category P1 contingencies in SIPC	x
2021_P2_Explicit_SIPC.con	Specified category P2 contingencies in SIPC	x
2021_P4_Explicit_SIPC.con	Specified category P4 contingencies in SIPC	x
ComEd_RTEP_Cat_P1.con	Specified category P1 contingencies in ComEd	x
ComEd_RTEP_Cat_P2-P7.con	Specified category P2-P7 contingencies in ComEd	x

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Model Data

B.1 Power Flow Model Data

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B.2 Dynamic Model Data

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B.3 2021 Slider Diagrams

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Reactive Power Requirement Analysis Results (FERC Order 827)

Table C-1: Reactive Power Requirement Analysis Results

Project #	HV Side Bus #	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?
J432	61031	96.22	31.04	0.943	Y	95.68	-72.30	-0.798	Y
J460	61042	196.43	74.56	0.935	Y	196.00	-90.25	-0.908	Y
J475	65730	196.82	71.39	0.940	Y	196.64	-104.78	-0.883	Y
J488	60418	147.10	34.80	0.973	N	145.50	-133.00	-0.738	N
J489	61418	148.00	35.60	0.972	N	146.70	-130.00	-0.748	N
J490	71041	58.93	24.42	0.924	Y	58.67	-45.26	-0.792	Y
J493	60215	146.70	40.30	0.964	N	146.20	-112.40	-0.793	
J495	61536	192.89	94.81	0.897	Y	192.08	-92.46	-0.901	Y
J504	71089	49.70	17.70	0.942	Y	49.60	-40.10	-0.778	Y
J506	15010	192.50	94.02	0.899	Y	191.52	-95.84	-0.894	Y
J523	97996	49.60	21.40	0.918	Y	49.60	-27.10	-0.878	Y
J524	96996	99.30	42.50	0.919	Y	99.20	-54.90	-0.875	Y
J525	68920	49.90	20.90	0.922	Y	49.80	-41.00	-0.772	Y
J526	72032	293.04	122.06	0.923	Y	290.53	-215.01	-0.804	Y
J527	65201	244.20	101.37	0.924	Y	242.34	-173.30	-0.813	Y
J528	65301	195.34	81.90	0.922	Y	193.65	-143.53	-0.803	Y
J530	75731	244.20	101.37	0.924	Y	242.34	-173.30	-0.813	Y
J531	61140	195.42	88.17	0.912	Y	194.29	-129.17	-0.833	Y
J534	66001	244.20	101.37	0.924	Y	242.34	-173.30	-0.813	Y

Reactive Power Requirement Analysis Results (FERC Order 827)

Project #	HV Side Bus #	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?
J535	66202	205.82	90.16	0.916	Y	204.96	-136.07	-0.833	Y

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2021 Summer Peak Contingency Analysis Results

2021 Summer Peak (SPK) Constraints

Table D-1: 2021 SPK System Intact Thermal Constraints

Table D-2: 2021 SPK System Intact Voltage Constraints

Table D-3: 2021 SPK Category P1 Thermal Constraints

Table D-4: 2021 SPK Category P1 Voltage Constraints

Table D-5: 2021 SPK Category P2-P7 Thermal Constraints

Table D-6: 2021 SPK Category P2-P7 Voltage Constraints

Table D-7: 2021 SPK Non-Converged Category P2-P7 Contingencies

Table D-8: 2021 SPK Non-Converged Category P2-P7 Contingencies DCCC Results

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2021 Summer Shoulder Contingency Analysis Results

E.1 Thermal Constraints Identified in 1st Round Contingency Analysis with Fictitious SVCs

Table E-1: 2021 SH System Intact Thermal Constraints in 1st Round ACCC

Table E-2: 2021 SH Category P1 Thermal Constraints in 1st Round ACCC

Table E-3: 2021 SH Category P2-P7 Thermal Constraints in 1st Round ACCC

Table E-4: 2021 SH Non-Converged Category P2-P7 Contingencies in 1st Round ACCC

Table E-5: 2021 SH Non-Converged Category P2-P7 Contingencies DCCC Results in 1st Round ACCC

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E.2 Backbone Network Upgrades Justification Results

E.2.1 Project Justification Details in Big Stone Area

Table E-6: Project Evaluation Details in Big Stone Area

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
Big Stone South-Alexandria 345 kV	95	Split Rock-White 345 kV: -153.6 Brookings Co-White 345 kV #1: -122.8 Brookings Co-White 345 kV #2: -212.6 Helena-Scott Co 345 kV: -129.8 Hazel Crk-Minn Valley 230 kV: -43.7 Minn Valley-Granite Falls 230 kV: -33.2 Panther-McLeod 230 kV: -42.1 Johnson Jct-Ortonville 115 kV: -73.2 Johnson Jct-Morris 115 kV: -73.0 Canby-Granite Falls 115 kV: -46.6 Big Stone-Highway 12 115 kV: -99.4 Big Stone 115-230-13.8 kV xfmr: -102.3 Highway 12-Ortonville 115 kV: -98.8 Big Stone-Browns Valley 230 kV: -182.0 Big Stone-Blair 230 kV: -357.3 Browns Valley-Hankinson 230 kV: -181.6 Hankinson-Wahpeton 230 kV: -212.1 Blair-Watertown 230 kV: -197.2 Ellendale-Aberdeen Jct 115 kV: -36.6	None	None

2021 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
Big Stone South-Hazel Creek 345 kV	80	Split Rock-White 345 kV: -183.7 Brookings Co-White 345 kV #1: -48.1 Brookings Co-White 345 kV #2: -258.5 Helena-Scott Co 345 kV: -129.8 Minn Valley-Granite Falls 230 kV: -147.8 Johnson Jct-Ortonville 115 kV: -72.7 Johnson Jct-Morris 115 kV: -72.4 Canby-Granite Falls 115 kV: -68.1 Big Stone-Highway 12 115 kV: -108.9 Big Stone 115-230-13.8 kV xfmr: -121.9 Highway 12-Ortonville 115 kV: -108.2 Big Stone-Browns Valley 230 kV: -160.4 Big Stone-Blair 230 kV: -472.0 Browns Valley-Hankinson 230 kV: -159.8 Hankinson-Wahpeton 230 kV: -170.0 Blair-Watertown 230 kV: -266.0 Ellendale-Aberdeen Jct 115 kV: -50.1	J506 POI-Raun 345 kV: 42.4 Helena-Scott Co 345 kV: 57.0	1) Hazel Creek 345-230 kV xfmr is loaded at 97.4% flo Lyon Co-Hazel Crk 345 kV

2021 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Constraints to be Relieved (MW)	Constraints to be aggravated (MW)	Potential New Constraints Caused by the NU (MW)
Big Stone South-Hazel Crk-Lyon County 345 kV	105	Split Rock-White 345 kV: -197.0 Brookings Co-White 345 kV #1: -45.3 Brookings Co-White 345 kV #2: -268.1 Minn Valley-Granite Falls 230 kV: -139.6 Johnson Jct-Ortonville 115 kV: -74.5 Johnson Jct-Morris 115 kV: -74.2 Canby-Granite Falls 115 kV: -69.0 Big Stone-Highway 12 115 kV: -111.1 Big Stone 115-230-13.8 kV xfmr: -124.2 Highway 12-Ortonville 115 kV: -110.3 Big Stone-Browns Valley 230 kV: -166.6 Big Stone-Blair 230 kV: -483.9 Browns Valley-Hankinson 230 kV: -166.0 Hankinson-Wahpeton 230 kV: -176.2 Blair-Watertown 230 kV: -280.2 Ellendale-Aberdeen Jct 115 kV: -52.0	Hazel Crk-Minn Valley 230 kV: 151.2 Helena-Scott Co 345 kV: 65.5	Helena-Scott Co 345 kV: 63.0
Big Stone South-Brookings 2nd 345 kV	NA	Johnson Jct-Ortonville 115 kV: -81.1 Johnson Jct-Morris 115 kV: -80.9 Canby-Granite Falls 115 kV: -70.9 Big Stone-Highway 12 115 kV: -118.5 Big Stone 115-230-13.8 kV xfmr: -129.2 Highway 12-Ortonville 115 kV: -117.6 Big Stone-Browns Valley 230 kV: -194.1 Big Stone-Blair 230 kV: -561.8 Browns Valley-Hankinson 230 kV: -193.1 Hankinson-Wahpeton 230 kV: -196.6 Blair-Watertown 230 kV: -391.6 Ellendale-Aberdeen Jct 115 kV: -65.9	Split Rock-White 345 kV: 21.7 Brookings Co-White 345 kV #1: 90.7 Helena-Scott Co 345 kV: 10.9 Hazel Crk-Minn Valley 230 kV: 20.8	Split Rock-White 345 kV: 52.9 Brookings Co-White 345 kV #1: 98.3

Table E-7: Projects Not Evaluated in Details in Big Stone Area

Transmission NUs	Miles	Reasons for Not Testing
Lakefield Jct-Lyon County 345 kV	100	1) Lake Field Jct-Lyon Co 345 kV flow is only 78 MVA if this line is added alone. 2) It is too expensive to add Lakefield Jct-Lyon Co to BSS-Hazel-Lyon; The flow on Lyon Co-Lake Field is only 101 MVA; No significant benefit; Overload of Hazel Crk-Minn Valley 230 kV under P7 contingency is not completely mitigated.

Table E-8: Loss Reduction Comparison by Projects in Big Stone Area

	Before NU		After BSS-Brookings			After BSS-Hazel-Lyon			After BSS-Alexandria	
Area	To Losses		To Losses	Loss Chng		To Losses	Loss Chng		To Losses	Loss Chng
600	637.7		638.2	0.5		647.1	9.4		632.1	-5.6
XEL	6066.3		6067.1	0.8		6164.3	98.0		6011.8	-54.5
608	141.9		141.0	-0.9		139.9	-2.0		143.0	1.1
MP	1606.9		1601.6	-5.3		1594.5	-12.4		1608.3	1.4
615	61.8		61.6	-0.2		61.6	-0.2		62.3	0.5
GRE	1260.6		1259.7	-0.9		1260.3	-0.3		1263.3	2.7
620	224.7		211.3	-13.4		195.0	-29.7		208.7	-16.0
OTP	1761.8		1678.9	-82.9		1569.1	-192.7		1713.5	-48.3
652	216.7		214.7	-2.0		204.3	-12.4		206.4	-10.3
WAPA	2079.5		2072.3	-7.2		1980.0	-99.5		2007.7	-71.8
661	18.8		18.4	-0.4		18.4	-0.4		18.9	0.1
MDU	156.0		151.1	-4.9		150.2	-5.8		153.2	-2.8
Total Loss Changes		P Loss		-16.4			-35.3			-30.2
		Q Loss		-100.4			-212.7			-173.3

E.2.2 Project Justification Details in Central-East Iowa Area

Table E-9: Project Evaluation Details in Central-East Iowa Area

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Blkhwk-Morgan & Beverly-Sub 92 1 - New Hartford-Morgan Valley Section; 2 - Morgan Valley Sub-Beverly Sub 3 - Beverly-Sub 92 345 kV	115	1) Significantly reduce loadings on J531 POI-Hazleton 345 kV, 498 POI-Grimes 345 kV, Montezuma-J530 POI-Hills 345 kV, Hazleton-Arnold 345 kV, Hazleton-Hickory Crk 345 kV, Hazleton-Blackhawk 345 kV, Arnold-Morgan Valley 345 kV, Ottumwa-Montezuma 345 kV, Morgan Valley-Tiffin 345 kV, Bondurant-Montezuma 345 kV, Hills-Tiffin 345 kV, Hills-Sub 92 345 kV, Adams 345-161 kV xfmr, Tiffin 161-69 kV xfmr, Rosehollow-Bertram 161 kV, Hayward-Glenworth 161 kV, Hazleton S-Dundee 161 kV, Arnold 345-161 kV xfmr, Jasper-Laurel 161 kV, Poweshiek-Reasnor 161 kV, Bondurant-Sycamore 345 kV	1) Increase contingency loading on Sub 56-Walcott 345 kV (149 MVA), Walcott-Sub 92 345 kV (150 MVA), Oak Grove-Sub 93 345 kV (134.2 MVA). If these are DPP constraints, solutions will be identified in the final ACCC analysis. 2) Blackhawk-Bremer Co 161 kV is loaded at 99.3% flo J416 POI-Hartford 345 kV 3) Sub 56-Walcott 345 kV is loaded at 100.5% flo Oak Grove-Sub 93 345 kV 4) Walcott-Sub 92 345 kV is loaded at 111.1% flo Oak Grove-Sub 93 345 kV
Blkhwk-Mntzuma_Tama-Morgan_Beverly-Sub 92 1 - New north-south 345 kV connecting Blackhawk, or new 345 kV switching station due north of Montezuma (New Hartford), to Montezuma 345 kV; 2 - New 345 kV "Tama County" switching station to Morgan Valley 345 kV 3 - Beverly-Sub 92 345 kV	155	1) Significantly reduce loadings on J531 POI-Hazleton 345 kV, 498 POI-Grimes 345 kV, Montezuma-J530 POI-Hills 345 kV, Hazleton-Arnold 345 kV, Hazleton-Hickory Crk 345 kV, Hazleton-Blackhawk 345 kV, Arnold-Morgan Valley 345 kV, Ottumwa-Montezuma 345 kV, Morgan Valley-Tiffin 345 kV, Bondurant-Montezuma 345 kV, Hills-Tiffin 345 kV, Hills-Sub 92 345 kV, J438 POI-Parnell 161 kV, Adams 345-161 kV xfmr, Tiffin 161-69 kV xfmr, Rosehollow-Bertram 161 kV, Hayward-Glenworth 161 kV, Hazleton S-Dundee 161 kV, Parnell-Hills 161 kV, Arnold 345-161 kV xfmr, Poweshiek-Reasnor 161 kV, Bondurant-Sycamore 345 kV	1) Increase contingency loading on Sub 56-Walcott 345 kV (149 MVA), Walcott-Sub 92 345 kV (150 MVA), Oak Grove-Sub 93 345 kV (134.2 MVA). If these are DPP constraints, solutions will be identified in the final ACCC analysis. 2) Blackhawk-Bremer Co 161 kV is loaded at 102.0% flo J416 POI-Hartford 345 kV 3) Sub 56-Walcott 345 kV is loaded at 100.5% flo Oak Grove-Sub 93 345 kV 4) Walcott-Sub 92 345 kV is loaded at 111.1% flo Oak Grove-Sub 93 345 kV

2021 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Blkhwk-Mntzuma_Tama-Morgan_Beverly-E Calamus 1 - New north-south 345 kV connecting Blackhawk, or new 345 kV switching station due north of Montezuma (New Hartford), to Montezuma 345 kV; 2 - New 345 kV "Tama County" switching station to Morgan Valley 345 kV 5 - Beverly to East Calamus 345 kV	168	1) Similar performance with the above NU option (Blkhwk-Mntzuma_Tama-Morgan_Beverly-Sub 92) 2) Reduce contingency loadings on Sub 56-Walcott 345 kV (-117 MVA), Walcott-Sub 92 345 kV (-118 MVA)	1) Increase contingency loading on Arnold-Morgan Valley 345 kV (81 MVA) 2) Rock Crk West-Dewitt 161 kV is loaded at 96.0% flo E Calamus-Sub 56 345 kV 3) Beverly 345-161 kV xfmr is loaded at 100.6% flo Arnold-Morgan Valley 345 kV 4) E Calamus-Dewitt 161 kV is loaded at 108.5% flo E Calamus-Sub 56 345 kV
Franklin-Marshalltown-Montezuma 345 kV, Marshalltown-Morgan Valley-Sub 92 345 kV 2 - Franklin-Marshalltown 345 kV 4 - Marshalltown-Montezuma 345 kV 3 - Marshalltown-Morgan Valley 345 kV 5 - Morgan Valley-Sub 92 345 kV	264	1) Similar performance with the above ITCM NU option (Blkhwk-Mntzuma_Tama-Morgan_Beverly-Sub 92)	1) Similar performance with the above NU option (Blkhwk-Mntzuma_Tama-Morgan_Beverly-Sub 92) 2) More expensive than the above NU option
Blkhwk-Mntzuma 1- New north-south 345 kV connecting Blackhawk, or new 345 kV switching station due north of Montezuma (New Hartford), to Montezuma 345 kV	70	1) Significantly reduce loadings on J531 POI-Hazleton 345 kV, 498 POI-Grimes 345 kV, Hazleton-Arnold 345 kV, Hazleton-Hickory Crk 345 kV, Hazleton-Blackhawk 345 kV, Arnold-Morgan Valley 345 kV, Morgan Valley-Tiffin 345 kV, Bondurant-Montezuma 345 kV, Hills-Tiffin 345 kV, Rosehollow-Bertram 161 kV, Hayward-Glenworth 161 kV, Jasper-Laurel 161 kV. 2) Reduce contingency loadings on Adams 345-161 kV xfmr (-65 MVA), Tiffin 161-69 kV xfmr (-21 MVA), Hazleton S-Dundee 161 kV (-42 MVA), Ottumwa-Montezuma 345 kV (-59MVA), Grimes-Sycamore 345 kV (-72 MVA), Bondurant-Sycamore 345 kV (-129 MVA).	1) Increase contingency loading on J530 POI-Hills 345 kV (284 MVA), J530 POI-Montezuma 345 kV (82 MVA) 2) Need to combine with ITCM (Tama-Morgan Valley) to be effective.

2021 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Blkhwk-Mntzuma & Tama-Morgan 1-New north-south 345 kV connecting Blackhawk, or new 345 kV switching station due north of Montezuma (New Hartford), to Montezuma 345 kV 2-New 345 kV "Tama County" switching station to Morgan Valley 345 kV	110	1) Significantly reduce loadings on J531 POI-Hazleton 345 kV, 498 POI-Grimes 345 kV, Montezuma-J530 POI 345 kV, Hazleton-Arnold 345 kV, Hazleton-Hickory Crk 345 kV, Hazleton-Blackhawk 345 kV, Arnold-Morgan Valley 345 kV, Ottumwa-Montezuma 345 kV, Bondurant-Montezuma 345 kV, J438 POI-Parnell 161 kV, Adams 345-161 kV xfmr, Rosehollow-Bertram 161 kV, Hayward-Glenworth 161 kV, Parnell-Hills 161 kV, Arnold 345-161 kV xfmr, Poweshiek-Reasnor 161 kV. 2) Reduce contingency loadings on Hazleton S-Dundee 161 kV (-63 MVA), Jasper-Laurel 161 kV (-35 MVA), Bondurant-Sycamore 345 kV (-77 MVA).	1) Increase contingency loading on J530 POI-Hills 345 kV (192 MVA), Morgan Valley-Tiffin 345 kV (425 MVA), Hills-Tiffin 345 kV (379 MVA), Hills-Sub 92 345 kV (65 MVA) 2) Need to combine with ITCM (Bertram – Sub 92, or Bertram – E Calamus) to be effective.
Beverly-Sub 92	45	1) Significantly reduce loadings on J530 POI-Hills 345 kV, Hazleton-Hickory Crk 345 kV, Morgan Valley-Tiffin 345 kV, Hills-Tiffin 345 kV, Hills-Sub 92 345 kV, Rosehollow-Bertram 161 kV. 2) Reduce contingency loadings on Tiffin 161-69 kV xfmr (-58 MVA), Bondurant-Montezuma 345 kV (-136 MVA).	1) Significantly increase contingency loading on Arnold-Morgan Valley 345 kV (108 MVA) 2) Increase contingency loading on Hazleton-Arnold 345 kV, Sub 56-Walcott 345 kV, Walcott-Sub 92 345 kV, Oak Grove-Sub 93 345 kV. 3) Beverly 345-161 kV xfmr is loaded at 105.9% flo Arnold-Morgan Valley 345 kV
Beverly-East Calamus	58	1) Significantly reduce loadings on Hazleton-Hickory Crk 345 kV, Morgan Valley-Tiffin 345 kV, Hills-Tiffin 345 kV, Hills-Sub 92 345 kV, Sub 56-Walcott 345 kV, Walcott-Sub 92 345 kV. 2) Reduce contingency loadings on Rosehollow-Bertram 161 kV (-41 MVA), Mt Vernon-Bertram 161 kV (-50 MVA), Mt Vernon-Wyoming 161 kV (-50 MVA), Bondurant-Montezuma 345 kV (-136 MVA).	1) Increase contingency loading on Arnold-Morgan Valley 345 kV. 2) Rock Crk West-Dewitt 161 kV is loaded at 96.0% flo E Calamus-Sub 56 345 kV 3) Beverly 345-161 kV xfmr is loaded at 100.6% flo Arnold-Morgan Valley 345 kV 4) E Calamus-Dewitt 161 kV is loaded at 108.5% flo E Calamus-Sub 56 345 kV

2021 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Beverly-E Calamus-Rock Crk - 5-Beverly to East Calamus 345 kV - 6-East Calamus-Rock Creek 345 kV	80	1) Significantly reduce loadings on Hazleton-Hickory Crk 345 kV, Morgan Valley-Tiffin 345 kV, Hills-Tiffin 345 kV, Hills-Sub 92 345 kV, Sub 56-Walcott 345 kV, Walcott-Sub 92 345 kV, Rosehollow-Bertram 161 kV. 2) Reduce contingency loadings on Mt Vernon-Bertram 161 kV (-65 MVA), Mt Vernon-Wyoming 161 kV (-64 MVA).	1) Significantly increase contingency loading on Arnold-Morgan Valley 345 kV (173 MVA), Hazleton-Arnold 345 kV (124 MVA), Quad City-MEC Cordova 345 kV (138 MVA). 2) Increase contingency loading on Arnold 345-161 kV xfmr. 3) Beaver Channel 65-Beaver Channel 5 161 kV is loaded at 97.7% flo Quad City-Rock Crk 345 kV 4) Beverly 345-161 kV xfmr is loaded at 111.6% flo Arnold-Morgan Valley 345 kV. 5) Expensive and less effective
Morgan-Sub 92 - 5-Morgan Valley-Sub 92 345 kV	65	1) Performance is similar to ITCM NU (Beverly-Sub 92) 2) Significantly reduce loadings on J530 POI-Hills 345 kV, Hazleton-Hickory Crk 345 kV, Morgan Valley-Tiffin 345 kV, Hills-Tiffin 345 kV, Hills-Sub 92 345 kV. 2) Reduce contingency loadings on Bondurant-Montezuma 345 kV (-123 MVA).	1) More expensive than ITCM NU (Beverly-Sub 92) 2) Significantly increase contingency loading on Arnold-Morgan Valley 345 kV (110 MVA) 2) Increase contingency loading on Hazleton-Arnold 345 kV, Sub 56-Walcott 345 kV, Walcott-Sub 92 345 kV, Oak Grove-Sub 93 345 kV. 3) Beverly 345-161 kV xfmr is loaded at 100.0% flo Arnold-Morgan Valley 345 kV

2021 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Franklin-Marshalltown-Mntzuma - 2-Franklin-Marshalltown 345 kV - 4-Marshalltown-Montezuma 345 kV	123	1) Performance is similar to ITCM NU (Blkhwk-Mntzuma) 2) Significantly reduce loadings on J531 POI-Hazleton 345 kV, 498 POI-Grimes 345 kV, Hazleton-Arnold 345 kV, Hazleton-Hickory Crk 345 kV, Hazleton-Blackhawk 345 kV, Arnold-Morgan Valley 345 kV, Morgan Valley-Tiffin 345 kV, Bondurant-Montezuma 345 kV, Hills-Tiffin 345 kV, Hayward-Glenworth 161 kV, Poweshiek-Reasnor 161 kV. 2) Reduce contingency loadings on Adams 345-161 kV xfmr (-62 MVA), Rosehollow-Bertram 161 kV (-44 MVA), Tiffin 161-69 kV xfmr (-19 MVA), Hazleton S-Dundee 161 kV (-42 MVA), Arnold 345-161 kV xfmr (-39 MVA), Jasper-Laurel 161 kV (-49 MVA), Grimes-Sycamore 345 kV (-97 MVA), Bondurant-Sycamore 345 kV (-167 MVA).	1) More expensive than ITCM NU (Blkhwk-Mntzuma) 2) Significantly increase contingency loading on J530 POI-Hills 345 kV (267 MVA), J530 POI-Montezuma 345 kV (151 MVA). 3) Increase contingency loading on Killdeer 345-161 kV xfmr, Ottumwa-Montezuma 345 kV 4) Blackhawk-Bremer Co 161 kV is loaded at 100.7% flo Franklin-J416 POI 345 kV.
Franklin-Marshalltown-Morgan - 2-Franklin-Marshalltown 345 kV - 3-Marshalltown-Morgan Valley 345 kV	145	1) Performance is not as good as ITCM NU (Blkhwk-Mntzuma & Tama-Morgan) 2) Significantly reduce loadings on J531 POI-Hazleton 345 kV, 498 POI-Grimes 345 kV, J530 POI-Hills 345 kV, Hazleton-Arnold 345 kV, Hazleton-Hickory Crk 345 kV, Hazleton-Blackhawk 345 kV, Arnold-Morgan Valley 345 kV, Bondurant-Montezuma 345 kV, Adams 345-161 kV xfmr, Hayward-Glenworth 161 kV, Arnold 345-161 kV xfmr. 2) Reduce contingency loadings on Montezuma-J530 POI 345 kV (-64 MVA), Arnold-Downtown Ind 161 kV (-61 MVA)	1) More expensive than ITCM NU (Blkhwk-Mntzuma & Tama-Morgan) 2) Significantly increase contingency loading on Killdeer 345-161 kV xfmr (110 MVA), Morgan Valley-Tiffin 345 kV, Hills-Tiffin 345 kV (172 MVA) 3) Increase contingency loading on Hills-Sub 92 345 kV (70 MVA), Hills-Sub T HSK 345 kV (75 MVA) 4) Tiffin 345-161 kV is loaded at 100.1% flo Hills-Tiffin 345 kV 5) Tiffin-Coral Ridge 161 kV is loaded at 104.5% flo Hills-Tiffin 345 kV 6) Beverly-PCI 161 kV is loaded at 108.2% flo Morgan Valley-Tiffin 345 kV 7) Need to combine with Marshalltown-Montezuma, Morgan Valley-Sub 92 to be effective.

2021 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Franklin-Marshalltown-Morgan-Sub 92 - 2-Franklin-Marshalltown 345 Kv - 3-Marshalltown-Morgan Valley 345 kV - 5-Morgan Valley-Sub 92 345 kV	210	1) Performance is not as good as ITCM NU (Blkhwk-Mntzuma_Tama-Morgan_Beverly-Sub 92) 2) Significantly reduce loadings on J531 POI-Hazleton 345 kV, 498 POI-Grimes 345 kV, Montezuma-J530 POI-Hills 345 kV, Hazleton-Arnold 345 kV, Hazleton-Hickory Crk 345 kV, Hazleton-Blackhawk 345 kV, Arnold-Morgan Valley 345 kV, Morgan Valley-Tiffin 345 kV, Hills-Tiffin 345 kV, Hills-Sub 92 345 kV, Adams 345-161 kV xfmr, Rosehollow-Bertram 161 kV, Hayward-Glenworth 161 kV, Arnold 345-161 kV xfmr. 3) Reduce contingency loadings on Tiffin 161-69 kV xfmr (-46 MVA), Hazleton S-Dundee 161 kV (-56 MVA), Arnold-Downtown Ind 161 kV (-50 MVA).	1) More expensive than NU (Blkhwk-Mntzuma_Tama-Morgan_Beverly-Sub 92) 2) Increase contingency loading on Killdeer 345-161 kV xfmr (126 MVA), Sub 56-Walcott 345 kV (140 MVA), Walcott-Sub 92 345 kV (141 MVA), Oak Grove-Sub 93 345 kV (125 MVA). If these are DPP constraints, solutions will be identified in the final ACCC analysis. 3) Sub 56-Walcott 345 kV is loaded at 99.3% flo Oak Grove-Sub 93 345 kV
Webster-Franklin - 1-Webster-Franklin 345 kV	50	1) Significantly reduce loadings on J531 POI-Hazleton 345 kV, 498 POI-Grimes 345 kV, J530 POI-Hills 345 kV, Killdeer 345-161 kV xfmr, Emery-Colby 161 kV, Emery-Floyd 161 kV, Bondurant-Montezuma 345 kV, Franklin-Wall Lake 161 kV. 2) Reduce contingency loadings on Montezuma-J530 POI 345 kV (-58 MVA), Adams 345-161 kV xfmr (-59 MVA), Grimes-Sycamore 345 kV (-83 MVA), Bondurant-Sycamore 345 kV (-78 MVA).	1) Significantly increase contingency loading on Hazleton-Blackhawk 345 kV (175 MVA) 2) Increase contingency loading on J506 POI-Raun 345 kV (79 MVA), Hazleton-Arnold 345 kV (67 MVA), Hazleton-Hickory Crk 345 kV (75 MVA), Blackhawk 345-161 kV xfmr (71 MVA) 3) Blackhawk-Bremer Co 161 kV is loaded at 99.1% flo Franklin-Blackhawk 345 kV 2) Need to combine with Franklin-Marshalltown and other options to be effective.

Table E-10: Projects Not Evaluated in Details in Central-East Iowa Area

Transmission NUs	Miles	Reasons for Not Testing
Beverly-Bertram - 1-Beverly-Bertram 345 kV	9	Since the flow on new Beverly-Bertram line and Bertram xfmr is only 56.1 MVA, this NU was not tested
Beverly-Bertram-Sub 92 - 1-Beverly-Bertram 345 Kv - 2-Bertram-Sub 92 345 kV line	47	Since the flow on new Bertram xfmr is only 77.0 MVA, this NU was not tested. Beverly-Sub 92 should be better
Beverly-Bertram-E Calamus - 1-Beverly-Bertram 345 Kv - 4-Bertram to East Calamus 345 kV	60	Since the flow on new Beverly xfmr is only 61.7 MVA, this NU was not tested. Beverly-East Calamus should be better

Table E-11: Loss Reduction Comparison by Projects in Big Stone Area

	Before NU		After Blkhwk-Morgan & Beverly-Sub	
Area	To Losses		To Losses	Loss Chng
222	532.7		539.1	6.4
CE	9475.1		9565	89.9
357	200		204.3	4.3
AMIL	2498.5		2535.3	36.8
600	637.7		632.2	-5.5
XEL	6066.3		6017.4	-48.9
627	448.3		421.6	-26.7
ALTW	3813.9		3648.5	-165.4
633	1.1		1.2	0.1
MPW	16.1		16.5	0.4
635	457.4		455.1	-2.3
MEC	4975.1		5035.8	60.7
652	216.7		214.5	-2.2

2021 Summer Shoulder Contingency Analysis Results

	Before NU		After Blkhwk-Morgan & Beverly-Sub	
Area	To Losses		To Losses	Loss Chng
WAPA	2079.5		2060.7	-18.8
680	108		103.2	-4.8
DPC	544.3		521.3	-23.0
694	123.7		119.8	-3.9
ALTE	1025.8		999	-26.8
Total Loss Changes		P Loss		-34.6
		Q Loss		-95.1

E.2.3 Project Justification Details in West-South-North Iowa Area

Table E-12: Project Evaluation Details in West-South-North Iowa Area

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Ida Co-Avoca 345 kV	76	1) Significantly reduce loadings on 498 POI-Grimes 345 kV, Raun-S 3451 345 kV, Raun-Tekamah 161 kV.	1) Significantly increase contingency loading on Council Bluffs-S3456 345 KV (250 MVA) 2) Increase contingency loading on J506 POI-Raun 345 kV (54 MVA)
Madison-Ottumwa-Sub T Haskins 345 kV	187	1) Significantly reduce loadings on J530 POI-Montezuma 345 kV, J530 POI-Hills 345 kV, Ottumwa-Montezuma 345 kV, Morgan Valley-Tiffin 345 kV, Bondurant-Sycamore 345 kV, Bondurant-Montezuma 345 kV, J438 POI-Parnell 161 kV, Parnell-Hills 161 kV, Poweshiek-Reasnor 161 kV. 2) Reduce contingency loadings on Council Bluffs-S3456 345 KV (-93 MVA), Grimes-Sycamore 345 kV (-120 MVA).	1) Increase loadings on Grimes-J498 POI, Walcott-Sub 92 345 kV, Oak Grove-Sub 93 345 kV. 2) Bondurant-Montezuma 345 kV is loaded at 125.5% flo Madison-Ottumwa 345 kV
A new Winterset area 345 kV to Ottumwa to Sub T	165	1) Similar performance with Madison-Ottumwa-Sub T Haskins 2) Cheaper than Madison-Ottumwa-Sub T Haskins	1) Similar performance with Madison-Ottumwa-Sub T Haskins
Madison-Ottumwa 345 kV	105	1) Significantly reduce loadings on Poweshiek-Reasnor 161 kV, Ottumwa-Montezuma 345 kV, Morgan Valley-Tiffin 345 kV, Bondurant-Sycamore 345 kV, Bondurant-Montezuma 345 kV, Grimes-Sycamore 345 kV. 2) Reduce contingency loadings on Council Bluffs-S3456 345 KV (-70 MVA).	1) Increase loadings on Montezuma-J530 POI-Hills 345 kV, Grimes-J498 POI, Wapello-Ottumwa 161 kV. 2) Need to combine with Ottumwa-Sub T Haskins 345 kV to be effective.

2021 Summer Shoulder Contingency Analysis Results

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Ottumwa-Sub T Haskins 345 kV	82	1) Significantly reduce loadings on Montezuma-J530 POI-Hills 345 kV, Hills-Sub 92 345 kV, J438 POI-Parnell 161 kV, Parnell-Hills 161 kV. 2) Reduce contingency loadings on Sub 56-Walcott 345 kV (-75 MVA), Walcott-Sub 92 345 kV (-78 MVA)	1) Significantly increase contingency loading on Ottumwa-Montezuma 345 kV (190 MVA) 2) Increase contingency loading on J506 POI-Raun 345 kV, Bondurant-Sycamore 345 kV, Bondurant-Montezuma 345 kV. 3) Oak Grove-Sub 93 345 kV is loaded at 109.7% flo Walcott-Sub 92 345 kV. 4) Walcott-Sub 92 345 kV is loaded at 99.0% flo Oakgrove-Sub 93 345 kV.
Colby-Mitchell County 345 kV	55	1) Significantly reduce loadings on Adams 345-161 kV xfmr, Hayward-Glenworth 161 kV, Emery-Colby 161 kV, Killdeer 345-161 kV xfmr, Emery-Floyd 161 kV. 2) Reduce contingency loadings on J498 POI-Grimes 345 kV (-97 MVA), Helena-Scott Co 345 kV (-94 MVA), Bondurant-Montezuma 345 kV (-75 MVA)	1) Significantly increase contingency loading on J531 POI-Hazleton 345 kV (253 MVA) 2) Increase contingency loading on Hazleton-Hickory Crk 345 kV (87 MVA) 3) J531 POI-Mitchell Co 345 kV is loaded at 124.2% flo J416 POI-Blackhawk 345 kV
Colby-Mitchell County-Rochester 345 kV - 1-Colby-Mitchell County 345 kV - 2-Mitchell County-Rochester 345 kV - 3-Convert Mitchell County to breaker and ½ configuration and configure as 2 separate Mitchell buses with normally open breakers between the two buses	95	1) Significantly reduce loadings on J531 POI-Hazleton 345 kV, Adams 345-161 kV xfmr, Hayward-Glenworth 161 kV	1) With this NU, flows go as a circle to Hazleton in the following: Colby->Mitchell_1->Adams->Pleasant Valley->Byron->NROC->Mitchell->J531->Hazleton. Not much help 2) Adams 345-161 kV xfmr is loaded at 108% flo Adams-Pleasant Valley 345 kV 3) Adams-Mitchell Co 345 kV is loaded at 96.3% flo J416 POI-Blakhawk 345 kV

Table E-13: Projects Not Evaluated in Details in West-South-North Iowa Area

Transmission NUs	Miles	Reasons for Not Testing
Mitchell County-Rochester 345 kV	45	Flow on the Mitchell Co-NROC 345 kV line is only 52.3 MVA

E.2.4 Project Justification Details in Xcel Area

Table E-14: Project Evaluation Details in Xcel Area

Transmission NUs	Miles	Project Advantages	Project Disadvantages
Briggs Road Series Compensation		1) Has good loading relief on some critical facilities 2) Relatively cheaper and easier to implement 4) Reduce contingency loadings on King-Eau Claire 345 kV (-71 MVA), Hazleton-Hickory Crk 345 kV (-63 MVA), J531 POI-Hazleton 345 kV (-84 MVA), Hazleton-Arnold 345 kV (-66 MVA), Hazleton-Blackhawk 345 kV (-54 MVA), Wabaco-Alma 161 kV (-65 MVA).	1) Significantly increase loading on Briggs Rd 345-161 kV xfmr 2) Briggs Rd-Mayfair 161 kV is loaded at 103.8% flo Lacrosse Tap-Lacrosse 161 kV

Table E-15: Projects Not Evaluated in Details in Xcel Area

Transmission NUs	Miles	Reasons for Not Testing
Huntley-Wilmarth 345 kV	NA	MTEP Appendix A project; Included in the base case.
Helena - Chub Lake 2nd CKT	NA	The need of this NU can be verified once the final ACCC is performed

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E.3 Constraints Identified in 2nd Round Contingency Analysis with Backbone NUs

Table E-16: 2nd Round System Intact Thermal Constraints

Table E-17: 2nd Round System Intact Voltage Constraints

Table E-18: 2nd Round Category P1 Thermal Constraints

Table E-19: 2nd Round Category P1 Voltage Constraints

Table E-20: 2nd Round Category P2-P7 Thermal Constraints

Table E-21: 2nd Round Category P2-P7 Voltage Constraints

Table E-22: 2nd Round Non-Converged Category P2-P7 Contingencies

Table E-23: 2nd Round Non-Converged Category P2-P7 Contingencies DCCC Results

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E.4 Worst Voltage Constraints in 2nd Round Contingency Analysis with Backbone NUs

Table E-24: 2021 SH Category P0 Worst Voltage Constraints

Table E-25: 2021 SH Category P1 Worst Voltage Constraints

Table E-26: 2021 SH Category P2-P7 Worst Voltage Constraints

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Affected System Contingency Analysis Results

F.1 CIPCO/MPC Companies LPC Analysis Results

Table F-1: 2021 SH CIPCO/MPC LPC Analysis Results

Table F-2: 2021 SPK CIPCO/MPC LPC Analysis Results

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F.2 J525 GRE Local Planning Criteria Analysis

Below is the GRE local planning criteria analysis report provided by GRE.

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F.3 PJM Affected System Study Results

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F.4 SPP Affected System Study Results

Below is the SPP affected system study report provided by SPP.

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Appendix

G

Transient Stability Results

G.1 2021 Summer Shoulder Stability Results Summary

Stability simulation was performed in the 2021 summer shoulder case with MWEX reactive power NUs (Table 4-14) and additional voltage constraints NUs (Table 4-15) modeled.

All simulations are transiently stable, transient period voltage criteria are met, oscillations are well damped. No additional Network Upgrades are required in stability analysis.

Stability study results summary is in Table G-1.

Table G-1: 2021 Summer Shoulder Stability Analysis Results Summary

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G.2 2021 Summer Shoulder Stability Plots

Plots of stability simulations for 2021 summer shoulder study case are in separate files which are listed below:

AppendixG2_2021SH_DPP 2016Feb-West_Study_Plots.zip

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MWEX Voltage Study Details

Below is the MWEX voltage stability study report provided by ATC.

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Short Circuit Analysis

- I.1 J432 Short Circuit Study Performed by Siemens PTI
- I.2 J460 Short Circuit Study Performed by Siemens PTI
- I.3 J475 Short Circuit Study Performed by MEC
- I.4 J485 Short Circuit Study Performed by Siemens PTI
- I.5 J488 & J489 Short Circuit Study Performed by OTP
- I.6 J490 Short Circuit Study Performed by Siemens PTI
- I.7 J493 and J510 Short Circuit Study Performed by OTP
- I.8 J495 Short Circuit Study Performed by ITCM
- I.9 J498 Short Circuit Facilities Study Performed by MEC
- I.10 J499 Short Circuit Facilities Study Performed by MEC
- I.11 J500 Short Circuit Facilities Study Performed by MEC
- I.12 J501 Short Circuit Facilities Study Performed by MEC
- I.13 J504 Short Circuit Facilities Study Performed by ITCM
- I.14 J506 Short Circuit Facilities Study Performed by Siemens PTI
- I.15 J514 Short Circuit Facilities Study Performed by ITCM
- I.16 J523 Short Circuit Facilities Study Performed by ITCM
- I.17 J524 Short Circuit Facilities Study Performed by Siemens PTI
- I.18 J525 Short Circuit Facilities Study Performed by GRE
- I.19 J526 Short Circuit Facilities Study Performed by OTP

- I.20 J527 Short Circuit Facilities Study Performed by Siemens PTI**
- I.21 J528 Short Circuit Facilities Study Performed by Siemens PTI**
- I.22 J529 Short Circuit Facilities Study Performed by MEC**
- I.23 J530 Short Circuit Facilities Study Performed by Siemens PTI**
- I.24 J531 Short Circuit Facilities Study Performed by ITCM**
- I.25 J534 Short Circuit Facilities Study Performed by Siemens PTI**
- I.26 J535 Short Circuit Facilities Study Performed by Siemens PTI**

J432 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J432 generating facility (49 Gamesa 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the Xcel system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "NSP_03202017.GDB"

The J432 short circuit benchmark model was developed as follows:

- Added Big Stone South-Ellendale 345 kV line and Ellendale 345/230/34.5 kV transformer.
- Added J460 project and turned it online.

The J432 short circuit study model was developed starting from the above benchmark case as follows:

- The J432 generating facility with its sequence parameters was added at the POI on Brookings County 345 kV Substation. The associated 345 kV Gentie, 345-34.5-13.2 kV transformer ($Z=9\%$ at 60 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.69 kV pad mount transformer ($Z=11.6\%$ at 115.15 MVA, D-YG) were also added.
- The J432 generator was modeled as a generator with impedances listed below. These values represent the DFIG configuration with the maximum crowbar resistance connected in the rotor.
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): $0.71474+j0.3344$ pu
 - o Negative sequence impedance (Z_2): $0.04899+j0.1626$ pu
 - o Zero sequence impedance (Z_0): $j999999$ pu (zero sequence currents are blocked by the GSU transformers (Wye-Delta connection, therefore, this value does not have any impact on the short circuit results))

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J432 project) and benchmark case (without the J432 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J432 (study case) and without J432 (benchmark case) are listed in Table 1. With the addition of the J432 generating facility, the maximum increase in the 3PH short circuit current is 0.74% (126.1 A), and the maximum increase in the SLG short circuit current is 2.19% (382.1 A), which are both observed at the Brookings County 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

As shown in the Table 1, the changes in fault current at buses around the POI are relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J432 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J432

Substation	Faulted Bus			Benchmark Case without J432 Project		Study Case with J432 Project		Ampere Change with J432 vs Benchmark		% Change with J432 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J432	61031	J432_POI	345	-	-	17514.8	16541.2	-	-	-	-
J432	61032	J432-COL	34.5	-	-	17756.7	11618.6	-	-	-	-
J432	61034	J432_COL	34.5	-	-	8518.4	8546.4	-	-	-	-
J432	61035	J432_TR_TER	13.2	-	-	0.0	21383.1	-	-	-	-
BROOKINGS COUNTY (NS	113200	BOK_345KV_1	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113201	BOK_345KV_2	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113202	BOK_IB_BSS	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113203	BOK_IB_HNL	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113204	BOK_IB_TR09	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113205	BOK_IB_TR10	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113206	BOK_IB_WH1	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113207	BOK_IB_WH12	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
WHITE (WAPA)	225600	WHI_345KV_1	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225601	WHI_345KV_2	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225602	WHI_IB_BOK1	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225603	WHI_IB_BOK2	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225604	WHI_IB_SPK	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225605	WHI_IB_TR01	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225606	WHI_IB_WTT	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	300094	WHI_IB_DCR	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
DEER CREEK (WAPA)	300095	DCR_345KV	345	17140.7	16552.6	17476.8	16669.9	336.1	117.3	1.96%	0.71%
J460	61041	J460_POI	345	11512.8	12497.8	11577.8	12538.5	65.0	40.7	0.56%	0.33%
BROOKINGS COUNTY (NS	113208	BOK_115KV_1	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113209	BOK_115KV_2	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113210	BOK_IB_TR10	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113211	BOK_IB_YNK1	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113212	BOK_IB_YNK2	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
J460	61042	J460	345	11150.9	11940.4	11210.8	11977.2	59.9	36.8	0.54%	0.31%
HAWKS NEST (WIND)	300068	HNL_345KV_1	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
HAWKS NEST (WIND)	300069	HNL_345KV_2	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
HAWKS NEST (WIND)	300070	HNL_BOK	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
RED PINE (WIND)	300071	HNL_GEN	345	9658.6	11499.7	9693.3	11527.4	34.7	27.7	0.36%	0.24%
WATERTOWN (WAPA)	221500	WTT_345KV	345	8739.3	9643.6	8760.8	9660.8	21.5	17.2	0.25%	0.18%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J432

Substation	Faulted Bus			Benchmark Case without J432 Project		Study Case with J432 Project		Ampere Change with J432 vs Benchmark		% Change with J432 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
BIG STONE SOUTH (OTP)	109600	BST_345KV	345	8289.1	7472.0	8302.6	7481.3	13.5	9.3	0.16%	0.12%
WHITE (WAPA)	225607	WHI_115KV	115	12163.7	10520.0	12191.8	10532.5	28.1	12.5	0.23%	0.12%
YANKEE (NSPM)	229500	YNK_115KV_1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229501	YNK_115KV_2	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229502	YNK_IB_BOK1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229503	YNK_IB_BOK2	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229504	YNK_IB_BRI	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229505	YNK_IB_RB1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229506	YNK_IB_TR01	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229507	YNK_IB_TR02	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	300448	YNK_REAC1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
WATERTOWN (WAPA)	221501	WTT_230KV	230	11808.2	13524.8	11825.4	13539.9	17.2	15.1	0.15%	0.11%
LYON COUNTY (NSPM)	168300	LYC_345KV_1	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168301	LYC_345KV_2	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168302	LYC_IB_BOK1	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168303	LYC_IB_CMT2	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168304	LYC_IB_TR09	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
BIG STONE SOUTH (OTP)	300906	BSS_230KV	230	16323.8	12559.0	16340.7	12569.7	16.9	10.7	0.10%	0.09%
BIG STONE (OTP)	109601	BST_230KV	230	17057.9	12766.6	17074.9	12777.1	17.0	10.5	0.10%	0.08%
SPLIT ROCK (NSPM)	206100	SPK_345KV_1	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206101	SPK_345KV_2	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206102	SPK_IB_NOB	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206103	SPK_IB_SOC	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206104	SPK_IB_WHI	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%

J460 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J460 generating facility (100 Vestas V110 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the Xcel system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "NSP_03202017.GDB"

The J460 short circuit benchmark model was developed as follows:

- Added Big Stone South-Ellendale 345 kV line and Ellendale 345/230/34.5 kV transformer.
- Added J432 project and turned it online

The J460 short circuit study model was developed starting from the above benchmark case as follows:

- The J460 generating facility with its sequence parameters was added at the POI on Brookings County – Hawks Nest (H081) 345 kV Substation. The associated 345 kV Gentie, two 345-34.5-13.2 kV transformers ($Z=6\%$ at 100 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z+=7.8\%$, $Z0=7.3\%$ at 105 MVA, D-YG) were also added.
- The J460 generation was represented using the conventional generator model with the impedance of the machines set as follows per the data provided in the Vestas manual.
 - Positive saturated sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.1669 p.u.
 - Negative saturated sequence impedance ($Z2$): j0.1669 p.u. (same as positive sequence impedances)
 - Zero sequence saturated impedance ($Z0$): j0.1669 p.u.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J460 project) and benchmark case (without the J460 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J460 (study case) and without J460 (benchmark case) are listed in Table 1. With the addition of the J460 generating facility, the maximum increase in the 3PH short circuit current is 6.58% (715.2 A), and the maximum increase in the SLG short circuit current is 16.76% (1399.6 A), which are both observed at the Hawks Nest (H081) 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 11577 A (increased by 715.2 A) and SLG fault current is 9751 A (increased by 1399.6 A) at the Hawks Nest 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J460 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J460	61041	J460_POI	345	-	-	11577.8	12538.5	-	-	-	-
J460	61042	J460	345	-	-	11210.8	11977.2	-	-	-	-
J460	61043	J460_COL1	34.5	-	-	38909.0	29079.0	-	-	-	-
J460	61046	J460_COL2	34.5	-	-	38436.0	28683.7	-	-	-	-
J460	61047	J460_COL1	34.5	-	-	33279.5	26558.9	-	-	-	-
J460	61048	J460_COL2	34.5	-	-	27091.6	23334.5	-	-	-	-
J460	61049	J460_TER1	13.2	-	-	0.0	53996.6	-	-	-	-
J460	61050	J460_TER2	13.2	-	-	0.0	54574.1	-	-	-	-
HAWKS NEST (WIND)	300068	HNL_345KV_1	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
HAWKS NEST (WIND)	300069	HNL_345KV_2	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
HAWKS NEST (WIND)	300070	HNL_BOK	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
RED PINE (WIND)	300071	HNL_GEN	345	8310.2	10818.4	9693.3	11527.4	1383.1	709.0	16.64%	6.55%
BROOKINGS COUNTY (NS)	113200	BOK_345KV_1	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113205	BOK_IB_TR10	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113206	BOK_IB_WH11	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113201	BOK_345KV_2	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113204	BOK_IB_TR09	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113207	BOK_IB_WH12	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113202	BOK_IB_BSS	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113203	BOK_IB_HNL	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
WHITE (WAPA)	225600	WHI_345KV_1	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	300094	WHI_IB_DCR	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225601	WHI_345KV_2	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225604	WHI_IB_SPK	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225606	WHI_IB_WTT	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225605	WHI_IB_TR01	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225603	WHI_IB_BOK2	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225602	WHI_IB_BOK1	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
J432	61031	J432_POI	345	16705.3	15910.4	17514.8	16541.2	809.5	630.8	4.85%	3.96%
DEER CREEK (WAPA)	300095	DCR_345KV	345	16729.8	16056.2	17476.8	16669.9	747.0	613.7	4.47%	3.82%
LYON COUNTY (NSPM)	168300	LYC_345KV_1	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168301	LYC_345KV_2	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168304	LYC_IB_TR09	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
LYON COUNTY (NSPM)	168302	LYC_IB_BOK1	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168303	LYC_IB_CMT2	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
BROOKINGS COUNTY (NS)	113210	BOK_IB_TR10	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113211	BOK_IB_YNK1	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113208	BOK_115KV_1	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113209	BOK_115KV_2	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113212	BOK_IB_YNK2	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
HAZEL CREEK (NSPM)	148702	HZC_IB_LYC	345	6884.9	7977.3	6974.8	8087.0	89.9	109.7	1.31%	1.38%
HAZEL CREEK (NSPM)	148700	HZC_345KV_1	345	6884.9	7977.3	6974.8	8087.0	89.9	109.7	1.31%	1.38%
CEDAR MOUNTAIN (GRE)	119104	CMT_IB_LYC2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119100	CMT_345KV_1	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119103	CMT_IB_HNA2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119101	CMT_345KV_2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119105	CMT_IB_TR01	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119102	CMT_IB_HNA1	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
WATERTOWN (WAPA)	221500	WTT_345KV	345	8698.8	9561.6	8760.8	9660.8	62.0	99.2	0.71%	1.04%
LYON COUNTY (NSPM)	168305	LYC_115KV_1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168306	LYC_115KV_2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168312	LYC_IB_CLO	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168307	LYC_CAP	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168316	LYC_IB_TR09	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168311	LYC_IB_CAP2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168310	LYC_IB_CAP1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168315	LYC_IB_MSH	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168313	LYC_IB_LAY1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168314	LYC_IB_LAY2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168308	LYC_CAP1	115	17035.5	16216.2	17208.7	16346.0	173.2	129.8	1.02%	0.80%
LYON COUNTY (NSPM)	168309	LYC_CAP2	115	17035.5	16216.2	17208.7	16346.0	173.2	129.8	1.02%	0.80%
BIG STONE SOUTH (OTP)	109600	BST_345KV	345	8261.4	7429.0	8302.6	7481.3	41.2	52.3	0.50%	0.70%
WATERTOWN (WAPA)	221501	WTT_230KV	230	11771.8	13446.7	11825.4	13539.9	53.6	93.2	0.46%	0.69%
YANKEE (NSPM)	229500	YNK_115KV_1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229501	YNK_115KV_2	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229506	YNK_IB_TR01	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
YANKEE (NSPM)	229507	YNK_IB_TR02	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229504	YNK_IB_BRI	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229502	YNK_IB_BOK1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	300448	YNK_REAC1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229505	YNK_IB_RB1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229503	YNK_IB_BOK2	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
HAZEL CREEK (NSPM)	148705	HZC_IB_MNV	230	11431.9	11656.2	11497.3	11730.0	65.4	73.8	0.57%	0.63%
HAZEL CREEK (NSPM)	148703	HZC_230KV_1	230	11431.9	11656.2	11497.3	11730.0	65.4	73.8	0.57%	0.63%
WHITE (WAPA)	225607	WHI_115KV	115	12120.3	10468.6	12191.8	10532.5	71.5	63.9	0.59%	0.61%
MINNESOTA VALLEY (NS	176501	MNV_230KV_1	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176502	MNV_230KV_2	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176503	MNV_230KV_5	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176504	MNV_230KV_6	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176500	MNVT_GRA_PTH	230	13506.2	12925.2	13571.7	13002.5	65.5	77.3	0.48%	0.60%
RED PINE (WIND)	300072	HNL_GEN1_034	34.5	15362.3	15293.5	15653.7	15384.2	291.4	90.7	1.90%	0.59%
RED PINE (WIND)	300073	HNL_GEN2_034	34.5	15362.3	15293.5	15653.7	15384.2	291.4	90.7	1.90%	0.59%
GRANITE FALLS (WAPA)	145100	GRA_230KV_1	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145101	GRA_230KV_2	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145102	GRA_IB_BAR	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145104	GRA_IB_KV8A	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145108	GRA_IB_WMR	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145105	GRA_IB_MNV1	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145107	GRA_IB_MOS	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145103	GRA_IB_KV3A	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145109	GRA_IB_WTT	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145106	GRA_IB_MNV2	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
MARSHALL SW STATION	170600	MSH_115KV	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170603	MSH_IB_SEA	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170601	MSH_IB_LYC	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170602	MSH_IB_NSV	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
BIG STONE SOUTH (OTP	300906	BSS_230KV	230	16282.6	12508.0	16340.7	12569.7	58.1	61.7	0.36%	0.49%
BIG STONE (OTP)	109601	BST_230KV	230	17015.1	12715.8	17074.9	12777.1	59.8	61.3	0.35%	0.48%
NORTH SEVENTH STREET	183200	NSV_115KV_1	115	8659.1	11161.9	8690.8	11212.5	31.7	50.6	0.37%	0.45%

CRITICAL ENERGY INFRASTRUCTURE INFORMATION NOTICE

The materials contained in this document include Critical Energy Infrastructure Information (CEII). All materials designated as CEII must be handled and protected per the requirements in FERC CEII Policy. There may be additional requirements for CEII materials in the future.

J475 and J555 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J475, a proposed 200 MW wind farm, and J555, a proposed 140 MW expansion, and nearby substations both with and without the Interconnection Customer interconnected. J475 and J555 were assumed to interconnect at the MidAmerican and ITC Midwest jointly-owned Montezuma Substation. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in DPP study cycles of the MISO generation interconnection process, where the system impact studies are not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 13,008 Amps without the Interconnection Customer included and 14,334 Amps with the full 340 MW from the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results, especially the 34.5 kV results that should be calculated by the Interconnection Customer. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J475 and J555

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison					3 Ph Fault Current Comparison				
						Base SLG w/o wind farms J475/J555	SLG with new J475 wind farm	SLG Difference w/ J475 wind farm vs Base	SLG with J475 & J555 wind farm	SLG Difference w/ J475 & J555 wind farm vs Base	Base 3PH w/o wind farms J475/J555	3PH with new J475 wind farm	3PH Difference w/ J475 wind farm vs Base	3PH with J475 & J555 wind farm	3PH Difference w/ J475 & J555 wind farm vs Base
635730	MNTZUMA3	Montezuma	345	635	MEC	9,020	10,083	1,063	11,078	2,058	13,008	13,705	697	14,334	1,326
		J475 Collector Bus	34.5			19,371	32,795	13,424	32,856	13,485	16,744	27,713	10,969	27,756	11,012
		J555 Collector Bus	34.5			19,371	19,371	-	29,325	9,954	16,744	16,744	-	24,307	7,563
635680	BONDRNT3	Bondurant	345	635	MEC	17,016	17,131	115	17,223	207	19,055	19,188	133	19,299	244
631143	OTTUMWA3	Ottumwa	345	627	ITCM	9,031	9,193	162	9,322	291	10,891	11,055	164	11,194	303
	J530 POI	J530 POI	345	635	MEC	10,293	10,542	249	10,742	449	12,902	13,146	244	13,354	452
635700	SYCAMOR3	Sycamore	345	635	MEC	19,005	19,070	65	19,122	117	20,106	20,193	87	20,265	159
635690	GDMEC	GDMEC	345	635	MEC	18,566	18,638	72	18,697	131	18,624	18,711	87	18,783	159
344000	7ADAIR	Adair/Zachary 345	345	356	AMMO	8,920	8,934	15	8,946	26	12,465	12,494	29	12,517	52
631115	OTTUMWA5	Ottumwa	161	627	ITCM	27,848	27,929	81	27,993	145	23,967	24,035	68	24,091	124
636400	HILLS 3	Hills	345	635	MEC	16,822	16,902	80	16,966	144	19,937	20,054	117	20,151	214
		J530 Collector Bus	34.5			30,409	30,480	71	30,537	128	26,522	26,574	52	26,617	95
635600	GRIMES 3	Grimes	345	635	MEC	18,245	18,292	47	18,330	85	20,460	20,535	75	20,597	137
635701	SYCAMOR5	Sycamore	161	635	MEC	33,003	33,067	64	33,119	116	32,925	33,003	78	33,068	143
635650	SE POLK3	SE Polk	345	635	MEC	17,992	18,057	65	18,110	118	18,356	18,438	82	18,505	149
345435	7MAYWOOD	Maywood	345	356	AMMO	8,222	8,226	4	8,228	6	11,420	11,429	9	11,436	16
344006	5ADAIR3	Adair/Zachary 161	161	356	AMMO	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
631110	WAPELLO5	Wapello	161	627	ITCM	10,727	10,735	8	10,741	14	11,096	11,108	12	11,117	21
631118	LUCAS 5	Lucas Co	161	627	ITCM	3,221	3,222	1	3,222	1	3,339	3,340	1	3,341	2
631116	BRDGPRT5	Bridgeport	161	627	ITCM	15,772	15,794	22	15,811	39	17,111	17,144	33	17,170	59
631134	TRICNTY5	Tri County	161	627	ITCM	15,937	15,960	23	15,978	41	17,358	17,391	33	17,418	60
635880	PLSNT CRNR 5	Pleasant Corner	161	635	MEC	9,837	9,846	9	9,853	15	12,403	12,420	17	12,434	31
636420	TIFFIN 3	Tiffin	345	635	MEC	11,251	11,275	24	11,295	44	14,815	14,864	49	14,905	90
636620	WALCOTT3	Walcott	345	635	MEC	11,076	11,083	7	11,088	12	15,648	15,667	19	15,682	34
636640	LOUISA 3	Louisa	345	635	MEC	19,061	19,085	24	19,105	44	19,077	19,111	34	19,139	62
636645	SUB T 3	Sub T Haskins	345	635	MEC	11,232	11,252	20	11,269	37	15,712	15,759	47	15,797	85
636401	HILLS 5	Hills	161	635	MEC	19,245	19,278	33	19,304	59	19,685	19,723	38	19,755	70
631034	ROSEHLW5	Rose Hollow	161	627	ITCM	5,802	5,803	2	5,804	3	6,551	6,554	3	6,556	5

J485 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J485 generating facility on the adequacy of existing circuit breakers and related equipment in the Project area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the project area was provided by MISO. This starting short circuit model is listed below:

- Short Circuit Model: "RPUcApe2014Base_2016.03.01.dxt"

The J485 short circuit study model was developed as follows:

- The J485 gas generating facility with its sequence parameter was added at the West Side 161 kV bus via a 13.8 kV/161 kV transformer ($Z=7.5\%$ at 39 MVA, $X/R=24$, D-YG)
- The J485 gas project was modeled at the West Side 13.8 kV bus with MVA base of 58.38 MVA. All impedance values are set using saturated reactance data.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J485 project) and benchmark case (without the J485 project). The following fault simulation options were used in the short circuit analysis.

- The prefault voltage was set to "from a linear network solution"

Three-phase-to-ground (3PH) and single-phase-to-ground (SLG) faults were simulated on buses within the Study Project area. The results of short circuit currents with J485 (study case) and without J485 (benchmark case) are listed in Table 1. With the addition of the J485 generating facility, the maximum increase in the 3PH short circuit current is 3.89% (643.1 amps), and the maximum increase in the SLG short circuit current is 13.91% (1632.1 amps), which are both observed at the West Side 161 kV substation bus.

1.4 Summary of J485 Short Circuit Analysis

The study results show that the 3PH fault current is 17159 kA (increased by 643.1 A) and SLG fault current is 13367 A (increased by 1632.1 A) at the West Side 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J485 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J485

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J485 Project		Study Case with J485 Project		Amps Change with J485 vs Benchmark		% Change with J485 vs Benchmark	
						3PH Fault SC Amps	SLG Fault SC Amps	3PH Fault SC Amps	SLG Fault SC Amps	3PH	SLG	3PH	SLG
16	WS 161	161	WESTSIDE	1	1	16515.8	11734.8	17158.9	13366.9	643.1	1632.1	3.89%	13.91%
98	Q9 T	161	IBM TAP	1	1	17794.7	13096.4	18337.7	14273.9	543.0	1177.5	3.05%	8.99%
61906	ML 161 TAP	161	MAPLE LE	1	1	16233.6	11439.9	16695.2	12412.6	461.6	972.7	2.84%	8.50%
61907	ML 161	161	MAPLE LE	1	1	16233.6	11439.8	16695.1	12412.6	461.5	972.8	2.84%	8.50%
110	IBM 161	161	IBM STAT	1	1	17536.3	12929.2	17993.9	13817.2	457.6	888.0	2.61%	6.87%
13	NH 161	161	NORTHERN	1	1	16699.4	12107.8	17112.2	12889.9	412.8	782.1	2.47%	6.46%
7	CC 161	161	CASCADE	1	1	18754.4	14433.5	19157.9	15116.2	403.5	682.7	2.15%	4.73%
14	CRT 161	161	CROSSTOW	1	1	18179.8	13912.0	18540.8	14496.0	361.0	584.0	1.99%	4.20%
5	SLS 161	161	SILVER L	1	1	18040.5	13804.6	18390.4	14365.0	349.9	560.4	1.94%	4.06%
61948	BYR 161	161	BYRON	1	1	20626.7	16352.2	21002.8	16955.3	376.1	603.1	1.82%	3.69%
8	ZR 161	161	ZUMBRO R	1	1	13178.6	8931.1	13394.3	9261.4	215.7	330.3	1.64%	3.70%
15	CHR 161	161	CHESTER	1	1	18106.0	13270.1	18385.3	13647.1	279.3	377.0	1.54%	2.84%
69547	ROCH 161	161	ROCHESTE	1	1	18074.4	13239.1	18351.8	13612.9	277.4	373.8	1.53%	2.82%
9	BV 161	161	BAMBER V	1	1	13681.0	9389.3	13870.4	9642.4	189.4	253.1	1.38%	2.70%
6	WC 161	161	WILLOW C	1	1	13109.0	8984.5	13272.7	9202.0	163.7	217.5	1.25%	2.42%
100	NR 161	161	NORTH RO	1	1	18282.8	17302.0	18407.0	17482.5	124.2	180.5	0.68%	1.04%
61950	BYR 345	345	BYRON	1	1	16573.2	12891.0	16675.2	13082.7	102.0	191.7	0.62%	1.49%
103	NR 345	345	NORTH RO	1	1	19418.7	15499.6	19513.8	15647.9	95.1	148.3	0.49%	0.96%
63032	PV 345	345	PLEASANT	1	1	16912.7	15416.4	16982.7	15480.9	70.0	64.5	0.41%	0.42%
63033	PV 161	161	PLEASANT	1	1	19807.0	4795.5	19875.0	4831.7	68.0	36.2	0.34%	0.75%
69549	WABACO 161	161	WABACO	1	1	10457.2	7464.1	10484.0	7490.9	26.8	26.8	0.26%	0.36%
60102	ADAMS 345	345	ADAMS	1	1	13874.1	11875.5	13900.4	11890.9	26.3	15.4	0.19%	0.13%

A short circuit analysis was performed to determine the impact of the proposed generation on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Otter Tail Power Company's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J488/J489 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Otter Tail equipment in the area are not exceeded.

With the proposed J488/J489 project additions, the fault currents are estimated to be roughly 8.6 kA at the Twin Brooks 345 kV bus, 11.2 kA at the Big Stone South 345 kV bus, 19.7 kA at the Big Stone South 230 kV bus, and 33.6 kA at the Big Stone South 34.5 kV bus. There does not appear to be any short circuit related upgrades needed for J488/J489.

A general protective relay coordination review will need to be performed by OTP when the projects approach completion and before energizing the generation source.

J490 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J490 generating facility (24 Siemens 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the MDU system and its neighboring systems was provided by the MDU. This starting short circuit model is listed below:

- Short Circuit Model: "MDU model 032917.olr"

The J490 short circuit study model was developed as follows:

- Turned on synchronous generators which were nearby the POI but were off line in the starting short circuit model (Table 1-1).

Table 1-1: Synchronous Generators Which Were Turn On

ASPEN Bus Name	kV	Starting Case Status	J490 Case Status
COY MDU9	24.kV	Offline	Online
WILLCT 9	12.5kV	Offline	Online
WILL DT9	12.5kV	Offline	Online
MILCYCT9	13.8kV	Offline	Online
TESORO G4	4.16kV	Offline	Online
ST.JOE GEN	0.48kV	Offline	Online
HESKET 3G	0.48kV	Offline	Online
PARSHAL6	69.kV	Offline	Online
SPROLE Gen	0.48kV	Offline	Online
Ellen WF gen	0.675kV	Offline	Online
Bowdle WF G	0.675kV	Offline	Online
Ell WF singl	0.675kV	Offline	Online
Bow WF singl	0.675kV	Offline	Online
GascWF singl	0.675kV	Offline	Online
GascWF gen	0.675kV	Offline	Online
EDG-WF singl	0.675kV	Offline	Online
DICK DSU GEN	0.48kV	Offline	Online

ASPEN Bus Name	kV	Starting Case Status	J490 Case Status
LINTON GEN 1	13.8kV	Offline	Online
LINTON GEN 2	13.8kV	Offline	Online
LINTON GEN 3	13.8kV	Offline	Online
CAPITOL G2	4.16kV	Offline	Online
CAPITOL G3	4.16kV	Offline	Online
St A-Hosp	12.5kV	Offline	Online
CAPITOL G4	4.16kV	Offline	Online
CAPITOL G1	4.16kV	Offline	Online
BisWP Gen	4.16kV	Offline	Online
WATFORD-S	4.16kV	Offline	Online
RAY D	4.16kV	Offline	Online

- The J490 generating facility with its sequence parameters was added at the POI on McIntosh 69 kV Substation. The J490 generator was modeled as a generator with impedances listed below.
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): j0.6973 pu
 - o Negative sequence impedance (Z_2): j0.6973 pu (same as positive sequence impedances)
 - o Zero sequence impedance (Z_0): j9999 pu (zero sequence currents are blocked by the GSU transformers (Wye-Delta connection, therefore, this value does not have any impact on the short circuit results)

1.3 Short Circuit Analysis

The following fault simulation options were used in the short circuit analysis, as suggested in the ASPEN OneLiner documentation:

- The prefault voltage was set to “from a linear network solution”
- Current limited generator was set to “Ignore current limits”
- All loads, shunts and transformer magnetizing branches were ignored.

Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated on buses within the Study Project area. The results of short circuit currents with J490 (study case) and without J490 (benchmark case) are listed in the Table 1. With the addition of the J490 generating facility, the maximum increase in the 3PH short circuit current is 35.28% (357.5 A) at the MCLGHLN5 115 kV bus, and the maximum increase in the SLG short circuit current is 6.77% (60.5 amps) at the MCLGHLN5 115 kV bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 1370.8 A (increased by 357.5 A) and SLG fault current is 954.6 A (increased by 60.5 A) at the MCLGHLN5 115 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J490 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J490

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J490 Project		Study Case with J490 Project		Amps Change with J490 vs Benchmark		% Change with J490 vs Benchmark	
						3PH Fault Short Circuit Amps	SLG Fault Short Circuit Amps	3PH Fault Short Circuit Amps	SLG Fault Short Circuit Amps	3PH	SLG	3PH	SLG
0	J490 G	0.69	MCINTSH5	3	1	0	0	182035.3	180122.9	182035.3	180122.9	N/A	N/A
0	J490 COL1	34.5	MCINTSH5	3	1	0	0	3779.1	2338.7	3779.1	2338.7	N/A	N/A
0	J490 COL2	34.5	MCINTSH5	3	1	0	0	3763.9	2307.9	3763.9	2307.9	N/A	N/A
0	J490	115	MCINTSH5	3	1	0	0	1351.9	922.6	1351.9	922.6	N/A	N/A
1286	MCINTSH5	115	MCINTSH5	3	1	1013.3	894.1	1370.8	954.6	357.5	60.5	35.28%	6.77%
1288	MCLGHLN5	115	MCLGHLN5	51	1	1501.8	1733.6	1823.5	1845.8	321.7	112.2	21.42%	6.47%
0	WWBISON5	115	McLaugh	1	1	1358.3	1438	1615.3	1512.7	257	74.7	18.92%	5.19%
509	MCINTSH8	41.6	MCINTSH8	1	1	1091.1	1252.9	1203.7	1289.2	112.6	36.3	10.32%	2.90%
0	MCINTSH-8	41.6	MCINTSH-	3	1	1091.2	1253	1203.8	1289.3	112.6	36.3	10.32%	2.90%
1220	MOBRIDG5	115	MOBRIDG5	7	1	2651.8	2555	2922.8	2623.7	271	68.7	10.22%	2.69%
0	MCLAUGHLT	13.8	0	1	1	7440.5	0	8195.8	0	755.3	0	10.15%	N/A
2968	MCINTOH8	41.6	MCINTOH8	3	1	1052.5	1163.6	1156.9	1194.1	104.4	30.5	9.92%	2.62%
0	MCINTOSH RT	41.6	MCINTOSH	52	1	1012.8	1073	1109.2	1098.2	96.4	25.2	9.52%	2.35%
3646	MCLAGHN6	69	MCLAGHN6	51	1	1489.2	1909.8	1630.5	1973.1	141.3	63.3	9.49%	3.31%
0	WAKPALA RTP	69	WAKPALA	1	1	1313.8	1435.4	1422.1	1468.7	108.3	33.3	8.24%	2.32%
0	HL412ST217r+	41.6	HL412ST2	1	1	876.7	820.1	947.4	832.8	70.7	12.7	8.06%	1.55%
0	L EAGLE RTP	69	L EAGLE	1	1	1200.4	1197.2	1289.7	1218.9	89.3	21.7	7.44%	1.81%
0	SW5009	41.6	SW5009	1	1	782.9	678.6	838.2	686.1	55.3	7.5	7.06%	1.11%
2815	WATAUGA8	41.6	WATAUGA8	3	1	782.3	677.8	837.5	685.2	55.2	7.4	7.06%	1.09%
2814	HL412ST108r+	41.6	HL412ST1	1	1	729.9	607.8	777.3	613.2	47.4	5.4	6.49%	0.89%
1212	GLENHAM5	115	GLENHAM5	7	1	4110	4900.7	4362.6	5004.9	252.6	104.2	6.15%	2.13%
0	GLENHAM-5	115	GLENHAM-	1	1	4112.9	4904.6	4365.4	5008.8	252.5	104.2	6.14%	2.12%
3651	SIOUX 6	69	SIOUX 6	51	1	1115.9	1045.4	1180.8	1058.5	64.9	13.1	5.82%	1.25%
3694	SELFDRG6	69	SELFDRG6	1	1	1115.8	1045.3	1180.6	1058.3	64.8	13	5.81%	1.24%
3658	TRAILCY6	69	TRAILCY6	52	1	905.6	737.5	954.2	743.8	48.6	6.3	5.37%	0.85%
2813	SW5036/5035	41.6	SW5036/5	1	1	625.3	483.8	658.8	486.1	33.5	2.3	5.36%	0.48%
0	MORRISTOWN	41.6	MORRISTO	3	1	623	481.3	656.2	483.5	33.2	2.2	5.33%	0.46%
0	MORRISTWN RT	41.6	MORRISTW	52	1	623	481.3	656.2	483.5	33.2	2.2	5.33%	0.46%
0	WAKPALA R	69	WAKPALA	1	1	868.1	715.2	912.7	721.1	44.6	5.9	5.14%	0.82%
0	TRAIL CITY R	69	TRAIL CI	1	1	846.9	674.7	889	679.6	42.1	4.9	4.97%	0.73%
2809	KLDRN R8	41.6	KLDRN R8	52	1	571.2	428.1	598.5	429.4	27.3	1.3	4.78%	0.30%
3633	FT YATE6	69	FT YATE6	1	1	995.4	896.6	1041.2	904.4	45.8	7.8	4.60%	0.87%
2969	KELDRON8	41.6	KELDRON8	3	1	548.5	405.9	573.5	406.8	25	0.9	4.56%	0.22%
3657	TIMBRK6	69	TIMBRK6	1	1	726.4	532.8	756.4	535.1	30	2.3	4.13%	0.43%
2808	THNDRHK8	41.6	THNDRHK8	3	1	488.4	349.5	507.5	349.7	19.1	0.2	3.91%	0.06%
0	FYatesCASINO	69		1	1	879.1	805.9	909	810.6	29.9	4.7	3.40%	0.58%
0	BOWDLE WF TP	115		7	1	2435.5	2319.5	2514.8	2338.2	79.3	18.7	3.26%	0.81%
0	GLENHAM -8	41.6	GLENHAM	1	1	5019.7	1138.6	5161.9	1138.1	142.2	-0.5	2.83%	-0.04%
505	GLENHAM8	41.6	GLENHAM8	7	1	5018	1138.5	5160.1	1138	142.1	-0.5	2.83%	-0.04%
0	GLENHAM Z	41.6	GLENHAM	1	1	5016.3	1138.5	5158.3	1138	142	-0.5	2.83%	-0.04%
0	GLENHAM 8	41.6	GLENHAM8	1	1	5016.3	1138.4	5158.3	1137.9	142	-0.5	2.83%	-0.04%
3678	CANNBALL	69	CANNBALL	1	1	824.9	794	848.1	797.9	23.2	3.9	2.81%	0.49%
0	GLENHAM-4	230	GLENHAM-	1	1	4318.1	4158.7	4432.6	4189.1	114.5	30.4	2.65%	0.73%
1526	GLENHAM4	230	GLENHAM4	7	1	4324.8	4163.5	4439.2	4193.9	114.4	30.4	2.65%	0.73%
2810	SHAMRCK8	41.6	SHAMRCK8	63	1	4313.1	1065.6	4415.7	1064.9	102.6	-0.7	2.38%	-0.07%
0	MOBRIDGE	12.5	MOBRIDGE	7	1	6843.4	7545.5	7004.4	7588.7	161	43.2	2.35%	0.57%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J490

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J490 Project		Study Case with J490 Project		Amps Change with J490 vs Benchmark		% Change with J490 vs Benchmark	
						3PH Fault Short Circuit Amps	SLG Fault Short Circuit Amps	3PH Fault Short Circuit Amps	SLG Fault Short Circuit Amps	3PH	SLG	3PH	SLG
0	MOBRIDGE D10	12.5		1	1	6833.8	7533.9	6994.4	7577	160.6	43.1	2.35%	0.57%
0	MOBRIDGE D20	12.5		1	1	6833.8	7533.9	6994.4	7577	160.6	43.1	2.35%	0.57%
0	MOBRIDGE D30	12.5		1	1	6833.8	7533.9	6994.4	7577	160.6	43.1	2.35%	0.57%
3652	SOLEN 6	69	SOLEN 6	1	1	777	824	794.7	827.7	17.7	3.7	2.28%	0.45%
0	NB St Anthy	69	St Anthy	1	1	769.2	836.5	786	840.1	16.8	3.6	2.18%	0.43%
1203	BOWDLE 5	115	BOWDLE 5	7	1	1750.4	1769.9	1787.5	1779.2	37.1	9.3	2.12%	0.53%



Interconnection Facility Study Report

Generation Interconnection
MISO Projects J493 and J510
Prepared for Midcontinent ISO

April 25, 2017

This report contains CEII and Trade Secret information and should not be released without consent from Otter Tail Power Company

2. Short Circuit Study

A short circuit analysis was performed to determine the impact of the proposed generation projects on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Transmission Owner's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J493 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded.

With the proposed J493 project additions, the fault currents are estimated to be roughly 16 kA at the Big Stone South 230 kV bus and the faults on the 345 kV bus were below these values. There do not appear to be any short circuit related upgrades needed for J493.

With the proposed J493 and J510 project additions, the fault currents are estimated to be roughly 16.6 kA at the Big Stone South 230 kV bus. There do not appear to be any short circuit related upgrades needed for J493 and J510.

A general protective relay coordination review will need to be performed by Transmission Owner when the projects approach completion and before energizing the generation source.

This report contains CEII and Trade Secret information and should not be released without consent from Otter Tail Power Company



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J495

7-17-2017

Summary

Generator Interconnection Project J495 is a 200 MW wind powered generating facility with proposed interconnection to the 345 kV bus at a new switch station tapping ITC Midwest's Ledyard to Colby 345 kV line. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- Two 100 MW equivalent generators
- Two 34.5kV to 690V equivalent transformers
- Two 34.5kV to 345 kV step up transformers
- 1.3 mile 345 kV interconnection line

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

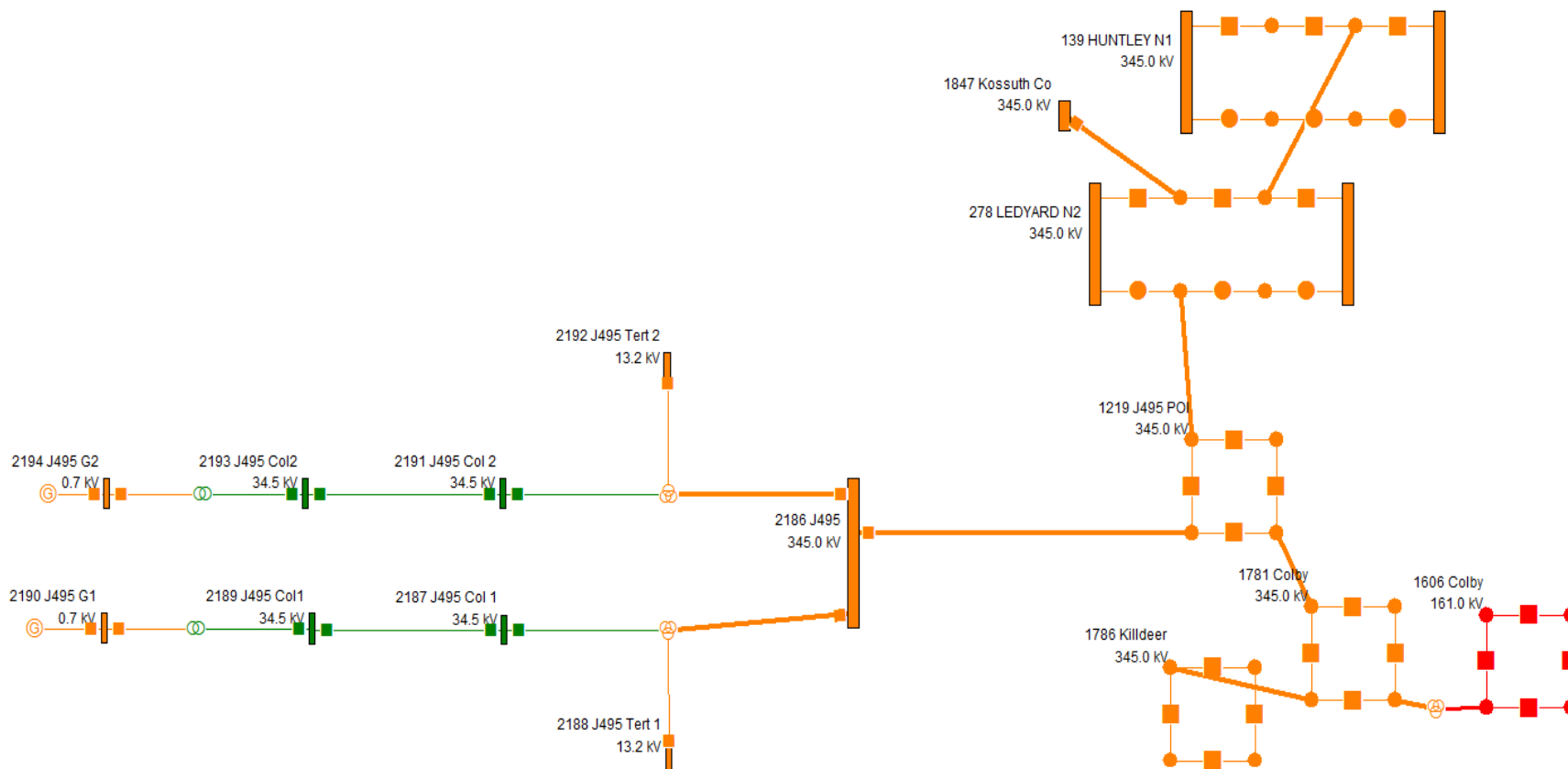


Figure1



Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J495.

J495	Ledyard 345	Colby 345	Colby 161	Huntley 345	Killdeer 345	Kossuth
SLG-pre (A)	9121.93	9313.45	20243.8	8496.7	9803.83	10370
3Ph-pre (A)	13062	11044.5	22145.6	11436.2	10822.5	14788.5
SLG-post (A)	10356.4	9530.15	20410.2	8795.42	9901.52	10673.7
3Ph-post (A)	13094.9	11059	22153.7	11449.9	10833	14804.2
Contribution SLG (A)	1234.47	216.7	166.4	298.72	97.69	303.7
Contribution 3Ph (A)	32.9	14.5	8.1	13.7	10.5	15.7

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J495. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. New circuit breakers at the POI switch station were assumed to have 50kA interrupting capability.

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J498 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J498, a proposed 340 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J498 was assumed to interconnect off the Lehigh-Grimes 345 kV line. MidAmerican is the operating agent of this jointly-owned line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 11,055 Amps without the Interconnection Customer included and 12,498 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J498

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12083	J498_TAP	J498 POI	345	MEC	7,799	9,576	1,776	11,055	12,498	1,443
11798	J498_COLHI	J498 Collector Bus	34.5		NA	31,416	NA	NA	28,289	NA
636010	LEHIGH 3	Lehigh	345	MEC	9,962	10,784	822	12,680	13,501	821
635600	GRIMES 3	Grimes	345	MEC	18,987	19,347	360	21,429	21,860	431
636000	WEBSTER3	Webster	345	MEC	10,304	10,683	379	12,492	12,946	454
6201	J411 POI	Ida Co	345	MEC	7,133	7,197	65	9,527	9,621	94
635700	SYCAMOR3	Sycamore	345	MEC	19,630	19,888	259	20,933	21,270	337
12084	J499_TAP	J499 POI	345	MEC	9,223	9,253	31	11,409	11,462	53
635630	BOONVIL3	Booneville	345	MEC	15,390	15,501	111	18,010	18,186	176
635601	GRIMES 5	Grimes	161	MEC	23,008	23,148	140	24,022	24,182	160
635369	KOSSUTH 3	Kossuth Co	345	MEC	12,312	12,389	77	14,979	15,093	113
636001	WEBSTER5	Webster	161	MEC	17,538	17,777	239	18,015	18,222	207
635200	RAUN 3	Raun	345	MEC	27,097	27,157	60	25,447	25,509	61
635680	BONDRNT3	Bondurant	345	MEC	17,504	17,629	125	19,708	19,907	200
635701	SYCAMOR5	Sycamore	161	MEC	33,629	33,864	235	33,694	33,974	281
635590	FALLOW 3	Fallow Ave	345	MEC	8,708	8,725	17	10,215	10,244	29
12085	J500_TAP	J500 POI	345	MEC	13,423	13,454	31	17,020	17,083	63
635635	MADISON3	Madison Co	345	MEC	13,218	13,279	61	16,259	16,374	115
635631	BOONVIL5	Booneville	161	MEC	18,692	18,754	63	21,728	21,825	97
635602	142ST 5	142nd St	161	MEC	17,682	17,762	80	21,324	21,447	123
635614	100 54 5	100th St & 54th Ave	161	MEC	20,682	20,778	97	25,122	25,280	158
635609	141 44 5	Hwys 141 & 44	161	MEC	15,237	15,286	49	19,093	19,182	89

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J499 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J499, a proposed 340 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J499 was assumed to interconnect off the Fallow-Grimes 345 kV line. MidAmerican is the operating agent for this jointly-owned line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,017 Amps without the Interconnection Customer included and 11,462 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J499

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12084	J499_TAP	J499 POI	345	MEC	7,553	9,253	1,700	10,017	11,462	1,445
11802	J499_COLHI	J499 Collector Bus	34.5		NA	31,310	NA	NA	28,180	NA
635590	FALLOW 3	Fallow Ave	345	MEC	7,984	8,725	741	9,476	10,244	768
635600	GRIMES 3	Grimes	345	MEC	18,785	19,347	562	21,164	21,860	695
635589	ECLIPSE3	Eclipse	345	MEC	6,642	7,086	443	7,914	8,428	514
635014	POTTCO 3	Overland Trail	345	MEC	24,626	24,800	173	24,686	24,909	222
635700	SYCAMOR3	Sycamore	345	MEC	19,481	19,888	408	20,727	21,270	543
12083	J498_TAP	J498 POI	345	MEC	9,543	9,576	33	12,434	12,498	64
635630	BOONVIL3	Booneville	345	MEC	15,317	15,501	185	17,889	18,186	297
635601	GRIMES 5	Grimes	161	MEC	22,931	23,148	217	23,928	24,182	255
635000	CBLUFFS3	Council Bluffs	345	MEC	31,629	31,829	199	28,966	29,192	226
635015	POTTCO 5	Overland Trail	161	MEC	25,444	25,496	52	26,439	26,516	77
635680	BONDRNT3	Bondurant	345	MEC	17,430	17,629	199	19,583	19,907	324
635701	SYCAMOR5	Sycamore	161	MEC	33,503	33,864	361	33,534	33,974	440
636010	LEHIGH 3	Lehigh	345	MEC	10,764	10,784	20	13,464	13,501	37
12085	J500_TAP	J500 POI	345	MEC	13,393	13,454	61	16,957	17,083	126
635635	MADISON3	Madison Co	345	MEC	13,176	13,279	102	16,177	16,374	197
635631	BOONVIL5	Booneville	161	MEC	18,650	18,754	105	21,664	21,825	161
635602	142ST 5	142nd St	161	MEC	17,637	17,762	124	21,252	21,447	196
635614	100 54 5	100th St & 54th Ave	161	MEC	20,629	20,778	149	25,031	25,280	249
635609	141 44 5	Hwys 141 & 44	161	MEC	15,211	15,286	76	19,043	19,182	139

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J500 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J500, a proposed 500 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J500 was assumed to interconnect off the Booneville-Atchison County and Rolling Hills-Madison County 345 kV lines. MidAmerican is the sole owner or operating agent for these line facilities. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 14,936 Amps without the Interconnection Customer included and 17,083 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J500

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12085	J500_TAP	J500 POI	345	MEC	10,841	13,454	2,613	14,936	17,083	2,147
11806	J500_COLHI	J500 Collector Bus	34.5		NA	31,689	NA	NA	28,412	NA
635100	RLHILLS3	Rolling Hills	345	MEC	9,295	10,160	865	12,196	12,955	759
635017	ATCHSN 3	Atchison Co	345	MEC	11,181	11,313	132	13,827	14,022	195
635630	BOONVIL3	Booneville	345	MEC	14,857	15,501	645	17,368	18,186	818
635635	MADISON3	Madison Co	345	MEC	12,711	13,279	568	15,633	16,374	741
635013	PNYCRK 3	Pony Creek	345	MEC	27,296	27,490	194	26,333	26,579	246
640139	COOPER 3	Cooper	345	NPPD	22,334	22,450	116	21,758	21,912	154
635600	GRIMES 3	Grimes	345	MEC	18,989	19,347	358	21,312	21,860	548
635631	BOONVIL5	Booneville	161	MEC	18,471	18,754	284	21,493	21,825	332
635640	NORWLK3	Norwalk	345	MEC	13,834	14,137	304	15,932	16,365	433
635016	STHLND 3	Southland	345	MEC	27,266	27,459	193	26,317	26,562	245
635000	CBLUFFS3	Council Bluffs	345	MEC	31,608	31,829	221	28,936	29,192	256
645458	S3458 3	Sub 3458 (Neb City)	345	OPPD	17,438	17,478	40	18,418	18,483	65
640277	MOORE 3	Moore	345	NPPD	11,222	11,231	10	12,317	12,334	17
541199	ST JOE 3	St Joseph	345	GMO	12,113	12,120	8	14,541	14,557	16
300039	7FAIRPT	Fairport	345	AECI	8,838	8,844	6	10,509	10,522	13
640140	COOPER 5	Cooper	161	NPPD	11,210	11,222	12	9,931	9,945	13
635700	SYCAMOR3	Sycamore	345	MEC	19,566	19,888	323	20,790	21,270	480
12083	J498_TAP	J498 POI	345	MEC	9,552	9,576	24	12,447	12,498	51
12084	J499_TAP	J499 POI	345	MEC	9,211	9,253	43	11,377	11,462	85
635601	GRIMES 5	Grimes	161	MEC	22,956	23,148	192	23,917	24,182	266
635621	SUGRCRK5	Sugar Creek	161	MEC	14,764	14,883	119	19,689	19,914	225
635632	EARLHAM5	Earlham	161	MEC	6,936	6,958	22	8,482	8,516	35
635641	NORWLK5	Norwalk	161	MEC	21,027	21,255	228	22,740	23,041	301
635650	SE POLK3	SE Polk	345	MEC	18,175	18,463	288	18,592	18,985	393

CRITICAL ENERGY INFRASTRUCTURE INFORMATION NOTICE

The materials contained in this document include Critical Energy Infrastructure Information (CEII). All materials designated as CEII must be handled and protected per the requirements in FERC CEII Policy. There may be additional requirements for CEII materials in the future.

J501 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J501, a proposed 500 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J501 was assumed to interconnect at the Kossuth County 345 kV substation. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 12,944 Amps without the Interconnection Customer included and 15,093 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J501

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
635369	KOSSUTH 3	Kossuth Co	345	MEC	9,961	12,389	2,429	12,944	15,093	2,149
		J501 Collector Bus	34.5		NA	31,524	NA	NA	28,295	NA
631197	LEDYARD3	Ledyard	345	ITCM	8,782	9,556	774	12,083	13,059	976
12110	J529_TAP	J529 POI	345	MEC	8,749	9,523	774	10,666	11,528	862
636000	WEBSTER3	Webster	345	MEC	10,336	10,683	347	12,530	12,946	416
631193	HUNTLEY3	Huntley	345	ITCM	7,990	8,288	298	10,627	11,079	453
12078	J495_TAP	J495 POI	345	ITCM	NA	NA	NA	NA	NA	NA
12099	J455_TAP	J455 POI	345	MEC	9,087	9,423	336	10,735	11,170	436
636010	LEHIGH 3	Lehigh	345	MEC	10,607	10,784	177	13,428	13,501	73
636001	WEBSTER5	Webster	161	MEC	17,546	17,777	231	18,023	18,222	199
631138	LAKEFLD3	Lakefield	345	ITCM	15,076	15,295	220	16,507	16,816	309
631045	HUNTLEY5	Huntley	161	ITCM	NA	NA	NA	NA	NA	NA
631198	COLBY3	Colby	345	ITCM	8,117	8,218	101	9,606	9,760	154
635368	OBRIEN 3	O'brien Co	345	MEC	11,692	11,983	291	14,112	14,534	422
6201	J411 POI	Ida Co	345	MEC	7,173	7,197	24	9,580	9,621	42
12083	J498_TAP	J498 POI	345	MEC	9,505	9,576	71	12,379	12,498	119
656353	HOPE5	Hope	161	CBPC	6,875	6,946	71	8,821	8,859	37
636050	WRIGHT 5	Wright	161	MEC	8,375	8,402	27	9,940	9,976	37
636020	FT.DODG5	Sub T Fort Dodge	161	MEC	9,605	9,642	37	11,003	11,050	47
636025	HAYES 5	Hayes	161	MEC	9,181	9,214	33	11,257	11,305	48
656369	WEBSTER8	Webster	69	CBPC	9,874	9,896	22	9,522	9,542	19



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J504

7-17-2017

Summary

Generator Interconnection Project J504 is a 50 MW solar powered generating facility with proposed interconnection to the 161 kV bus at ITC Midwest's Duane Arnold Energy Center (DAEC) Substation. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- A 50 MW equivalent current source
- 34.5kV to 600V equivalent transformer
- 34.5kV to 161 kV step up transformer

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

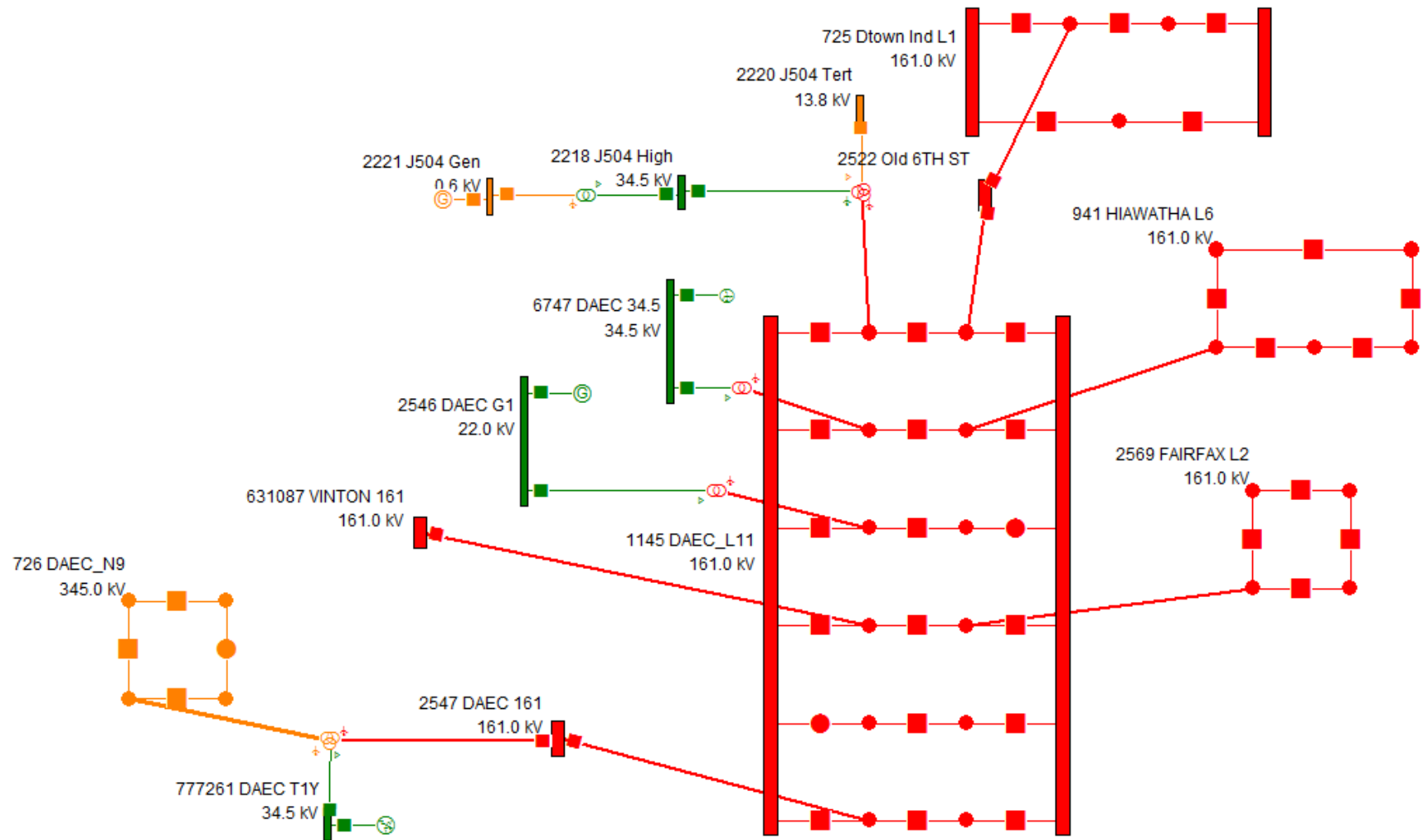


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J504.

J504	DAEC 161	DAEC 345	Hiawatha 161	Fairfax 161	DTI 161	Vinton 161
SLG-pre (A)	29067.2	11730.1	18425.9	11886.1	13286	8448.33
3Ph-pre (A)	25349.7	13413.1	14772.8	12818.1	14390.9	10401.1
SLG-post (A)	29391.4	11771.3	18450.6	11898.1	13301.5	8454.19
3Ph-post (A)	25461.9	13428.7	14795.8	12834.3	14410.7	10408.3
Contribution SLG (A)	324.2	41.2	24.7	12	15.5	5.86
Contribution 3Ph (A)	112.2	15.6	23	16.2	19.8	7.2

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J504. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. There are several existing breakers at DAEC 161 kV substation that are expected to be replaced as part of planned work at this location, as well as new breakers being installed with the bus expansion. 40kA interrupting capability was assumed for the new and replaced breakers.

J506 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J506 generating facility (100 Vestas V110 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J506 short circuit study model was developed as follows:

- Added J524, J527, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J506 generating facility with its sequence parameters at approximately 35 miles from Raun on Raun – Lakefield Jct 345 kV line. The associated 345 kV interconnection generation tie line (3 miles), two 345-34.5-13.2 kV transformers ($Z_+=9\%$, $Z_0=8.01\%$ at 75 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z_+=9.79\%$, $Z_0=9.07\%$ at 105 MVA, D-YG) were also added.
- The J506 generation was represented using the conventional generator model with the impedance of the machines set as follows per the data provided in the Vestas manual.
 - Positive saturated sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): j0.1669 p.u.
 - Negative saturated sequence impedance (Z_2): j0.1669 p.u. (same as positive sequence impedances)
 - Zero sequence saturated impedance (Z_0): j0.1669 p.u.

The J506 short circuit benchmark model was developed as follows:

- Turned off J506 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J506 project) and benchmark case (without the J506 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J506 (study case) and without J506 (benchmark case) are listed in Table 1. With the addition of the J506 generating facility, the maximum increase in the 3PH short circuit current is 7.07% (806.3 A), and the maximum increase in the SLG short circuit current is 20.57% (1698.8 A), which are both observed at the J506 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 12205.7 A (increased by 806.3 A) and SLG fault current is 9955.5 A (increased by 1698.8 A) at the J506 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J506 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J506

Substation	Faulted Bus			Benchmark Case without J506 Project		Study Case with J506 Project		Ampere Change with J506 vs Benchmark		% Change with J506 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J506	16030	J506-COL2	34.5	-	-	16639.9	16313.0	-	-	-	-
J506	16020	J506_COL2	34.5	-	-	23877.6	18046.6	-	-	-	-
J506	16011	J506_TR2_TER	13.2	-	-	0.0	31964.5	-	-	-	-
J506	15030	J506-COL1	34.5	-	-	16639.9	16313.0	-	-	-	-
J506	15020	J506_COL1	34.5	-	-	23877.6	18046.6	-	-	-	-
J506	15011	J506_TR1_TER	13.2	-	-	0.0	31964.5	-	-	-	-
J506	15010	J506	345	-	-	9166.4	11157.3	-	-	-	-
J506	65400	J506_POI	345	8256.7	11399.4	9955.5	12205.7	1698.8	806.3	20.57%	7.07%
SHD HIGHLAND WIND FA	14839	SHD GEN BUS1	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	10839	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	473	SHD GEN BUS3	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	472	SHD GEN BUS2	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	471	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	470	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	469	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	468	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
ROCK ISLAND CLEAN LI	10838	RICL-345	345	10711.0	13314.3	11019.8	13577.2	308.8	262.9	2.88%	1.97%
MVP SBC OBRIEN COUNT	500001	OBRIEN-345	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
SBC OBRIEN COLLECTOR	11830	COLECTOR-GEN	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
SBC OBRIEN COLLECTOR	10830	COLECTOR-345	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
RAU RAUN	10840	RAU-345-A	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	485	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	484	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	483	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	482	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	481	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	479	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	478	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	477	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	476	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	475	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	474	RAU-345-B	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
NLN NEAL NORTH	11542	NLN 3 345	345	26055.5	24691.2	26396.1	25004.5	340.6	313.3	1.31%	1.27%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J506

Substation	Faulted Bus			Benchmark Case without J506 Project		Study Case with J506 Project		Ampere Change with J506 vs Benchmark		% Change with J506 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
NL4 NEAL SOUTH	11207	NEAL 4 345	345	25102.2	23337.7	25384.8	23606.8	282.6	269.1	1.13%	1.15%
UPLAND PRARIE WIND F	500036	J455-345	345	9346.1	12092.4	9468.7	12228.8	122.6	136.4	1.31%	1.13%
RED ROCK NORTH STAR	500030	RED ROCK TAP	345	9632.6	11451.1	9689.0	11518.8	56.4	67.7	0.59%	0.59%



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J514

7-19-2017

Summary

Generator Interconnection Project J514 is an increase of 30MW summer and 60 MW winter to an existing generating facility interconnected to ITC Midwest's Marshalltown 161 kV substation. Three phase and single line to ground fault current values were evaluated and compared to the previous values from the original J233 interconnection study using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- Two 262 MVA generators (combustion turbines)
- One 306 MVA generator (steam turbine)
- Three 18kV to 161 kV step up transformers

Sutherland Steam units 1 and 3 have approved Attachment Y retirements, and were taken out of service in the analysis of J514 which changes the fault currents as well.

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

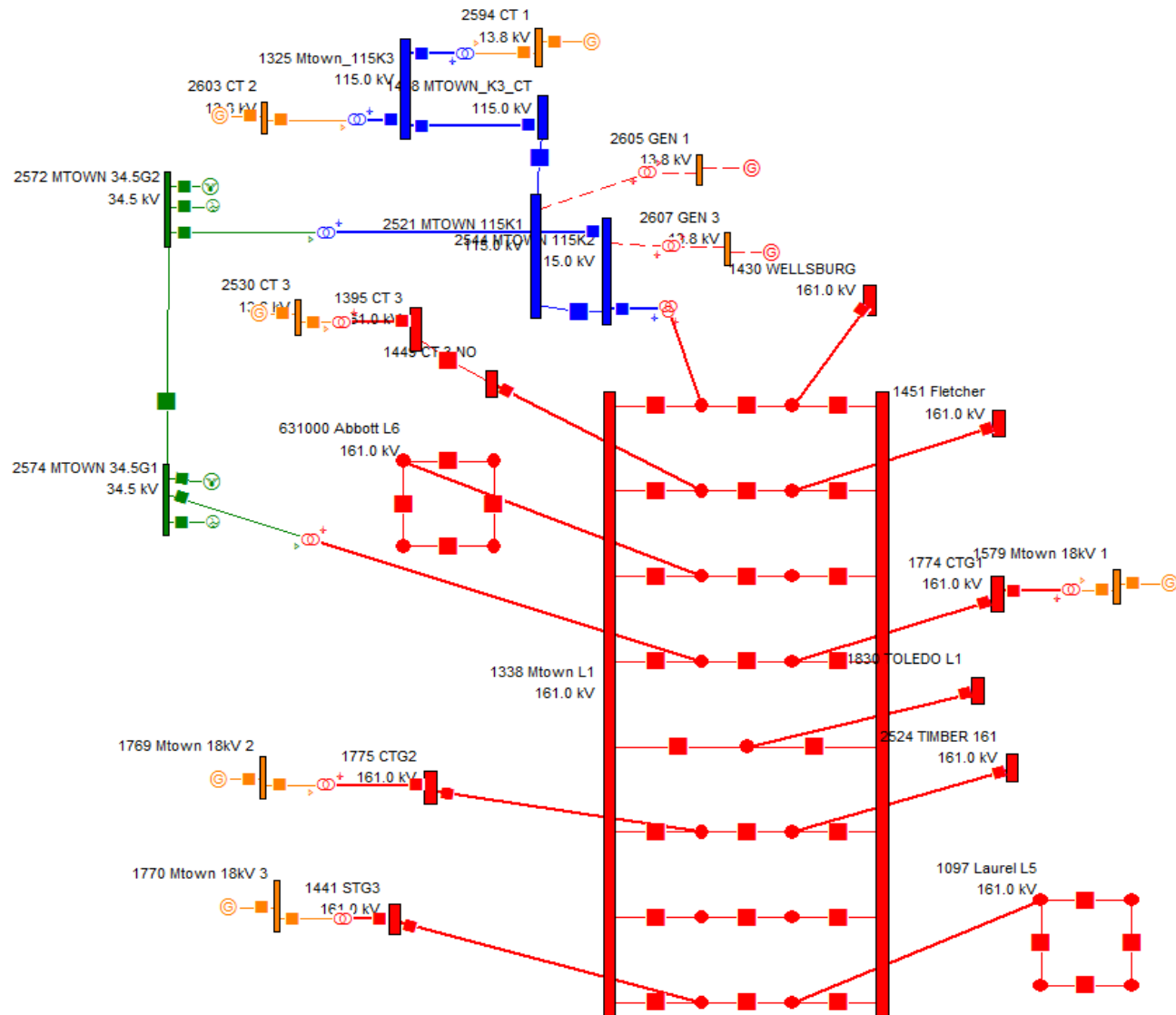


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents comparing the results from the J233 study and J514.

J514	Marshalltown 161	Marshalltown 115	Marshalltown 34.5	Wellsburg 161	Abbott 161	Toledo 161	Timber Creek 161	Laurel 161
SLG-J233 (A)	28504.7	19667.1	3372.16		7223.32		11473	9971.34
3Ph-J233 (A)	24795.4	16976.8	15773.3		10640.5		13556.6	12807.7
SLG-J514 (A)	28730.9	16514.7	3381.4	7462.9	9278.54	6641.68	12510.3	9955.3
3Ph-J514 (A)	24202.4	14536.9	15466.2	8312.51	10941.6	7754.12	13534.4	12145.9
Difference SLG (A)	226.2	-3152.4	9.24	N/A	2055.22	N/A	1037.3	-16.04
Difference 3Ph (A)	-593	-2439.9	-307.1	N/A	301.1	N/A	-22.2	-661.8

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J514. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. Circuit breaker 2680 at Marshalltown 115 kV was previously identified as a constraint for J233, but has since been removed.



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J523

7-17-2017

Summary

Generator Interconnection Project J523 is a 50 MW solar powered generating facility with proposed interconnection to the 161 kV bus at ITC Midwest's Adams Substation. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- A 50 MW equivalent current source
- 34.5kV to 600V equivalent transformer
- 34.5kV to 161 kV step up transformer
- 0.8 mile 161 kV interconnection line

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

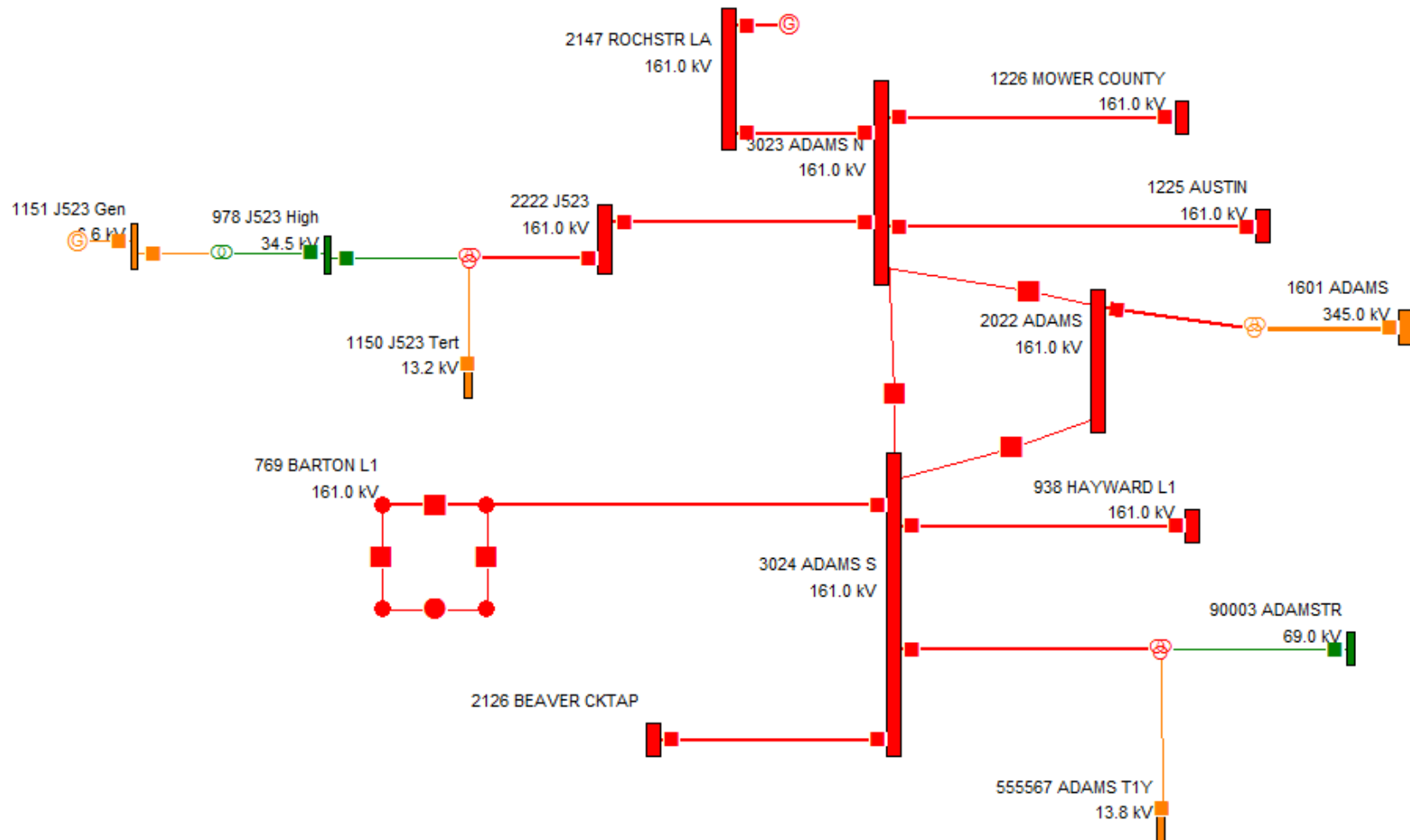


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J523.

J523	Adams 161	Adams 345	Adams 69	Mower Co 161	Rochester 161	Austin 161	Barton 161	Beaver Creek 161	Hayward 161
SLG-pre (A)	17100.7	9853.96	12922.1	8904.79	5240.59	10452.2	9788.32	6865.9	11568.2
3Ph-pre (A)	17353.5	10595.8	12845	9170.35	4482.69	12145.4	11117	8544.83	14138.6
SLG-post (A)	17790.5	9918.36	13011.4	8971.37	5244.7	10486.1	9803.9	6901.44	11584.3
3Ph-post (A)	17464	10607.2	12860.9	9197.78	4486.18	12161.8	11126	8559.99	14151.5
Contribution SLG (A)	689.8	64.4	89.3	66.58	4.11	33.9	15.58	35.54	16.1
Contribution 3Ph (A)	110.5	11.4	15.9	27.43	3.49	16.4	9	15.16	12.9

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J523. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

J524 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J524 Solar PV generating facility (40 SMA SC 2.5 MVA inverters) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J524 short circuit study model was developed as follows:

- Added J506, J527, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J524 generating facility with its sequence parameters at Webster 161 kV substation. The associated 161 kV interconnection generation tie line, 161-34.5-13.2 kV transformer ($Z_+=9.0\%$, $Z_0=8.0\%$ at 75 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.550 kV pad mount transformer ($Z_+=6\%$, $Z_0=5.4\%$ at 112 MVA, D-YG) were also added.
- Per CAPE's manual, solar generation project should be modeled the same way as Type 4 wind generation. The J524 solar project was modeled as Type 4 controlled power and current-limited generator. It is assumed that the project has the maximum short circuit contribution of 1.5 p.u. This corresponds to a maximum fault current limit of 598 amps at 161 kV side of the GSU transformer.

The J524 short circuit benchmark model was developed as follows:

- Turned off J524 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J524 project) and benchmark case (without the J524 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J524 (study case) and without J524 (benchmark case) are

listed in Table 1. With the addition of the J524 generating facility, the maximum increase in the 3PH short circuit current is 3.63% (1028.1 A) observed at Webster 13.8 kV bus. The maximum increase in the SLG short circuit current is 6.82% (1244.2 A) observed at the Webster 161 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 19095 A (increased by 580.4 A) and SLG fault current is 19487 A (increased by 1244.2 A) at the Webster 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J524 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J524	97000	J524_TR_TER	13.2	-	-	0.0	28842.8	-	-	-	-
J524	96998	J524-COL1	34.5	-	-	19577.9	14160.9	-	-	-	-
J524	96997	J524_COL1	34.5	-	-	20175.1	14679.4	-	-	-	-
J524	96996	J524	161	-	-	17620.2	17752.8	-	-	-	-
WEB WEBSTER	10663	WEB-13.8-9T1	13.8	0.0	28298.2	0.0	29326.3	0.0	1028.1	0.00%	3.63%
WEB WEBSTER	10652	WEB-161	161	18242.8	18515.0	19487.0	19095.4	1244.2	580.4	6.82%	3.13%
WEB WEBSTER	10642	WEB-69	69	9937.1	9570.4	10246.7	9837.1	309.6	266.7	3.12%	2.79%
WEB WEBSTER	10645	WEB-12.4-8T1	12.5	0.0	13579.2	0.0	13949.8	0.0	370.6	0.00%	2.73%
CBPC-HOPE	10600	CB HOPE 161	161	7023.7	8906.4	7234.1	9138.7	210.4	232.3	3.00%	2.61%
CBPC-VINCENT JCT	10362	VNCNT JCT 69	69	3659.4	5108.1	3754.8	5238.2	95.4	130.1	2.61%	2.55%
CBPC-SNELL	10364	SNELL 69	69	3485.2	4818.2	3573.7	4938.3	88.5	120.1	2.54%	2.49%
FDD SUB D FD	401100	FDD-13.8	13.8	2033.2	2833.5	2033.2	2903.9	0.0	70.4	0.00%	2.48%
FDT SUB T FD	4607	FDT-8T6-13.2	13.2	9677.2	9467.4	9917.5	9702.6	240.3	235.2	2.48%	2.48%
FDT SUB T FD	2101	FDT-8T2-13.2	13.2	8180.0	8272.4	8180.0	8477.9	0.0	205.5	0.00%	2.48%
FDT SUB T FD	1100	FDT-8T1-13.2	13.2	8026.7	8039.2	8026.7	8238.9	0.0	199.7	0.00%	2.48%
FDH HARVEST AVE	629	FDH-8T2-13.2	13.2	7765.5	7608.8	7765.5	7797.3	0.0	188.5	0.00%	2.48%
FDH HARVEST AVE	628	FDH-8T1-13.2	13.2	7765.5	7608.8	7765.5	7797.3	0.0	188.5	0.00%	2.48%
FTL TATE AND LYLE	6262	FTL-8T2-13.2	13.2	10892.3	10500.3	11162.1	10760.4	269.8	260.1	2.48%	2.48%
FTL TATE AND LYLE	6261	FTL-8T1-13.2	13.2	10892.3	10500.3	11162.1	10760.4	269.8	260.1	2.48%	2.48%
FDT SUB T FD	3610	FDT-8T5-13.8	13.8	0.0	7317.9	0.0	7498.9	0.0	181.0	0.00%	2.47%
FDT SUB T FD	610	FDT-69-BUS#1	69	6394.9	6300.0	6556.7	6455.8	161.8	155.8	2.53%	2.47%
FDQ SUB Q FD	2520	FDQ-6T2-13.2	13.2	9335.5	8543.8	9566.0	8754.8	230.5	211.0	2.47%	2.47%
FDQ SUB Q FD	620	FDQ-69-B2	69	5144.9	6305.1	5274.2	6460.8	129.3	155.7	2.51%	2.47%
FDQ SUB Q FD	448	FDQ-69-B1	69	5144.9	6305.1	5274.2	6460.8	129.3	155.7	2.51%	2.47%
FDQ SUB Q FD	1620	FDQ-6T1-13.2	13.2	9517.3	8695.5	9752.3	8910.2	235.0	214.7	2.47%	2.47%
FDQ SUB Q FD	2529	FDQ-FD-FOODS	13.2	2787.1	3775.1	2787.1	3868.3	0.0	93.2	0.00%	2.47%
FDN SUB N FD	402520	FDN-13.8	13.8	2540.2	3431.2	2540.2	3515.9	0.0	84.7	0.00%	2.47%
FHA HAYES	617	FHA-69	69	10652.1	9552.7	10920.9	9788.5	268.8	235.8	2.52%	2.47%
FHA HAYES	613	FHA-13.8-8T1	13.8	0.0	17504.2	0.0	17936.2	0.0	432.0	0.00%	2.47%
FHA HAYES	619	FHA-6T1-13.2	13.2	6516.1	6252.2	6516.1	6406.5	0.0	154.3	0.00%	2.47%
FDG SUB G FD	1618	FDG-6T1-13.2	13.2	5377.1	5145.8	5377.1	5272.7	0.0	126.9	0.00%	2.47%
FDB SUB B BARNUM FD	630	FDB-69	69	4444.3	5738.7	4554.9	5880.2	110.6	141.5	2.49%	2.47%
FDT SUB T FD	454	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
FDT SUB T FD	453	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	452	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	451	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	450	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	100	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDM SUB M FD	634	FDM-69	69	1923.9	3059.7	1971.5	3135.1	47.6	75.4	2.47%	2.46%
FDK SUB K FD	1615	FDK-6T1-13.2	13.2	9490.9	8883.6	9724.8	9102.5	233.9	218.9	2.46%	2.46%
FDG SUB G FD	618	FDG-69	69	6235.7	7467.3	6392.6	7651.3	156.9	184.0	2.52%	2.46%
FDK SUB K FD	615	FDK-69	69	8008.0	8632.3	8210.5	8844.9	202.5	212.6	2.53%	2.46%
FDT SUB T FD	611	FDT-69-BUS#2	69	8821.0	7517.1	9040.7	7702.1	219.7	185.0	2.49%	2.46%
FDT SUB T FD	455	FDT-69-TIE	69	8821.0	7517.1	9040.7	7702.1	219.7	185.0	2.49%	2.46%
FDT SUB T FD	1610	FDT-8T3-13.8	13.8	0.0	13494.4	0.0	13826.5	0.0	332.1	0.00%	2.46%
FTL TATE AND LYLE	626	FTL-161-B1	161	8793.0	10958.0	9040.3	11227.6	247.3	269.6	2.81%	2.46%
FTL TATE AND LYLE	449	FTL-161-B2	161	8793.0	10958.0	9040.3	11227.6	247.3	269.6	2.81%	2.46%
FDH HARVEST AVE	627	FDH-161	161	8866.0	11023.1	9115.5	11294.3	249.5	271.2	2.81%	2.46%
FDT SUB T FD	2510	FDT-8T4-13.8	13.8	0.0	13349.1	0.0	13677.5	0.0	328.4	0.00%	2.46%
FHA HAYES	616	FHA-161	161	9277.0	11381.1	9539.0	11660.8	262.0	279.7	2.82%	2.46%
FDK-FDT-XFR TAP POIN	612	FDK-FDT-XFR	69	7788.8	7755.9	7983.3	7946.5	194.5	190.6	2.50%	2.46%
XFR FRONTIER ETHANOL	641	XFR-69	69	1962.1	2737.5	2009.5	2803.3	47.4	65.8	2.42%	2.40%
CBPC-SNELL SW STR	10365	SNSLL SW 69	69	3517.3	4552.4	3602.1	4660.9	84.8	108.5	2.41%	2.38%
FDV SUB V FD TAP	10370	SUB V FD TAP	69	3204.4	4040.2	3279.7	4134.5	75.3	94.3	2.35%	2.33%
WRI WRIGHT	10653	WRI-161	161	8459.7	10040.3	8683.2	10274.3	223.5	234.0	2.64%	2.33%
CBPC-DUNCOMBE	10373	DUNCOMBE-69	69	2958.9	3824.5	3027.3	3912.2	68.4	87.7	2.31%	2.29%
CBPC-HOPE	10361	CB HOPE 69	69	9993.2	12653.3	10239.2	12942.3	246.0	289.0	2.46%	2.28%
HUM HUMBOLDT CENTRAL	11369	HUM-6T1-12.4	12.5	8088.4	7554.9	8088.4	7726.4	0.0	171.5	0.00%	2.27%
HUM HUMBOLDT CENTRAL	10369	HUM-69	69	4391.0	6691.8	4493.5	6843.3	102.5	151.5	2.33%	2.26%
HUMBOLDT CENTRAL TAP	10368	HUM-69-TAP	69	5496.9	7969.2	5626.2	8149.6	129.3	180.4	2.35%	2.26%
HUE HUMBOLDT EAST	11371	HUE-6T1-12.4	12.5	4996.5	4802.4	4996.5	4911.0	0.0	108.6	0.00%	2.26%
GMC GILMORE CITY	10322	GMC-69	69	1156.2	1846.3	1156.4	1888.0	0.2	41.7	0.02%	2.26%
BOD BODE	10326	BOD-69	69	1047.1	1700.2	1047.3	1738.6	0.2	38.4	0.02%	2.26%
CBPC-SWEAZY JCT	10366	SWEAZY J 69	69	3534.5	3928.2	3614.7	4016.9	80.2	88.7	2.27%	2.26%
HUE HUMBOLDT EAST	10371	HUE-69	69	4866.7	7192.7	4980.2	7355.0	113.5	162.3	2.33%	2.26%
RUTLAND TAP	10324	RUT-69-TAP	69	4728.5	7021.8	4838.7	7180.2	110.2	158.4	2.33%	2.26%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
RUTLAND JCT	10323	RUT-69-JCT	69	1951.2	3098.9	1995.9	3168.8	44.7	69.9	2.29%	2.26%
CBPC-BODE	10325	CB BODE 69	69	1258.4	2035.0	1258.7	2080.9	0.3	45.9	0.02%	2.26%
HUM HUMBOLDT CENTRAL	19369	HUM-TIE	69	3075.6	4760.1	3146.5	4867.4	70.9	107.3	2.31%	2.25%
CBPC-HOPE	10363	CB HOPE MP	99	40344.1	61437.5	41854.5	62819.5	1510.4	1382.0	3.74%	2.25%
WRI WRIGHT	10610	WRI-69	69	10152.2	9079.0	10384.8	9282.9	232.6	203.9	2.29%	2.25%
XFR FRONTIER ETHANOL	642	XFR-6T1-13.8	13.8	5782.8	4972.5	5782.8	5083.9	0.0	111.4	0.00%	2.24%
CBPC-PASSWATER	11360	PASSWATER-13	13.2	0.0	7630.5	0.0	7801.0	0.0	170.5	0.00%	2.23%
CBPC-PASSWATER	10360	PASSWATER-69	69	4012.3	4076.2	4102.1	4167.0	89.8	90.8	2.24%	2.23%
THO THOR	11332	THO-6T1-12.4	12.5	4423.3	4198.2	4423.3	4290.9	0.0	92.7	0.00%	2.21%
THO THOR	10332	THO-69	69	3006.0	4671.2	3073.4	4774.0	67.4	102.8	2.24%	2.20%
CBPC-WEBSTER CITY MU	10609	CB WCMU 69	69	3905.4	5206.5	3991.8	5320.8	86.4	114.3	2.21%	2.20%
FGF GOLDFIELD	11343	FGF-6T1-12.4	12.5	6290.3	5823.9	6290.3	5949.6	0.0	125.7	0.00%	2.16%
CBPC-WEBSTER	10608	CB WBSTR J69	69	4421.2	5190.7	4517.0	5302.4	95.8	111.7	2.17%	2.15%
FGF GOLDFIELD	10343	FGF-69	69	2891.7	4488.8	2954.7	4585.3	63.0	96.5	2.18%	2.15%
EAG EAGLE GROVE NORT	18372	EAG TO AGP	12.5	4810.0	5114.4	4810.0	5222.4	0.0	108.0	0.00%	2.11%
EAG EAGLE GROVE NORT	12372	EAG-6T2-12.4	12.5	7165.8	6669.5	7165.8	6810.3	0.0	140.8	0.00%	2.11%
EAG EAGLE GROVE NORT	19372	EAG TO CLR	12.5	6868.3	6414.7	6868.3	6550.1	0.0	135.4	0.00%	2.11%
EAG EAGLE GROVE NORT	11372	EAG-6T1-12.4	12.5	6869.4	6415.7	6869.4	6551.1	0.0	135.4	0.00%	2.11%
EAG EAGLE GROVE NORT	10372	EAG-69	69	3817.0	5544.9	3898.2	5661.5	81.2	116.6	2.13%	2.10%
CLR CLARION	10329	CLR-69	69	2036.3	3158.1	2036.5	3215.4	0.2	57.3	0.01%	1.81%
CBPC-LAKE CORNELIA T	10328	CORNELIA TAP	69	1959.1	3041.5	1959.2	3095.4	0.1	53.9	0.01%	1.77%
CBPC-LAKE CORNELIA	10341	LKE CORNELIA	69	1497.7	2375.9	1497.8	2418.0	0.1	42.1	0.01%	1.77%
GMC GILMORE CITY	10327	GMC-69-TIE	69	2272.0	3548.2	2272.4	3611.0	0.4	62.8	0.02%	1.77%
CBPC-GILMORE JUNCTIO	10318	CB GILMJ 69	69	2479.6	3864.7	2524.1	3933.1	44.5	68.4	1.79%	1.77%
CBPC-JEWEL	10619	JEWEL-69	69	3473.1	4304.5	3534.6	4380.5	61.5	76.0	1.77%	1.77%
CBPC-RENWICK TAP	10273	CB-RENWICK69	69	906.2	1476.0	906.3	1501.8	0.1	25.8	0.01%	1.75%
LVM MAPCO	13289	LVM MAPCO 69	69	1827.0	2950.4	1827.2	3001.9	0.2	51.5	0.01%	1.75%
LUV LUVERNE	10289	LUV-69	69	1827.0	2950.4	1827.2	3001.9	0.2	51.5	0.01%	1.75%
IPS TAP LIVERMORE /	10288	IPS TAP 69	69	2488.4	4000.5	2532.4	4070.3	44.0	69.8	1.77%	1.74%
CBPC-GARDEN CITY JCT	10612	GARDEN-69	69	3546.9	4555.6	3605.8	4630.9	58.9	75.3	1.66%	1.65%
CBPC-WILLIAMS	10614	WILLIAMS-69	69	3367.4	4309.1	3422.5	4379.6	55.1	70.5	1.64%	1.64%
CBPC-BOONDOCKS	10618	BOONDOCKS-69	69	3499.1	4496.5	3555.6	4569.1	56.5	72.6	1.61%	1.61%
CBPC-SHER HUMB TAP	636	SHER HUM TAP	69	1911.2	2516.1	1911.3	2556.1	0.1	40.0	0.01%	1.59%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
CBPC-SHER HUMB TAP	635	SUB M TAP 69	69	1911.2	2516.1	1911.3	2556.1	0.1	40.0	0.01%	1.59%
FDM SUB M FD	633	FDM-69-CBPC	69	1581.5	2167.6	1581.6	2202.0	0.1	34.4	0.01%	1.59%
CBPC-BLAIRSBURG	10615	BLAIRSRBG-69	69	4913.2	5931.3	4991.1	6025.2	77.9	93.9	1.59%	1.58%
FWL WALL LAKE WIND F	18625	FWL-13.8-8T3	13.8	0.0	15624.6	0.0	15869.3	0.0	244.7	0.00%	1.57%
FWL WALL LAKE WIND F	17625	FWL-69	69	9590.8	8876.4	9741.5	9015.2	150.7	138.8	1.57%	1.56%
FWL WALL LAKE WIND F	10625	FWL-161	161	8688.3	9811.2	8827.2	9964.6	138.9	153.4	1.60%	1.56%
CBPC-WILLEMSEN	10333	WILMSEN-69	69	4568.6	6076.6	4638.1	6168.7	69.5	92.1	1.52%	1.52%
CBPC-TWIN LAKES	10198	TWN LKS 69	69	1525.2	1902.3	1525.3	1931.0	0.1	28.7	0.01%	1.51%
CBPC-TWIN LAKES	10191	TWN LKS TAP	69	1965.9	2479.0	1966.0	2516.4	0.1	37.4	0.01%	1.51%
CBPC-GALBRAITH	10291	CB GALBR 69	69	2720.6	4306.7	2721.0	4369.8	0.4	63.1	0.01%	1.47%
CBPC-ALGONA SOUTH	10391	ALGONA SOUTH	69	2670.5	4205.3	2670.8	4264.0	0.3	58.7	0.01%	1.40%
CBPC-ALGONA SOUTH	10390	ALGONA SOUTH	69	2670.5	4205.3	2670.8	4264.0	0.3	58.7	0.01%	1.40%
WEB WEBSTER	10662	WEB-345	345	11464.8	13570.3	11813.0	13759.6	348.2	189.3	3.04%	1.39%
GAMESA POCAHONTAS PR	10649	GAMESA-161	161	7585.7	8951.0	7689.9	9072.6	104.2	121.6	1.37%	1.36%
PMY POMEROY 161KV	10648	PMY-161	161	7585.7	8951.0	7689.9	9072.6	104.2	121.6	1.37%	1.36%
ITC MIDWEST-COULTER	10309	COULTER TAP	69	1975.0	3011.2	1975.1	3051.5	0.1	40.3	0.01%	1.34%
CBPC-BURT	11128	CB BURT 161	161	4864.2	6360.1	4935.1	6445.0	70.9	84.9	1.46%	1.33%
CBPC-POC-POCAHONTAS	10320	POC-69	69	2764.2	3930.8	2764.3	3982.9	0.1	52.1	0.00%	1.33%
WIL WILLIAMS PIPELIN	10287	WIL-69	69	1624.5	2491.0	1624.5	2522.9	0.0	31.9	0.00%	1.28%
LATIMER TAP	10302	LAT-69-TAP	69	2091.6	3169.2	2091.6	3209.7	0.0	40.5	0.00%	1.28%
CBPC-ALGONA EAST	10395	ALGONA EAST	69	3294.4	4580.4	3336.6	4638.8	42.2	58.4	1.28%	1.27%
CBPC-ALGONA NORTH	10397	ALGONA NORTH	69	3768.9	4770.5	3816.6	4830.4	47.7	59.9	1.27%	1.26%
IPL-KOSSUTH	10398	IPL-KOSSUTH	69	3491.9	4509.1	3536.0	4565.7	44.1	56.6	1.26%	1.26%
LAT LATIMER	10301	LAT-69	69	2209.4	3328.5	2209.4	3369.5	0.0	41.0	0.00%	1.23%
CBPC-BURT	10248	CB BURT 69	69	6334.3	6064.2	6413.8	6138.7	79.5	74.5	1.26%	1.23%
CBPC-BURT	11127	CB BURT MP	99	4189.7	4151.6	4241.6	4202.0	51.9	50.4	1.24%	1.21%
CBPC-BOONE VALLEY JU	10263	CB-BV-JCT-69	69	1995.7	3123.9	1995.8	3161.3	0.1	37.4	0.01%	1.20%
CBPC-EAST	10197	CBPC EAST 69	69	1664.3	1875.2	1664.3	1897.1	0.0	21.9	0.00%	1.17%
CBPC-EAST	10192	CBPC EASTTAP	69	2472.2	2791.1	2472.2	2823.6	0.0	32.5	0.00%	1.16%
FVR FARNHAMVILLE	10190	FVR-69	69	1563.8	1781.6	1563.8	1802.3	0.0	20.7	0.00%	1.16%
PMY POMEROY 161KV	10651	PMY-69-A27	69	5020.2	4726.7	5078.4	4781.4	58.2	54.7	1.16%	1.16%
PMY POMEROY 161KV	10650	PMY-69	69	5020.2	4726.7	5078.4	4781.4	58.2	54.7	1.16%	1.16%
FPC POCAHONTAS WIND	10430	FPC-161	161	7940.4	9119.6	8032.3	9224.6	91.9	105.0	1.16%	1.15%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
CBPC-RINARD	10196	RINARD 69	69	2536.4	2861.3	2536.4	2893.8	0.0	32.5	0.00%	1.14%
PMY POMEROY 161KV	10654	PMY-34.5-AB	34.5	785.4	2604.7	785.4	2634.2	0.0	29.5	0.00%	1.13%
PMY POMEROY 161KV	10644	PMY-34.5	34.5	785.4	2604.7	785.4	2634.2	0.0	29.5	0.00%	1.13%
CBPC-WHITTEMORE JUNC	10274	WHITTEMORE J	69	3393.4	5207.1	3393.6	5261.7	0.2	54.6	0.01%	1.05%
FWL WALL LAKE WIND F	12625	FWL-34.5-2	34.5	15068.5	15726.4	15226.4	15890.8	157.9	164.4	1.05%	1.05%
FWL WALL LAKE WIND F	11625	FWL-34.5-1	34.5	14896.4	15653.0	15051.9	15816.1	155.5	163.1	1.04%	1.04%
CBPC-DOVER	10164	CB-DOVER-69	69	2461.9	3760.8	2461.9	3799.1	0.0	38.3	0.00%	1.02%
FWL WALL LAKE WIND F	14625	FWL-13.8-8T2	13.8	0.0	20321.1	0.0	20526.8	0.0	205.7	0.00%	1.01%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11487.0	14089.3	11674.5	14231.8	187.5	142.5	1.63%	1.01%
LEH LEHIGH	10775	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	188	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	187	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	186	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
FWL WALL LAKE WIND F	13625	FWL-13.8-8T1	13.8	0.0	20260.2	0.0	20464.9	0.0	204.7	0.00%	1.01%



Local Transient Stability Analysis Short Circuit Analysis

50 MW Solar Generation in Murray County, Minnesota

MISO # J525
Prepared for MISO

July 19, 2017
Great River Energy

Contact:
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Melissa Phillips, Substation Engineer
Great River Energy

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Local Transient Stability Analysis

Prepared by Transmission Planning Engineers

Executive Summary

Great River Energy has performed a transient stability analysis in the local area around the proposed interconnection of a 50 MW Solar PV project at the Lake Wilson substation. Information (maps, one-lines and power flow slider files) can be found in the Local System Impact Study report recently prepared by Great River Energy.¹

The purpose of this study was to determine whether there were concerns about the transient stability of the system in the area given the weak short-circuit strength of the system that may result from local contingencies. The study includes a description of the model development and the results of the analysis.

In summary, the results indicate that there is a possible instability of the inverter controls due to the low short-circuit-ratio (SCR) at the point of interconnection during a system fault on the Chander tap—Lake Wilson 69 kV line circuit that results in the tripping of the 69 kV line from Lake Wilson to Fenton. This leaves the 50 MW inverter-based generation on a long radial circuit from Lake Wilson to Lyon County through Tracy and results in low short-circuit ratio (approximately 1.96) at the point of interconnection.

Before the interconnection can be allowed to operate, the developer and/or system manufacturer must provide assurance that the generation can operate during the low SCR conditions.

¹Local Transmission System Impact Study, J525 Generation Interconnection, June 1, 2017

Section 1: Model Development

The “Backbone-NUs_StudyCase-DPP-FEB16-West_2021SH-Dynamics_v33_170605.sav” case was used for the study. This case is located in the “sav” folder of the “DPP-2016Feb_Backbone NUs_Stability Package_170608.zip” package. No sensitivities or modifications were made to or created from the case.

Section 2: Transient Stability Analysis

2.1.1 Faults

The following faults were performed as part of the transient stability study:

- Three phase fault near Lake Wilson on Lake Wilson-Fenton 69 kV line
- Single-line-ground fault near Lake Wilson on Lake Wilson-Fenton 69 kV line
- Three phase fault near Fenton on Fenton-Lake Wilson 69 kV line
- Single-line-ground fault near Fenton on Fenton-Lake Wilson 69 kV line
- Three phase fault near Lake Wilson on Lake Wilson-Lake Sarah 69 kV line
- Single-line-ground fault near Lake Wilson on Lake Wilson-Lake Sarah 69 kV line
- Three phase fault on Fenton 115 kV bus

2.1.2 Results

The simulation results indicate that there are voltage violations for the following faults:

- Three phase fault near Lake Wilson on Lake Wilson-Fenton 69 kV line
- Three phase fault near Fenton on Fenton-Lake Wilson 69 kV line
- Single-line-ground fault near Fenton on Fenton-Lake Wilson 69 kV line

A summary of all the transient stability simulation results can be found in the accompanying spreadsheet, “J525 Transient Stability Results Table”.

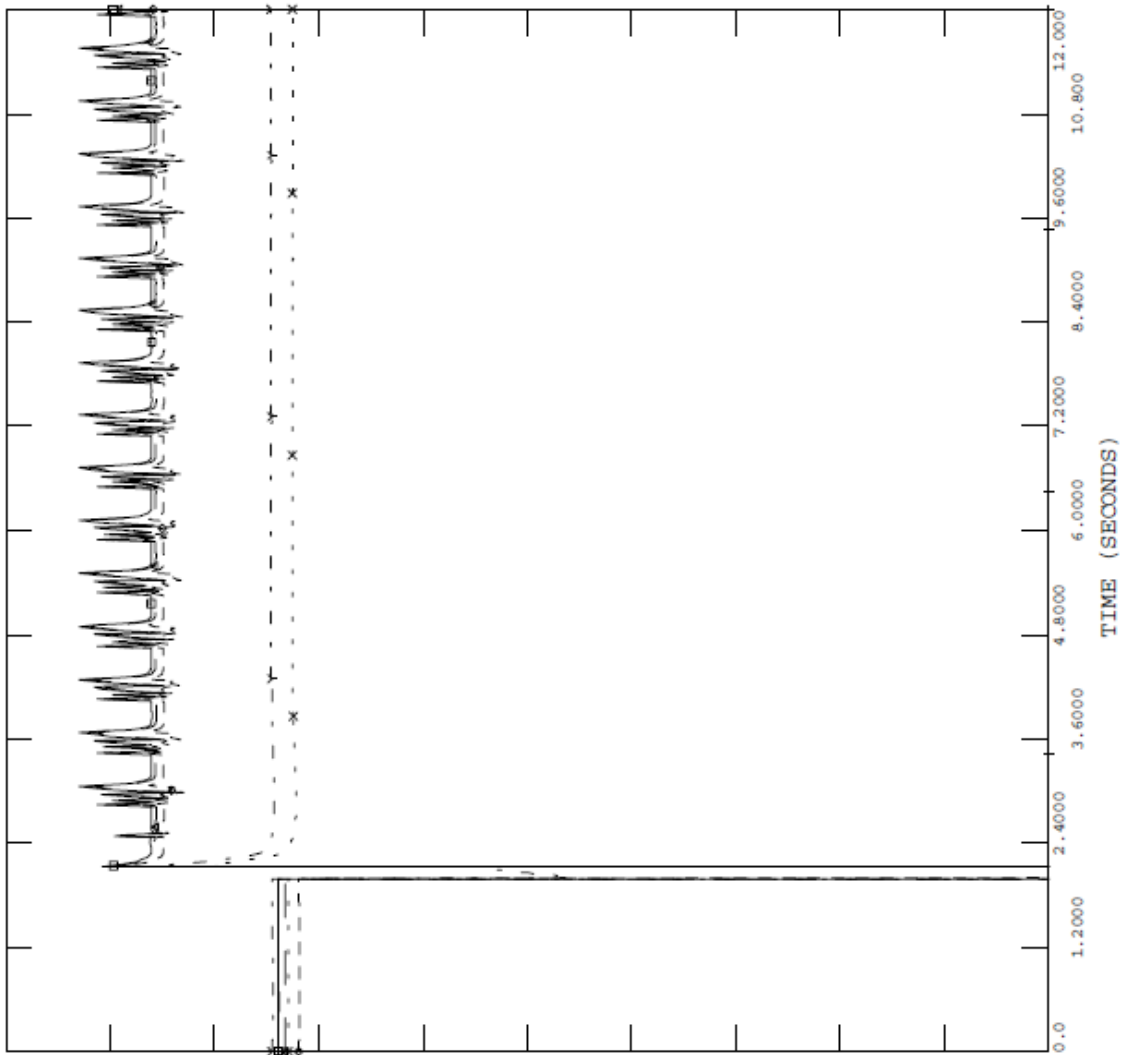
The voltage violations are identical across all of the faults identified above. An example of the voltage violation can be seen in Figure 1 Representative Voltage Violations which shows the results of the three-phase fault near Lake Wilson that clears the Lake Wilson—Fenton 69 kV line. Each of these faults results in the generation connected to the system via a long, 69 kV line from Lake Wilson to Lyon County through Tracy.



2021 SUMMER SHOULDER STUDY DYNAMICS CASE
2016 FEB DPP CYCLE; 2020SH90-MISO15-PASS4
CON_LKWILSN_3PH_CHNDLRT_69

FILE: Backbone-NUs_StudyCase-DPP-FEB16-West_2021SH-Dynamics_v33_170605_LK

1.1500	CHNL# 1292: [VOLT 603195 [FENTON 7 115.00]]	0.65000
1.1500	CHNL# 1291: [VOLT 605689 [FENTON8 69.000]]	0.65000
1.1500	CHNL# 1290: [VOLT 605192 [CHNDLRT8 69.000]]	0.65000
1.1500	CHNL# 1289: [VOLT 618918 [GRE-LKSRHTP869.000]]	0.65000
1.1500	CHNL# 1288: [VOLT 605213 [HADLEY 8 69.000]]	0.65000
1.1500	CHNL# 1287: [VOLT 618920 [GRE-LKWILSN869.000]]	0.65000



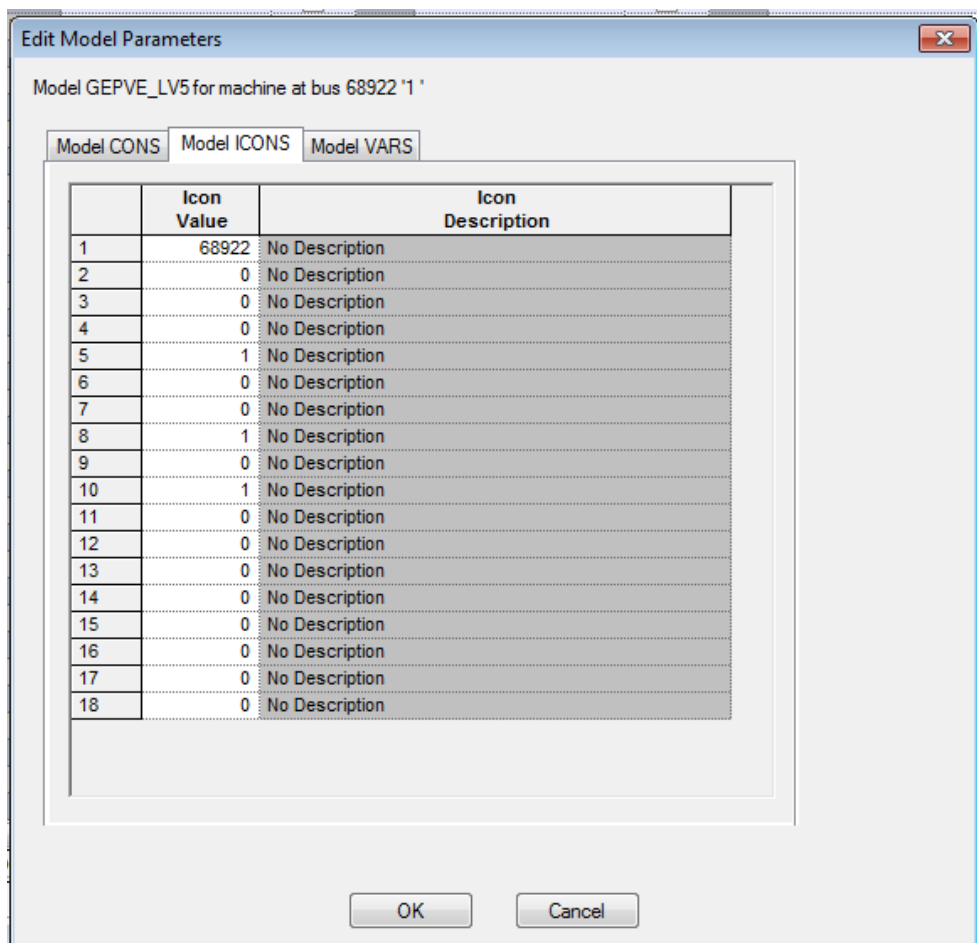
TUE, JUL 11 2017 9:08
J525 LOCAL VOLTAGES

Figure 1 Representative Voltage Violations

The following alerts are representative of what was captured in the simulation transcript files:

```
IN MODE 2 LVRT, Ip. ramp mode 0.0000000E+00
*** VinH 1
IN MODE 2, Iq LVRT ramp mode -1.4580846E-02
*** Enter HVRT: Pinh
```

There is no documentation in the user's manual provided with the J525 modeling data that indicates what these alerts reference. Additionally, these alerts seem to indicate that some sort of reactive current control was performed based on low-voltage ride through commands when the model data indicates that this control mode is disabled. Specifically, ICON (M+9) is set to 1 in the snapshot file used in the J525 study ("2020SH-MISO16Feb-Dynamics_170605.snp").



The user's manual for the GE LV5 model, "LV5 PSSE Model V3 User Manual Rev 002.pdf" states the following:

Electrical control of the GE PV plant using LV5 inverters during low voltage ride-through (LVRT) or high voltage ride-through (HVRT) events can be performed using two control strategies, Type 1 and Type 2 based on block diagram in Figure 3-4. The control strategy can be selected by setting flag ICON(M+8). Setting the flag to 1 selects LV5 Type 1 Voltage Ride-Through (VRT) strategy while setting the ICON(M+8) flag to 2 selects Type 2 VRT strategy. When set to Type 1 VRT, the inverter power bridges stop gating during LVRT and HVRT and do not produce current. With Type 2 VRT, the inverter continues gating and produces reactive current during LVRT and HVRT. Unless advised otherwise by GE, the inverter should be set for Type 1 VRT and flag ICON(M+8) should be set to 1.

The third option for voltage control regulation, type 0, depicted in Figure 3-3 is not used in LV5 inverters.

This language seems to indicate that when ICON (M+9) is set to 1, there should be no current control, and yet the transcript output from the simulation mentioned above seems to indicate that current control was active.

Section 3: Observations

Based on the observation of the stability plot shown in Figure 1 above and the control mode switching that is indicated in the transcript of the stability run, it is suspected that the generator controls may switching operating modes between two different low voltage ride-through modes. This is describe on page 13 of the manufacturer's documentation²

This condition is also described in a NERC Reliability Guideline "Integrating Inverter-Based Resources into Weak Power Systems"³. Below is graph from the page 16 of the NERC report showing this mode-cycling condition.

GRE has recently contacted the manufacturer regarding these results but no answers were available at the time this report was prepared (July 19,2017).

²LV5-SLR, LV5 PSS/E MODEL VERSION 3 USER MANUAL, GE Power Conversion, Revision 002 2015-05-15

³NERC Reliability Guideline "Integrating Inverter-Based Resources into Weak Power Systems, June, 2017

Section 4: Conclusions and Recommendations

Due to the issues mentioned above, GRE recommends the following before the solar PV project can be allowed to operate.

1. The Generation Owner provide feedback regarding the voltage violations mentioned identified in section 2.1.2
2. The Generation Owner and/or manufacturer provide input as to the meaning of the alerts identified in section 2.1.2
3. The Generation Owner and/or manufacturer provide input as to the discrepancies between the transcript alerts and the control mode identified in the PSSE snapshot file as discussed in section 2.1.2
4. The manufacturer provides assurance that the inverter controls can be adjusted to accommodate the low SCR that could exist on the local transmission system. The preference is that the manufacturer provide the output of simulations under similar low SCR conditions

Short Circuit Analysis

Prepared by Substation Engineer

Introduction

This report presents the results of the short circuit analysis for J525. The objective of the analysis is to determine the present fault current levels at the POI for J525 and determine the future fault current levels after the solar become commercially operational.

Background

This study will calculate the fault currents before and after the addition of J525's 50 MW of solar generation. Short circuit calculations are based on the impedances of the transmission lines, transformers, generators, and on the pre-fault voltages. This solar generation will be connected to the system at GRE's Lake Wilson/Moon Lake Substation.

Methodology

Aspen *One-liner* short circuit analysis software is used to determine the symmetrical fault currents. Great River Energy has an Aspen model already built that includes Lake Wilson/Moon Lake and surrounding substations.

The generator and transformer impedances for interconnection equipment were received from the customer. GRE models solar PV inverters based on a method recommended by Aspen. Solar generation can be simulated in Aspen *Oneliner* using a current limited generator and a linear network solution for fault calculations.

When modeling PV inverters in Aspen, the zero-sequence and negative sequence impedance are set to $j999$ pu. The positive sequence impedance is set to $j0.02$. The current limit for the generator is set to 150% of the nominal full load current of the inverter.

Data

	Fault	Fault Current (A) and angle	ANSI X/R	MVA
Before J525	1LG	2956@-54	5.68	368.7
	3LG	3779@-54	6.45	471.4
After J525	1LG	4067@-66	22.06	430.0
	3LG	4128@-63	18.88	436.5

Results

Fault currents were calculated before and after the addition of J525's 50 MW of solar generation. Table 1 summarizes the fault currents at Lake Wilson Substation.

A short circuit analysis was performed to determine the impact of the proposed generation on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Otter Tail Power Company's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J526 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Otter Tail equipment in the area are not exceeded.

With the proposed J526 project additions, the fault currents are estimated to be roughly 11.7 kA at the Big Stone South 345 kV bus, 20.6 kA at the Big Stone South 230 kV bus, and 37.1 kA at the Big Stone South 34.5 kV bus. There does not appear to be any short circuit related upgrades needed for J526.

A general protective relay coordination review will need to be performed by OTP when the projects approach completion and before energizing the generation source.

J527 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J527 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J527 short circuit study model was developed as follows:

- Added J506, J524, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J527 generating facility with its sequence parameters at approximately 33 miles from Booneville on Booneville - Cooper 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J527 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J527 short circuit benchmark model was developed as follows:

- Turned off J527 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J527 project) and benchmark case (without the J527 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J527 (study case) and without J527 (benchmark case) are listed in Table 1. With the addition of the J527 generating facility, the maximum increase in the 3PH short circuit current is 7.03% (1119.2 A), and the maximum increase in the SLG short circuit current is 18.94% (2339.9 A), which are both observed at the J527 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 17037 A (increased by 1119.2 A) and SLG fault current is 14696 A (increased by 2339.9 A) at the J527 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J527 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J527	65209	J527_TER2	13.8	-	-	0.0	39002.8	-	-	-	-
J527	65208	J527_TER1	13.8	-	-	0.0	39002.8	-	-	-	-
J527	65207	J527_COL2	34.5	-	-	27952.6	22515.4	-	-	-	-
J527	65206	J527_COL1	34.5	-	-	27952.6	22515.4	-	-	-	-
J527	65204	J527 COL2	34.5	-	-	31599.5	23596.0	-	-	-	-
J527	65202	J527 COL	34.5	-	-	31599.5	23596.0	-	-	-	-
J527	65201	J527	345	-	-	14510.2	16812.7	-	-	-	-
J527	65200	J527_POI	345	12356.5	15917.7	14696.4	17036.9	2339.9	1119.2	18.94%	7.03%
ORIENT WIND PROJECT	20921	J500-345-BT	345	14798.1	17781.6	16501.8	18769.8	1703.7	988.2	11.51%	5.56%
ORIENT WIND PROJECT	20911	J500-345	345	14798.1	17781.6	16501.8	18769.8	1703.7	988.2	11.51%	5.56%
J528	65300	J528_POI	345	13148.1	15972.6	14134.3	16673.3	986.2	700.7	7.50%	4.39%
J528	65301	J528	345	12987.8	15770.3	13941.5	16451.9	953.7	681.6	7.34%	4.32%
CRH ROLLING HILLS WI	21916	CRH-345-BT	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	20916	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	426	CRH-345-B1	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	425	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	424	CRH-345-B3	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	423	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	422	CRH-345-B2	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
DEB BOONEVILLE	20905	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	16	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	15	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	14	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	13	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	10	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DED MADISON COUNTY	20904	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	212	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	211	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	210	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DGN NORWALK	20955	DGN-345	345	14316.1	16542.0	14467.6	16737.6	151.5	195.6	1.06%	1.18%
DGG GRIMES	20920	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	54	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	53	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DGG GRIMES	52	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	51	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	50	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	49	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	48	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DAG SYCAMORE EC	20900	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	185	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	184	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	183	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	182	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	181	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAC SE POLK	20907	DAC-345	345	18605.3	19127.6	18747.6	19305.9	142.3	178.3	0.76%	0.93%
DGC GREATER DES MOIN	20893	GDMEC CT TR2	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20892	GDMEC CT TR1	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20891	GDMEC ST TR1	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20890	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	82	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	81	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	80	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	79	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	78	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DAH BONDURANT	20901	DAH-345	345	17701.5	20025.2	17815.0	20198.8	113.5	173.6	0.64%	0.87%
DEB BOONEVILLE	20831	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	21	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	20	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	19	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	18	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	17	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DAG SYCAMORE EC	20800	DAG-161-B2	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	504	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	503	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	502	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	501	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DAG SYCAMORE EC	500	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	499	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	498	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	497	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	496	DAG-161-B1	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%

J528 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J528 generating facility (80 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J528 short circuit study model was developed as follows:

- Added J506, J524, J527, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J528 generating facility with its sequence parameters at approximately 32 miles from Madison on Madison – Rolling Hills 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 66 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z=5.75\%$ at 112 MVA, D-YG) were also added.
- Modeled the J528 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J528 short circuit benchmark model was developed as follows:

- Turned off J528 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J528 project) and benchmark case (without the J528 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J528 (study case) and without J528 (benchmark case) are listed in Table 1. With the addition of the J528 generating facility, the maximum increase in the 3PH short circuit current is 5.50% (868.9 A), and the maximum increase in the SLG short circuit current is 14.20% (1757.2 A), which are both observed at the J528 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 16673 A (increased by 868.9 A) and SLG fault current is 14134 A (increased by 1757.2 A) at the J528 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J528 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J528

Substation	Faulted Bus			Benchmark Case without J528 Project		Study Case with J528 Project		Ampere Change with J528 vs Benchmark		% Change with J528 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J528	65309	J528_TER2	13.8	-	-	0.0	29537.6	-	-	-	-
J528	65308	J528_TER1	13.8	-	-	0.0	29537.6	-	-	-	-
J528	65307	J528_COL2	34.5	-	-	21600.0	17422.8	-	-	-	-
J528	65306	J528_COL1	34.5	-	-	21600.0	17422.8	-	-	-	-
J528	65304	J528 COL2	34.5	-	-	24201.3	18134.2	-	-	-	-
J528	65302	J528 COL1	34.5	-	-	24201.3	18134.2	-	-	-	-
J528	65301	J528	345	-	-	13941.5	16451.9	-	-	-	-
J528	65300	J528_POI	345	12377.1	15804.4	14134.3	16673.3	1757.2	868.9	14.20%	5.50%
ORIENT WIND PROJECT	20921	J500-345-BT	345	15219.2	18001.0	16501.8	18769.8	1282.6	768.8	8.43%	4.27%
ORIENT WIND PROJECT	20911	J500-345	345	15219.2	18001.0	16501.8	18769.8	1282.6	768.8	8.43%	4.27%
J527	65200	J527_POI	345	13924.2	16471.8	14696.4	17036.9	772.2	565.1	5.55%	3.43%
J527	65201	J527	345	13763.9	16263.3	14510.2	16812.7	746.3	549.4	5.42%	3.38%
DED MADISON COUNTY	20904	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	212	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	211	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	210	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
CRH ROLLING HILLS WI	21916	CRH-345-BT	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	20916	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	426	CRH-345-B1	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	425	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	424	CRH-345-B3	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	423	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	422	CRH-345-B2	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
DEB BOONEVILLE	20905	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	16	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	15	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	14	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	13	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	10	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DGN NORWALK	20955	DGN-345	345	14330.9	16572.9	14467.6	16737.6	136.7	164.7	0.95%	0.99%
DGG GRIMES	20920	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	54	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	53	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J528

Substation	Faulted Bus			Benchmark Case without J528 Project		Study Case with J528 Project		Ampere Change with J528 vs Benchmark		% Change with J528 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DGG GRIMES	52	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	51	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	50	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	49	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	48	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DAG SYCAMORE EC	20900	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	185	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	184	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	183	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	182	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	181	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAC SE POLK	20907	DAC-345	345	18629.9	19161.5	18747.6	19305.9	117.7	144.4	0.63%	0.75%
DGC GREATER DES MOIN	20893	GDMEC CT TR2	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20892	GDMEC CT TR1	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20891	GDMEC ST TR1	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20890	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	82	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	81	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	80	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	79	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	78	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DAH BONDURANT	20901	DAH-345	345	17726.3	20063.0	17815.0	20198.8	88.7	135.8	0.50%	0.68%
DEB BOONEVILLE	20831	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	21	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	20	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	19	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	18	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	17	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%

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The materials contained in this document include Critical Energy Infrastructure Information (CEII). All materials designated as CEII must be handled and protected per the requirements in FERC CEII Policy. There may be additional requirements for CEII materials in the future.

J529 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J529, a proposed 250 MW wind farm along with J590, a proposed 90 MW expansion at the same interconnection point, and nearby substations both with and without the Interconnection Customer interconnected. J529/J590 was assumed to interconnect off the Obrien County-Kossuth County 345 kV line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,000 Amps without the Interconnection Customer included and 11,705 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J529/J590

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
		J529J590 POI	345		7,660	9,703	2,043	10,000	11,705	1,705
		J529J590 Collector	34.5		23,730	37,957	14,227	20,331	32,060	11,730
55368	J455 POI	J455 POI	345	MEC	9,827	10,320	493	11,388	11,993	606
635369	KOSSUTH 3	Kossuth Co	345	MEC	8,502	9,587	1,085	11,804	12,852	1,048
635368	OBRIEN 3	O'Brien Co	345	MEC	11,489	11,804	315	13,740	14,192	452
631197	LEDYARD3	Ledyard	345	ITCM	8,258	8,705	448	11,552	12,112	560
636000	WEBSTER3	Webster	345	MEC	9,825	10,033	208	11,901	12,150	249
631138	LAKEFLD3	ITCM Lakefield	345	ITCM	15,041	15,234	192	16,469	16,736	267
635400	HIGHLND 3	Highland	345	MEC	10,112	10,283	171	12,330	12,588	258
631193	HUNTLEY3	Huntley	345	ITCM	7,823	8,019	196	10,432	10,725	293
631198	COLBY3	Colby	345	ITCM	7,974	8,039	65	9,386	9,484	98
636010	LEHIGH 3	Lehigh	345	MEC	9,802	9,910	109	12,385	12,541	156
636001	WEBSTER5	Webster	161	MEC	17,205	17,352	147	17,701	17,831	130
601034	NOBLES 3	Nobles Co	345	XCEL	5,670	5,687	16	8,543	8,587	43
601029	LKFLDXL3	Xcel Lakefield	345	XCEL	20,018	20,173	156	14,337	14,438	101
631041	LAKEFLD5	ITCM Lakefield	161	ITCM	17,185	17,286	101	18,941	19,081	140
635200	RAUN 3	Raun	345	MEC	27,244	27,345	101	25,743	25,839	95

J530 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J530 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J530 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J530 generating facility with its sequence parameters at approximately 26 miles from Montezuma on Montezuma – Hills 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J530 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J530 short circuit benchmark model was developed as follows:

- Turned off J530 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J530 project) and benchmark case (without the J530 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J530 (study case) and without J530 (benchmark case) are listed in Table 1. With the addition of the J530 generating facility, the maximum increase in the 3PH short circuit current is 9.32% (1118.4 A), and the maximum increase in the SLG short circuit current is 35.43% (2793.4 A), which are both observed at the J530 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 13118 A (increased by 1118.4 A) and SLG fault current is 10678 A (increased by 2793.4 A) at the J530 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J530 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J530

Substation	Faulted Bus			Benchmark Case without J530 Project		Study Case with J530 Project		Ampere Change with J530 vs Benchmark		% Change with J530 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J530	75739	J530_TER2	13.8	-	-	0.0	38565.2	-	-	-	-
J530	75738	J530_TER1	13.8	-	-	0.0	38565.2	-	-	-	-
J530	75737	J530_COL2	34.5	-	-	27565.4	22179.3	-	-	-	-
J530	75736	J530_COL1	34.5	-	-	27565.4	22179.3	-	-	-	-
J530	75734	J530 COL2	34.5	-	-	31050.5	23176.8	-	-	-	-
J530	75732	J530 COL1	34.5	-	-	31050.5	23176.8	-	-	-	-
J530	75731	J530	345	-	-	10605.1	12990.5	-	-	-	-
IOWA COUNTY WIND FAR	30143	J530-345	345	7884.2	11999.6	10677.6	13118.0	2793.4	1118.4	35.43%	9.32%
DBM MONTEZUMA	30150	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	3	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	2	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	1	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
IHL HILLS	30120	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	302	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	301	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	300	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	299	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	298	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
NORTH ENGLISH WIND F	30152	J555-345	345	7080.8	10350.4	7466.1	10612.6	385.3	262.2	5.44%	2.53%
ENGLISH FARMS WIND F	31151	J475-GEN-BT	345	7134.4	9894.0	7467.6	10123.3	333.2	229.3	4.67%	2.32%
ENGLISH FARMS WIND F	30151	J475-345	345	7134.4	9894.0	7467.6	10123.3	333.2	229.3	4.67%	2.32%
ICK SUB K IOWA CITY	30125	ICK 345	345	11112.0	14647.8	11315.5	14870.4	203.5	222.6	1.83%	1.52%
ICT SUB T HASKINS	30126	ICT-345	345	11124.8	15559.9	11284.2	15762.5	159.4	202.6	1.43%	1.30%
Q92 SUB 92	30118	SUB 92 345	345	12960.6	17700.9	13131.1	17926.8	170.5	225.9	1.32%	1.28%
OGS	602660	OGS 345	345	9121.7	10976.8	9226.6	11082.8	104.9	106.0	1.15%	0.97%
IHL HILLS	30282	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	309	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	308	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	307	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	306	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	305	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	304	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	303	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J530

Substation	Faulted Bus			Benchmark Case without J530 Project		Study Case with J530 Project		Ampere Change with J530 vs Benchmark		% Change with J530 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
Q93 SUB 93	30128	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	30127	Q93-345RATBT	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	317	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	316	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	315	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	314	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
IEN HYDROCARBON SUB	30288	IEN-161	161	11971.3	14732.5	12065.3	14825.5	94.0	93.0	0.79%	0.63%
ICP SUB P CORALVILLE	30296	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	313	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	312	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	311	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICJ SUB J IOWA CITY	30290	ICJ-161	161	10866.8	13294.4	10934.5	13368.6	67.7	74.2	0.62%	0.56%
ICJ SUB J IOWA CITY	332	ICJ-161-B2	161	10866.8	13294.4	10934.5	13368.6	67.7	74.2	0.62%	0.56%
ICU SUB U IOWA CITY	30294	ICU-161-B1	161	10425.2	13260.2	10485.8	13333.7	60.6	73.5	0.58%	0.55%
ICU SUB U IOWA CITY	297	ICU-161-B2	161	10425.2	13260.2	10485.8	13333.7	60.6	73.5	0.58%	0.55%
ICR CORAL RIDGE SUBS	30298	ICR-161	161	10158.2	12727.6	10216.0	12796.1	57.8	68.5	0.57%	0.54%
QWL WALCOTT	30117	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	293	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	292	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	291	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
ICK SUB K IOWA CITY	30303	ICK 161	161	10535.8	12157.9	10599.9	12222.4	64.1	64.5	0.61%	0.53%
ICE SUB E IOWA CITY	30302	ICE-161-B2	161	9279.8	12250.4	9335.8	12314.6	56.0	64.2	0.60%	0.52%
ICE SUB E IOWA CITY	331	ICE-161-B1	161	9279.8	12250.4	9335.8	12314.6	56.0	64.2	0.60%	0.52%
ICY SUB Y IOWA CITY	30301	ICY-161	161	8893.4	11843.9	8943.4	11903.5	50.0	59.6	0.56%	0.50%
ICG NORTHGATE IOWA C	30300	ICG-161	161	8990.9	11938.3	9039.1	11998.3	48.2	60.0	0.54%	0.50%



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J531

7-18-2017

Summary

Generator Interconnection Project J531 is a 200 MW wind powered generating facility with proposed interconnection to the 345 kV bus at a new switch station tapping ITC Midwest's portion of the Mitchell County to Hazleton 345 kV line. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- Two 100 MW equivalent generators
- Two 34.5kV to 690V equivalent transformers
- Two 34.5kV to 345 kV step up transformers
- 345 kV interconnection line, impedance from MISO-DPP-FEB16-West-2021SH model

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

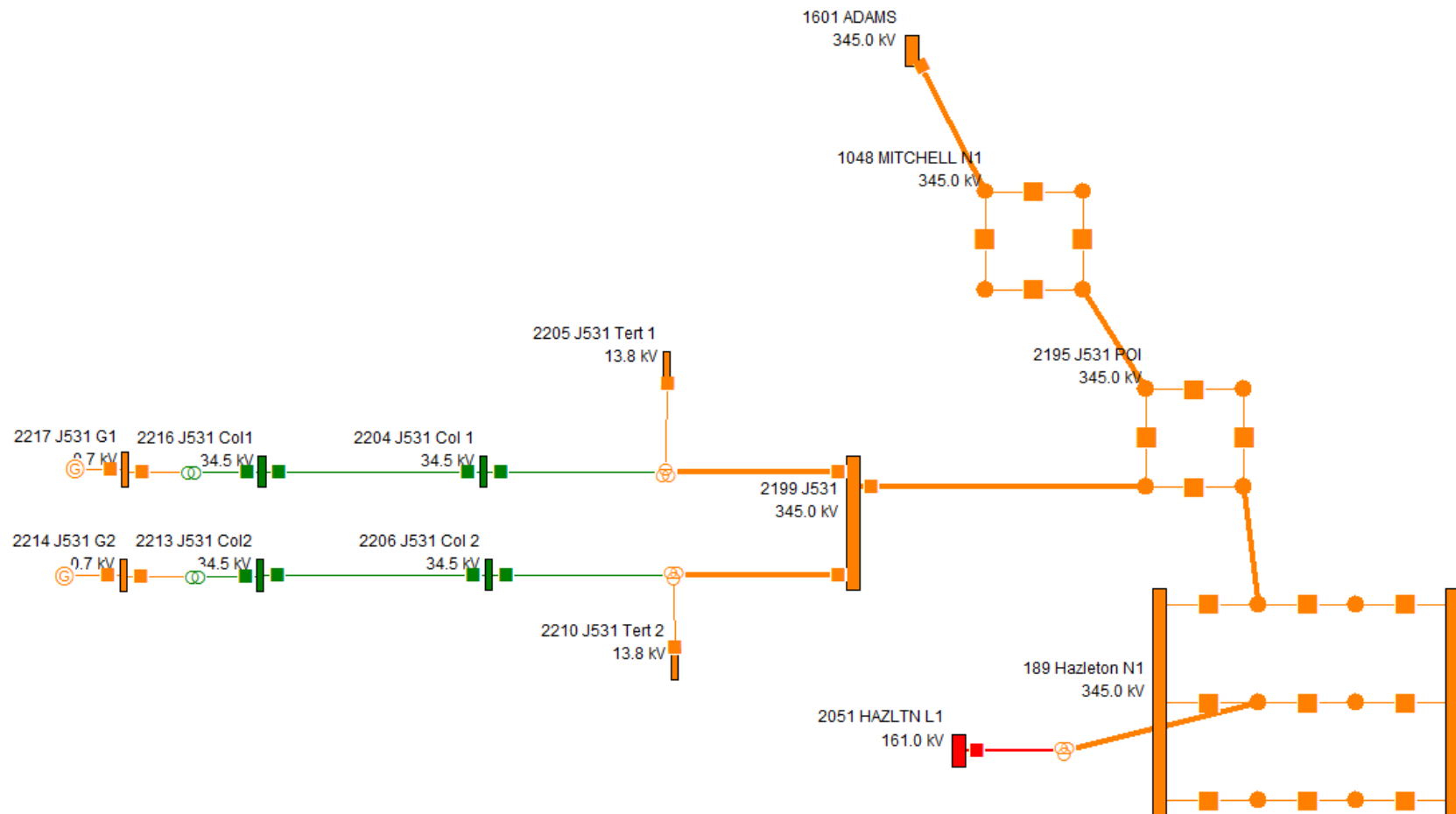


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J531.

J531	Mitchell Co 345	Hazleton 345	Hazleton 161	Adams 345
SLG-pre (A)	9510.96	13085.1	20235.4	9853.96
3Ph-pre (A)	10252.4	15229.3	21384.5	10595.8
SLG-post (A)	9940.54	13349.7	20422.6	10122.3
3Ph-post (A)	10282.3	15251.2	21395.8	10621.2
Contribution SLG (A)	429.58	264.6	187.2	268.34
Contribution 3Ph (A)	29.9	21.9	11.3	25.4

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J531. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. New circuit breakers at the POI switch station were assumed to have 50kA interrupting capability.

J534 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J534 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J534 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J530, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J534 generating facility with its sequence parameters at approximately 17 miles from Kossuth on Kossuth – Webster 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J534 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J534 short circuit benchmark model was developed as follows:

- Turned off J534 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J534 project) and benchmark case (without the J534 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J534 (study case) and without J534 (benchmark case) are listed in Table 1. With the addition of the J534 generating facility, the maximum increase in the 3PH short circuit current is 9.16% (1119.5 A), and the maximum increase in the SLG short circuit current is 26.78% (2413.0 A), which are both observed at the J534 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 13335 A (increased by 1119.5 A) and SLG fault current is 11425 A (increased by 2413.0 A) at the J534 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J534 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J534

Substation	Faulted Bus			Benchmark Case without J534 Project		Study Case with J534 Project		Ampere Change with J534 vs Benchmark		% Change with J534 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J534	66009	J534_TER2	13.8	-	-	0.0	38787.0	-	-	-	-
J534	66008	J534_TER1	13.8	-	-	0.0	38787.0	-	-	-	-
J534	66007	J534_COL2	34.5	-	-	27709.5	22309.0	-	-	-	-
J534	66006	J534_COL1	34.5	-	-	27709.5	22309.0	-	-	-	-
J534	66004	J534 COL2	34.5	-	-	31209.3	23318.2	-	-	-	-
J534	66002	J534 COL1	34.5	-	-	31209.3	23318.2	-	-	-	-
J534	66001	J534	345	-	-	11331.0	13204.1	-	-	-	-
J534	66000	J534_POI	345	9011.9	12215.5	11424.9	13335.0	2413.0	1119.5	26.78%	9.16%
MVP SCS KOSSUTH COUN	500103	KOSSUTH CO	345	12236.9	15105.6	13296.1	15802.7	1059.2	697.1	8.66%	4.61%
PLUM CREEK WIND FARM	10834	J501-345-BT	345	12236.9	15105.6	13296.1	15802.7	1059.2	697.1	8.66%	4.61%
WEB WEBSTER	10662	WEB-345	345	11283.7	13342.4	11813.0	13759.6	529.3	417.2	4.69%	3.13%
MVP SCS LEDYARD	500104	LEDYARD 345	345	9597.2	13156.1	9877.2	13444.6	280.0	288.5	2.92%	2.19%
PALO ALTO CNTY WIND	500132	J529-345-BT	345	10494.1	12549.1	10777.8	12822.8	283.7	273.7	2.70%	2.18%
PALO ALTO CNTY WIND	500032	J529-345	345	10494.1	12549.1	10777.8	12822.8	283.7	273.7	2.70%	2.18%
LEH LEHIGH	10775	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	188	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	187	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	186	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11453.1	13993.8	11674.5	14231.8	221.4	238.0	1.93%	1.70%
RED ROCK NORTH STAR	500030	RED ROCK TAP	345	9585.3	11390.5	9689.0	11518.8	103.7	128.3	1.08%	1.13%
MVP SCS HUNTLEY	500105	HUNTLEY 345	345	8367.7	11202.8	8463.1	11327.8	95.4	125.0	1.14%	1.12%
WEB WEBSTER	10652	WEB-161	161	19220.4	18918.6	19487.0	19095.4	266.6	176.8	1.39%	0.93%
J524	96996	J524	161	17407.6	17600.3	17620.2	17752.8	212.6	152.5	1.22%	0.87%
BEAVER CREEK WIND PR	11778	J498-345-BT	345	9868.2	12803.8	9943.1	12909.4	74.9	105.6	0.76%	0.82%
BEAVER CREEK WIND PR	10778	J498-345	345	9868.2	12803.8	9943.1	12909.4	74.9	105.6	0.76%	0.82%
RED ROCK WIND FARM J	500044	RR-9T3-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500043	RR-9T2-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500042	RR-9T1-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500041	RROCK-GEN-BT	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500040	RED ROCK-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
UPLAND PRARIE WIND F	500036	J455-345	345	9417.1	12147.5	9468.7	12228.8	51.6	81.3	0.55%	0.67%
NORTH STAR WIND FARM	500050	N STAR-345	345	5860.4	7247.0	5892.3	7295.0	31.9	48.0	0.54%	0.66%
MVP SBC OBRIEN COUNT	500001	OBRIEN-345	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J534

Substation	Faulted Bus			Benchmark Case without J534 Project		Study Case with J534 Project		Ampere Change with J534 vs Benchmark		% Change with J534 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
SBC OBRIEN COLLECTOR	11830	COLECTOR-GEN	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%
SBC OBRIEN COLLECTOR	10830	COLECTOR-345	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%

J535 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J535 generating facility (84 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J535 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J530, J534 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J535 generating facility with its sequence parameters at approximately 44 miles from Lehigh on Lehigh – J411 345 kV line. The associated 345 kV interconnection generation tie line, 345-34.5-13.8 kV transformer ($Z_+ = 9\%$, $Z_0 = 6.7\%$ at 150 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.69 kV pad mount transformer ($Z = 5.75\%$ at 235.2 MVA, D-YG) were also added.
- Modeled the J535 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): j0.2 p.u.
 - Negative sequence impedance (Z_2): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance (Z_0): j0.2 p.u.

The J535 short circuit benchmark model was developed as follows:

- Turned off J535 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J535 project) and benchmark case (without the J535 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J535 (study case) and without J535 (benchmark case) are listed in Table 1. With the addition of the J535 generating facility, the maximum increase in the 3PH short circuit current is 10.66% (950.0 A), and the maximum increase in the SLG short circuit current is 32.09% (2040.0 A), which are both observed at the J535 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 9858 A (increased by 950.0 A) and SLG fault current is 8398 A (increased by 2040.0 A) at the J535 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J535 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J535

Substation	Faulted Bus			Benchmark Case without J535 Project		Study Case with J535 Project		Ampere Change with J535 vs Benchmark		% Change with J535 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J535	66206	J535_TER	13.8	-	-	0.0	61330.5	-	-	-	-
J535	66205	J535_COL	34.5	-	-	44527.2	35055.8	-	-	-	-
J535	66203	J535 COL	34.5	-	-	47736.3	35775.1	-	-	-	-
J535	66202	J535	345	-	-	8352.8	9788.2	-	-	-	-
J535	66201	J535_POI	345	6358.0	8908.1	8398.0	9858.1	2040.0	950.0	32.09%	10.66%
FIC IDA CO SUB IDA G	11776	GEN BUS-345	345	7273.9	9733.8	8223.5	10299.9	949.6	566.1	13.05%	5.82%
FIC IDA CO SUB IDA G	10776	IDA CNTY-345	345	7273.9	9733.8	8223.5	10299.9	949.6	566.1	13.05%	5.82%
IDA CO WEST WIND FAR	10779	J412-345	345	7218.9	10014.2	7899.1	10479.1	680.2	464.9	9.42%	4.64%
LEH LEHIGH	10775	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	188	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	187	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	186	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11226.9	13891.5	11674.5	14231.8	447.6	340.3	3.99%	2.45%
WEB WEBSTER	10662	WEB-345	345	11624.2	13575.3	11813.0	13759.6	188.8	184.3	1.62%	1.36%
BEAVER CREEK WIND PR	11778	J498-345-BT	345	9798.7	12757.5	9943.1	12909.4	144.4	151.9	1.47%	1.19%
BEAVER CREEK WIND PR	10778	J498-345	345	9798.7	12757.5	9943.1	12909.4	144.4	151.9	1.47%	1.19%
RAU RAUN	10840	RAU-345-A	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	485	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	484	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	483	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	482	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	481	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	479	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	478	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	477	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	476	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	475	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	474	RAU-345-B	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
NLN NEAL NORTH	11542	NLN 3 345	345	26199.5	24808.5	26396.1	25004.5	196.6	196.0	0.75%	0.79%
NL4 NEAL SOUTH	11207	NEAL 4 345	345	25215.9	23438.7	25384.8	23606.8	168.9	168.1	0.67%	0.72%
J534	66000	J534_POI	345	11374.6	13268.4	11424.9	13335.0	50.3	66.6	0.44%	0.50%
J534	66001	J534	345	11281.7	13138.9	11331.0	13204.1	49.3	65.2	0.44%	0.50%

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2021 Cost Allocation Results

J.1 Distribution Factor (DF) and MW Contribution Results for Cost Allocation in 2021

Table J-1: Distribution Factor and MW Contribution on Constraints for Backbone NU Cost Allocation

Table J-2: Distribution Factor and MW Contribution on Constraints for Other Thermal NU Cost Allocation

Table J-3: Distribution Factor and MW Contribution on MWEX Constraints for NROC-Rocky Run NU Cost Allocation

Table J-4: Distribution Factor and MW Contribution on MWEX Constraints for Other NUs (except NROC-Rocky Run) Cost Allocation

Table J-5: Distribution Factor and MW Contribution on Voltage Constraints for Other Reactive Power NU Cost Allocation

Table J-6: SPP Suggested Mitigations within DPP 2016 February West Area Study

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J.2 Cost Allocation Details

Table J-7: Network Upgrades Cost Allocation in 2021

Monitored Element			English Name	Cost	J432	J460	J475	J485	J488	J489	J490	J493	J495	J498	J499	J500	J501	J504	J506	J510	J514	J523	J524	J525	J526	J527	J528	J529	J530	J531	J534	J535	Upgrade For																																																																																																																																																																																																																																																																																																																																																																															
Big Stone South-Alexandria 345 kv			Big Stone South-Alexandria 345 kv	\$197,400,000	\$11,088,653	\$22,594,651			\$0	\$29,088,417	\$29,088,417																						Upgrade for Alexandria 345 kv																																																																																																																																																																																																																																																																																																																																																																															
New Hartford-Morgan Valley-Beverly-Sub 92 345 kv			New Hartford-Morgan Valley-Beverly-Sub 92 345 kv	\$468,100,000	\$1,544,477	\$6,404,796	\$33,346,570		\$0	\$2,229,731	\$2,229,731	\$530,588	\$2,325,313	\$12,974,127	\$84,283,894	\$56,002,663	\$41,583,335	\$34,558,377		\$0	\$8,181,832		\$0			\$0	\$4,622,240	\$19,205,450	\$15,209,024	\$13,919,483	\$32,357,780	\$48,098,474	\$19,159,783	\$25,788,778	Backbone NU Big Stone South-Alexandria 345 kv																																																																																																																																																																																																																																																																																																																																																																													
N Rochester-Rocky Run 345 kv			N Rochester-Rocky Run 345 kv	\$798,787,928	\$31,551,797	\$65,782,035		\$0	\$57,188,398	\$57,188,398	\$19,012,701	\$50,424,832	\$33,302,155	\$35,378,401	\$22,427,562	\$30,533,022	\$89,694,530		\$0	\$34,613,691		\$0			\$0	\$102,391,430	\$15,190,934	\$12,116,470	\$46,026,263	\$0	\$23,567,885	\$42,519,500	\$28,453,753	New 345 kv Line North Roshester - Rocky Run																																																																																																																																																																																																																																																																																																																																																																														
Other Nus for mMex Voltage Stability			Other Nus for mMex Voltage Stability	\$227,412,672	\$9,202,638	\$10,349,253		\$0	\$16,414,337	\$16,414,337	\$5,217,113	\$14,651,349	\$0,329,678	\$10,657,108	\$6,319,874	\$8,614,761	\$25,853,219		\$0	\$16,087,492		\$0			\$0	\$29,720,925	\$4,285,039	\$3,413,733	\$13,381,941	\$0	\$4,464,662	\$12,229,565	\$8,152,260	Other Nus for mMex Voltage Stability																																																																																																																																																																																																																																																																																																																																																																														
2*50 Mvar cap at Zachary 161 kv (344018)			2*50 Mvar cap at Zachary 161 kv (344018)	\$1,000,000	\$8,743	\$17,267	\$133,900		\$0	\$12,381	\$12,381	\$5,366	\$13,077	\$18,514	\$74,457	\$86,489	\$118,703	\$61,607		\$0	\$25,483		\$0			\$0	\$25,930	\$60,662	\$48,570	\$30,431	\$167,350	\$5,671	\$33,158	\$34,890	Other Reactive Power NU																																																																																																																																																																																																																																																																																																																																																																													
5*50 Mvar SVC at Overton 161 kv (345409)			5*50 Mvar SVC at Overton 161 kv (345409)	\$25,000,000	\$473,755	\$928,682	\$158,405		\$0	\$690,231	\$690,231	\$350,749	\$713,132	\$732,522	\$1,853,830	\$2,284,772	\$3,775,465	\$2,305,291		\$0	\$1,246,724		\$0			\$0	\$1,417,031	\$1,842,591	\$1,475,715	\$1,199,839	\$26,162	\$302,858	\$1,178,807	\$1,264,247	Other Reactive Power NU																																																																																																																																																																																																																																																																																																																																																																													
2*150 Mvar cap at Hickory 345 kv (631191)			2*150 Mvar cap at Hickory 345 kv (631191)	\$8,000,000	\$146,511	\$303,155	\$236,166		\$0	\$221,299	\$221,299	\$77,728	\$222,951	\$501,643	\$475,857	\$353,759	\$485,996	\$965,020		\$0	\$301,358		\$0			\$0	\$444,972	\$245,089	\$195,638	\$467,617	\$343,621	\$952,781	\$469,419	\$308,162	Other Reactive Power NU																																																																																																																																																																																																																																																																																																																																																																													
1*150 Mvar cap at New Hartford 345 kv (631240)			1*150 Mvar cap at New Hartford 345 kv (631240)	\$4,000,000	\$27,837	\$148,479	\$31,282		\$0	\$102,938	\$102,938	\$36,611	\$109,038	\$577,205	\$219,702	\$83,309	\$114,496	\$709,162		\$0	\$186,509		\$0			\$0	\$216,262	\$57,353	\$45,742	\$368,032	\$39,701	\$163,163	\$364,938	\$168,013	Other Reactive Power NU																																																																																																																																																																																																																																																																																																																																																																													
1*150 Mvar cap at Blackhawk 345 kv (636199)			1*150 Mvar cap at Blackhawk 345 kv (636199)	\$4,000,000	\$43,858	\$82,980	\$48,387		\$0	\$49,804	\$49,804	\$23,837	\$62,431	\$534,039	\$318,156	\$201,206	\$274,134	\$742,881		\$0	\$184,540		\$0			\$0	\$121,467	\$138,100	\$110,089	\$344,245	\$61,408	\$0	\$358,115	\$204,093	Other Reactive Power NU																																																																																																																																																																																																																																																																																																																																																																													
1*150 Mvar cap at Montezuma 345 kv (635730)			1*150 Mvar cap at Montezuma 345 kv (635730)	\$4,000,000	\$37,008	\$71,745	\$485,219		\$0	\$51,278	\$51,278	\$24,883	\$55,039	\$60,618	\$357,662	\$437,774	\$607,533	\$255,018		\$0	\$117,242		\$0			\$0	\$108,903	\$309,472	\$247,920	\$127,539	\$235,959	\$41,667	\$140,465	\$164,400	Other Reactive Power NU																																																																																																																																																																																																																																																																																																																																																																													
2*150 Mvar cap at Hills 345 kv (636400)			2*150 Mvar cap at Hills 345 kv (636400)	\$8,000,000	\$55,784	\$114,729	\$1,931,108		\$0	\$84,968	\$84,968	\$30,276	\$85,072	\$188,945	\$480,965	\$624,773	\$882,508	\$348,843		\$0	\$144,504		\$0			\$0	\$169,018	\$448,470	\$359,348	\$170,131	\$1,963,719	\$309,915	\$175,201	\$214,360	Other Reactive Power NU																																																																																																																																																																																																																																																																																																																																																																													
2*150 Mvar cap at Morgan Valley 345 kv (631148)			2*150 Mvar cap at Morgan Valley 345 kv (631148)	\$8,000,000	\$192,143	\$412,440		\$0	\$312,338	\$312,338	\$86,476	\$298,409	\$764,687	\$305,560	\$138,500	\$175,753	\$1,162,102		\$0	\$286,584		\$0			\$0	\$599,879	\$88,100	\$69,930	\$550,097	\$0	\$1,383,381	\$538,620	\$240,146	Other Reactive Power NU																																																																																																																																																																																																																																																																																																																																																																														
51113 J438 POI			161 631885 PARNEL 5	161 1		\$200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0

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